Long Paper — Working Draft

Reasoning States and Expert Routing in Mixture-of-Experts Language Models

Author information to be completed*

Affiliations and corresponding-author details to be completed

Mixture-of-experts language models select a subset of feed-forward experts for each token. We examine what these selections reveal about labelled mathematical reasoning across seven checkpoints and six benchmarks. On a common population of 522 questions in 509 problem families, routing improves held-out reasoning-label prediction beyond a retrospective text baseline in every model, reducing cross-entropy by 0.079–0.149 nats per sentence. The original corrected intervals exclude zero, including under a lexical-null comparison. A separate next-class analysis finds predictive information beyond past text and history for GPT-OSS and Qwen3.6; its scope is limited to those two checkpoints. We also compare expert keep-sets learned on training families at 75% retention per layer. Across five folds covering 1,486 families, state-balanced selection improves minimum held-out class coverage over labelled-frequency selection in five models; frequency performs better in Qwen3.5 and Qwen3.6. Thus reasoning labels provide information about expert allocation, while the benefit of balancing them depends on the model. These results concern prediction and coverage of saved native routes. Accuracy after expert removal and realized resident memory savings remain unestablished.

1. Introduction

A mixture-of-experts (MoE) language model routes each token to a subset of its feed-forward experts. The resulting sequence of expert selections provides an internal view of how computation changes while the model solves a problem. A useful account must distinguish three questions: which reasoning states routing describes, which changes routing predicts, and whether changing that routing improves the solution or its cost. The answer to one question does not determine the others.

We examine seven checkpoints: GPT-OSS-20B, Nemotron-3.5-Lightning-30B-A3B, Qwen3.5-35B-A3B, Qwen3.6-35B-A3B, Gemma-4-26B-A4B, GLM-4.7-Flash and Qwen3-30B-A3B. The preserved corpus contains benchmark answers, routing captures and reasoning labels. Its common analysis supports model comparisons without equating expert indices or router score scales across architectures. Problem families define independent units; sentences, tokens and repeated answers supply measurements within those units.

* Draft author list; affiliation and contact details pending.

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Our main contribution has two seven-model results. First, routing adds held-out information about reasoning labels beyond the specified text baseline. Second, a training-only comparison of expert keep-sets shows that balancing reasoning classes improves minimum held-out coverage in five models, while pooled labelled frequency performs better in two. A separate two-model next-class analysis tests predictive information before the destination sentence. Together these results distinguish information about reasoning episodes from a useful rule for allocating experts. The deployment question remains whether a chosen keep-set can reduce resident expert memory while preserving answer accuracy; the saved-route analyses do not answer it.

The main text reports the comparable analysis first. Appendices preserve the historical annotation, probing, three-model correlations, interventions, current verification audit and full figure atlases. These studies retain their own populations and measurement rules. They are evidence for particular parts of the workflow, rather than repeated estimates of one common effect.

2. Reasoning Labels, Model Contracts and Benchmarks

2.1 Reasoning labels and related approaches

The episode vocabulary is Read, Analyze, Explore, Plan, Implement, Verify and Monitor. The annotation and external probing studies build on the released SAT-response annotations of **?**. The prompt selection procedure uses GEPA (**?**); the layerwise prospective probing design adapts **?**. These are labels assigned to text, not independently established modules inside a model. When a classifier predicts such a label from routing, the supported claim concerns the information in routing about that label.

The benchmark selection and repeat counts adapt the mathematical evaluation suite used by **?**. Expert intervention is related to the (de)activation approach of **?**, while our historical policies retain their own selection estimator, implementations and evaluation populations. We do not treat those policies as a complete replication of that method.

Expert pruning already has dedicated methods. EAC-MoE combines router calibration for quantization with pruning based on selection frequency (**?**). REAP instead uses routing weights together with expert-output magnitudes in its pruning criterion (**?**). Counterfactual routing analysis holds model parameters fixed and compares native routes with alternatives using equal computation (**?**). These works make frequency, activation-sensitive selection and alternative-route evaluation relevant comparisons. Our comparison adds reasoning-class balance to a controlled seven-model selection analysis using training families only. A new pruning claim requires a controlled comparison with relevant pruning methods.

2.2 A model-specific routing contract

Let $S_{\ell,t}$ be the selected expert set at routed layer ℓ and token position t, and let $k_\ell = |S_{\ell,t}|$ be its native size. Every expert identity includes checkpoint, layer and local index. A capture binds its token ownership, score space, selection rule and any shared-expert path. This matters for architectures with sigmoid scores, correction biases, grouped selection or learned scaling; a common raw-score manipulation need not implement a common action.

Model	Routed layers	Expert pool	Native k
GPT-OSS	24	32	4
Nemotron	23	128	6
Qwen3.5	40	256	8
Qwen3.6	40	256	8
Gemma	30	128	8
GLM	46	64	4
Qwen3-30B	48	128	8

Table 1
Captured routed-layer and selection contracts for the seven checkpoints. Routed layers exclude dense blocks; expert pools and local indices do not align across models. Exact checkpoints and quantization belong to the source manifest.

Table 2
Saved seven-model card-profile mathematical accuracy (%). Each model has 4,647 assigned attempts on the same 1,547 problems (32,529 attempts in total). AIME24/25 and AMC23 use 32 samples per problem; the other datasets use one. Entries are adjudicated accuracy among scored attempts. U counts unresolved attempts per model, retained separately. These checkpoints use different native decoding profiles and output limits; this table is a benchmark description, not an equal-budget architecture comparison.

Model Problems	AIME24 30	AIME25 30	AMC23 40	MATH500 500	Minerva 272	Olympiad 675	U
GPT-OSS	78.7	76.0	97.3	97.0	69.1	83.2	11
Nemotron	87.8	87.2	99.8	98.8	73.5	88.6	0
Qwen3.5	94.0	88.3	100.0	99.4	73.4	92.3	1
Qwen3.6	94.8	89.7	100.0	99.6	74.8	92.0	2
Gemma	90.8	86.8	99.7	99.0	75.6	90.8	15
GLM	93.2	89.6	99.4	98.4	73.1	90.5	2
Qwen3-30B	80.2	71.1	97.7	98.0	75.6	84.6	2

The source inventory retains strict scores, all denominators, completion-token costs, caps, and the original question-bootstrap 95% intervals. Unresolved outcomes total 33; repeated samples are not independent questions. The historical 32k campaign is a separate 32,529-attempt population.

2.3 Benchmark populations and answer scoring

The six benchmarks are AIME24, AIME25, AMC23, MATH-500, OlympiadBench and Minerva. The complete native benchmark corpus contains 4,647 attempts per model over 1,547 source questions: 32 attempts per AIME24/AIME25/AMC23 question and one on the other benchmarks. The routing analyses use their own captured subsets. In particular, the latest seven-model observational A corpus contains 1,547 attempts per model, one per source question; the repeated B cohort contains 895 attempts but only 51 distinct questions in the cross-model union. The historical benchmark run likewise contains 4,647 attempts for each of all seven models under its original deployment contract; the older report sections analyzed three-model subsets of that run. These populations must not be pooled into a new independent sample.

Accuracy tables retain their source scorer and missing-answer rules. Token-limit completions and scoring failures remain visible; a comparison on scorable answers is not accuracy over all assigned requests. Historical intervention comparisons were separately rescored under a common answer-equivalence rule, so their values do not overwrite observational labels. Mathematical benchmark scores describe the workload

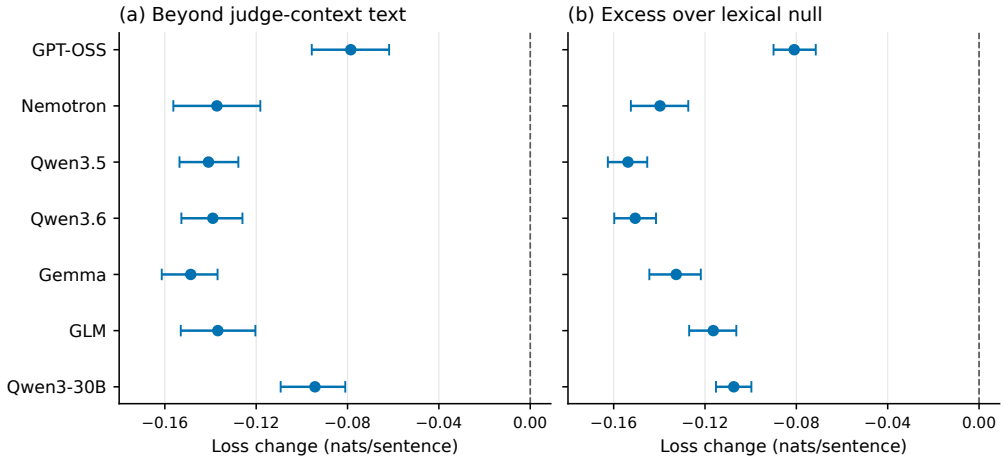


Figure 1
Held-out present-class readout on the common 522-question, 509-family population. (a) Routing-augmented minus retrospective judge-context loss. (b) Routing-augmented minus the mean lexical-null loss. Negative values favor routing. Bars copy the original model-specific seven-comparison 95% intervals; they are not a joint simultaneous test across all models and both panels. Questions receive equal weight; sentences within questions are repeated measurements. Exact values are in Appendix I.

and anchor utility evaluation; high native accuracy does not establish the effect of a proposed routing policy.

3. Routing Information About Reasoning States

The CLEAN present-class analysis covers 522 questions and 509 problem families for each checkpoint. It compares routing-augmented prediction with a retrospective baseline that already observes the annotation reader's text context. Families are held together during fitting and assessment. The saved seven-comparison simultaneous intervals for the loss difference exclude zero in all seven models (Figure 1). Lower cross-entropy favors routing. Separately reported lexical-null excess intervals also exclude zero under their saved procedure.

The readout uses multinomial ridge classification with five outer and three inner duplicate-family folds. The baseline includes token-identity features, position and sentence length, together with the original previous, current and next sentence text available to the annotation reader; a missing sentence stays missing. Routing adds top-1 expert histograms reduced to 64 components, depth-path surprise and temporal persistence. Learned transforms and regularization are fitted within training folds. Questions receive equal total weight. The saved uncertainty resamples whole duplicate families 5,000 times, retaining that question-weighted estimand. Five lexical-null draws permute token routing paths within token-identity and absolute-position strata and repeat the same fitting procedure. These controls test incremental information about the assigned label, not an online recognition policy.

This common direction grounds further reasoning-state and expert analyses: expert allocation contains information beyond the declared text baseline. The sizes of the gains are conditional on each model's generated text and label coverage. Comparing the rows

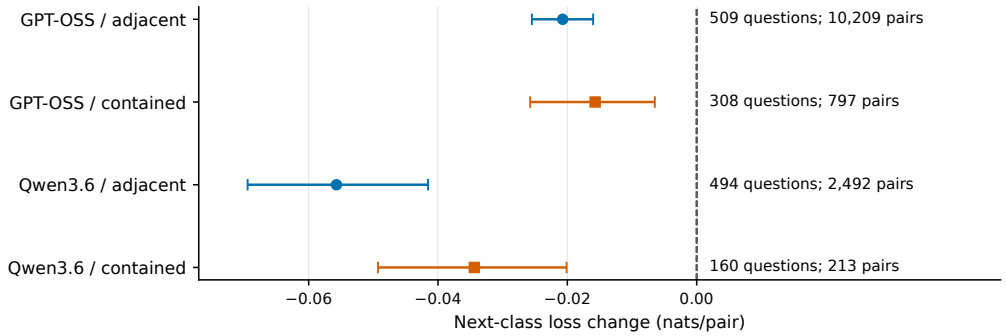


Figure 2
Next-class loss changes for the original 16-token routing composition endpoint and its contained-window sensitivity. Each contained source sentence has at least 32 tokens, keeping the declared trailing windows inside it. Intervals copy the saved model-specific six-comparison 95% procedure. Estimates average pairs within questions before averaging questions; whole-family resampling retains clustering. Only GPT-OSS and Qwen3.6 have completed equivalent analyses in this evidence set.

does not by itself identify a more interpretable model or a more effective intervention layer.

An independent part of the evidence concerns prospective hidden-state probes on externally supplied SAT responses. Five response-disjoint folds and inner-validation layer selection yield macro-AUROC 0.852 $[0.838, 0.868]$ for Qwen3.5, 0.780 $[0.761, 0.797]$ for GPT-OSS and 0.788 $[0.772, 0.805]$ for Gemma. Each exceeds the corresponding prefix-position baseline. These intervals resample 38 responses conditional on the saved folds and fits. The result supports held-out-response information on the external corpus; it does not establish a deployable detector on each model's own generations. The complete probe analysis remains in the appendix.

4. Next-Class Prediction

An adjacent-state prediction uses the original sentence order. Missing labels and sentence gaps break adjacency. Descriptive transition matrices for all seven models remain in Appendix I.5; they describe observed labelled adjacencies rather than complete trajectories.

The stricter historical next-class analysis asks whether routing observed before the destination adds to past text/history and a training-only estimate of the current class. GPT-OSS improves log loss by 0.02073 nats per pair on 10,209 adjacent pairs from 509 questions; Qwen3.6 improves it by 0.05570 on 2,492 pairs from 494 questions. The candidate-minus-baseline intervals are respectively $[-0.02548, -0.01601]$ and $[-0.06944, -0.04154]$ under the saved six-comparison correction. Contained-window sensitivities retain the direction, with substantially smaller support: 797 pairs for GPT-OSS and 213 for Qwen3.6 (Figure 2).

These results ground an anticipation hypothesis. They do not establish fine-grained trigger timing, and equivalent completed analyses are missing for the other five checkpoints. State-switch prediction is a separate endpoint: GPT-OSS's smaller switch improvement does not pass its historical practical gate. A deployable controller requires evidence about its particular decision and time horizon.

5. Family-Held-Out Expert Selection

5.1 Training-only selection and common controls

We compare state-balanced retention, pooled labelled-token frequency and the original fixed random control at 75% retained expert identities per routed layer. The layer quotas are 24, 96, 192, 192, 96, 48 and 96 experts for GPT-OSS, Nemotron, Qwen3.5, Qwen3.6, Gemma, GLM and Qwen3-30B, respectively. These are selections evaluated on saved native traces; no expert is removed during this analysis. Serialized checkpoint bytes and actual GPU residence are separate quantities.

Five common retrospective folds group the 1,547 historical questions into 1,486 duplicate families. A family has total weight one, shared equally among its member questions. For class c, layer ℓ and expert e, first average selected-slot counts over questions within family f, giving $C_{fc\ell e}$. The state-balanced score is

$$s^{\mathrm{bal}}_{\ell e} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathcal{F}_c|} \sum_{f \in \mathcal{F}_c} \frac{C_{fc\ell e}}{\sum_j C_{fc\ell j}}, \tag{1}$$

where $\mathcal{F}_c$ contains training families with observed class c. Every class requires training support. The frequency control pools the same family-weighted labelled counts, proportional to their observed token mass:

$$s^{\mathrm{freq}}_{\ell e} = \frac{\sum_{f,c} C_{fc\ell e}}{\sum_{f,c,j} C_{fc\ell j}}. \tag{2}$$

Both learned methods use training families only, identical layer/group quotas and original expert ID to break ties. Neither uses correctness to rank experts. Random priorities use seed 20261008 and fix one set per model; the comparison is conditional on those sets. The frequency arm uses labelled tokens, so it is not a comparison against every conventional all-token or output-aware pruning criterion.

For each held-out fold and class, coverage is the retained fraction of family-weighted native selected slots at each layer, averaged equally over routed layers. The primary descriptive summary takes the minimum over all seven classes. Every displayed fold has complete class coverage. It measures which original selections a keep-set contains, rather than what the model would do after rerouting.

5.2 Observed model-dependent coverage

The complete five-fold analysis covers all seven models (Figure 3). State balancing has higher minimum class coverage than pooled labelled frequency in every fold for GPT-OSS, Nemotron, Gemma, GLM and Qwen3-30B. Pooled labelled frequency is higher in every fold for Qwen3.5 and Qwen3.6. The arithmetic mean paired differences across folds are $+1.77$, $+4.58$, $+0.21$, $+2.79$ and $+0.59$ percentage points for GPT-OSS, Nemotron, Gemma, GLM and Qwen3-30B, respectively. They are -1.35 for Qwen3.5 and -0.55 for Qwen3.6. The fixed random masks have minimum class coverage between 73.4% and 77.2% across the model/fold cells. All five folds are shown, including the two models favoring frequency. Their spread describes partition stability, not a confidence interval; folds with overlapping training data are not independent replications.

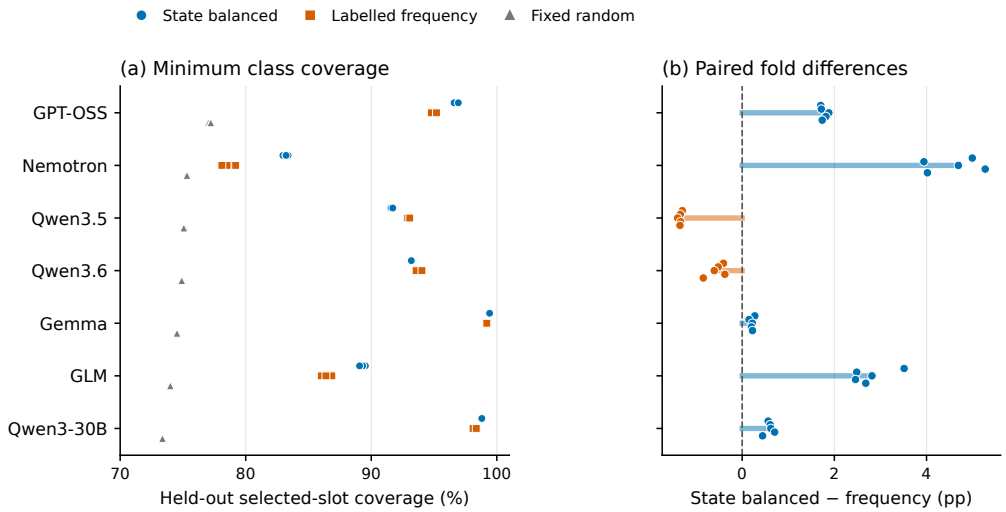


Figure 3
Minimum held-out reasoning-class coverage at 75% expert retention. (a) All five common duplicate-family folds for each selection method; horizontal lines connect the fold extrema. (b) All five paired state-balanced-minus-frequency differences; bars end at their descriptive fold mean. Coverage averages layers within class before taking the minimum. The comparison uses training-only selection and identical quotas. Neither fold ranges nor mean bars are inferential intervals.

The comparison shows that balancing labelled states can improve the least-covered class at a fixed identity budget, but the direction and magnitude depend on the model. Near-complete coverage in Gemma and Qwen3-30B also leaves little room for a large coverage improvement. These observations establish neither accuracy retention after physical removal nor a realized GPU-memory saving. Missing and unlabelled tokens remain outside this class-conditioned endpoint.

6. Interventions and Deployment Limits

Historical sparse interventions demonstrate that candidate experts can be engaged: pair-target inclusion changes by $+0.1376$ $[0.0763, 0.1882]$ across eight families in study A and $+0.2940$ $[0.1277, 0.4761]$ across thirteen in study B. This verifies an effect on routing. The larger FRESH verification comparison estimates a semantic success change of $+5.5$ percentage points against native, with simultaneous 95% interval $[-4.5, +15.5]$. Missing semantic judgments widen the identification range. The retained global-steering comparisons also show that some routing policies can substantially reduce accuracy; all reported policies and their uncertainty remain in the appendix.

For a user deciding whether to deactivate experts or constrain routes, usefulness is a paired change in answer quality and resource use on independent evaluation families. A nonsignificant accuracy difference does not establish retention. Such a claim needs a prespecified acceptable loss margin and an interval precise enough to assess it. Expert exclusion also needs matched random exclusions with comparable exposure, and actual measurement of the claimed deployment saving. Actual expert-memory reduction requires omitted expert weights to be absent from the measured resident model. Masking

an expert while retaining its weights is an intervention test, not evidence of a smaller memory footprint. The reported deployment objective must therefore pair memory measurements with accuracy and account for shared experts and the unchanged dense parameters. Such a reduced-memory deployment has not yet been measured here.

The separate verification study passed all 14 engineering checks for all seven checkpoints. Its source-bound observation snapshot reports model-specific recognition and control gates in Table H.2. A model that fails its full frozen recognition qualification stops before target nomination and treatment under that controller; its causal and utility cells remain unmeasured. These are limits of this controller and measurement pipeline, not evidence that all routing interventions fail. Every checkpoint retains its own gates; the full observed audits and the separate answer-grader limit are reported in Appendix H.

7. Discussion

Routing supplies additional information about automatically labelled reasoning episodes in all seven models under the declared text controls. The next-class results refine this finding for two checkpoints, where past routing adds to past text and history. Their effect sizes depend on the generated text, annotation coverage and prediction task. They neither establish latent cognitive modules nor identify a universal controller.

The held-out selection comparison supplies a further distinction: reasoning-class balance is useful for coverage in some architectures and less useful than labelled frequency in others. The common five-fold comparison makes that model dependence visible without using correctness to rank experts. Its endpoint is coverage of original selected slots; post-removal generation can change the routes and subsequent text. Thus even near-complete native coverage cannot establish preserved answer accuracy or realized memory savings.

Labels are measurements of visible text, with imperfect annotation and sparse coverage. Families define independent problem units, while saved uncertainty remains conditional on the original fits and selections. Exploratory same-cohort expert/correctness correlations and descriptive concentration curves are preserved in Appendix I with their original limitations. They do not select the main reported results or justify suppressing a class-associated expert. Historical intervention uncertainty, harmful policies and failed qualification gates also remain visible in the appendices.

The present evidence supports routing readout and model-dependent selection coverage. A deployment claim needs a separate controlled measurement of answer quality and resident memory after actual removal, including random and relevant pruning controls. That joint outcome remains unestablished for the seven-model comparison.

8. Observed continuation results

These results use the frozen seven-model study. Missing cells remain unavailable; qualification and scientific completion are reported separately. Accuracy uses the original strict scoring contract, with unknown-answer bounds. Intervals use 10,000 paired family resamples and equal benchmark/family weights. One replicate does not estimate run-to-run variability.

8.1 Accuracy and resident memory

The fixed evaluation has 420 questions in 400 families. Five pruning methods and both layer allocations are compared at 12.5, 25 and 50 percent removal, alongside native. Resident bytes come from qualified full-context fixtures with fixed KV capacity. Same-budget contrasts, paired uncertainty and unknown bounds are retained in the source-bound tables.

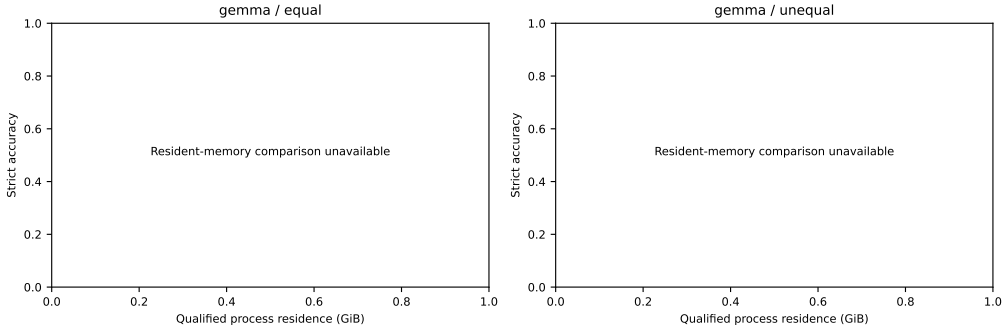


Figure 4
Accuracy versus measured resident memory; unavailable comparisons are explicit.

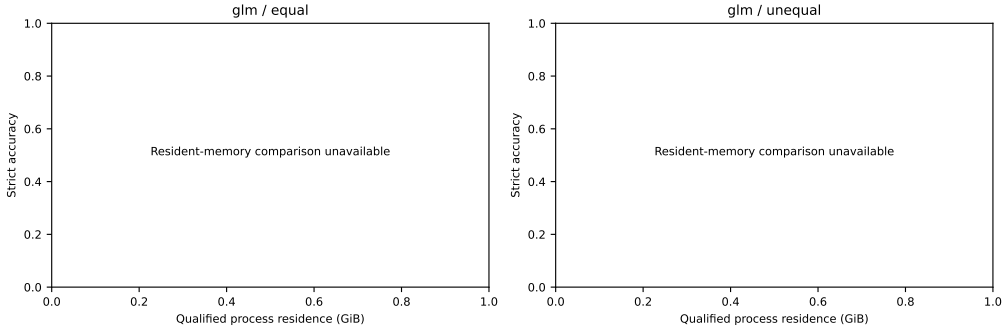


Figure 5
Accuracy versus measured resident memory; unavailable comparisons are explicit.

8.2 Central steering

The 129-family cohort retains all seven classes, three depths, both signs and native/random controls. First-complete-sentence response and post-pulse target occupancy share one simultaneous family over 1,176 contrasts and 2,352 bound components. Dwell, transitions, re-entry, recovery and target exits have separate supporting families; accuracy and tokens are reported jointly. Small or imprecise effects remain unresolved.

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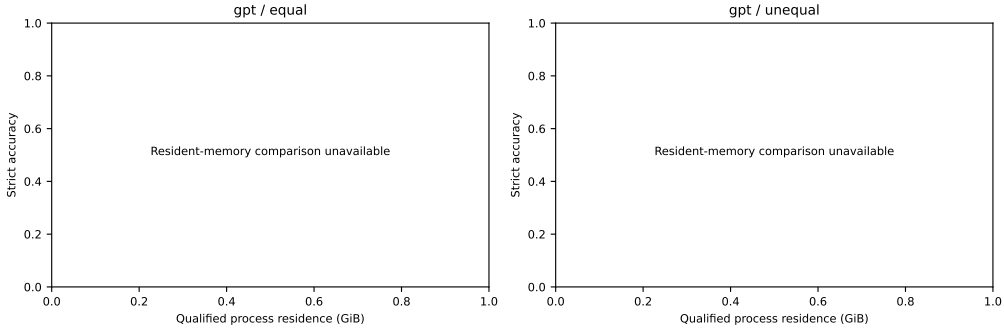


Figure 6
Accuracy versus measured resident memory; unavailable comparisons are explicit.

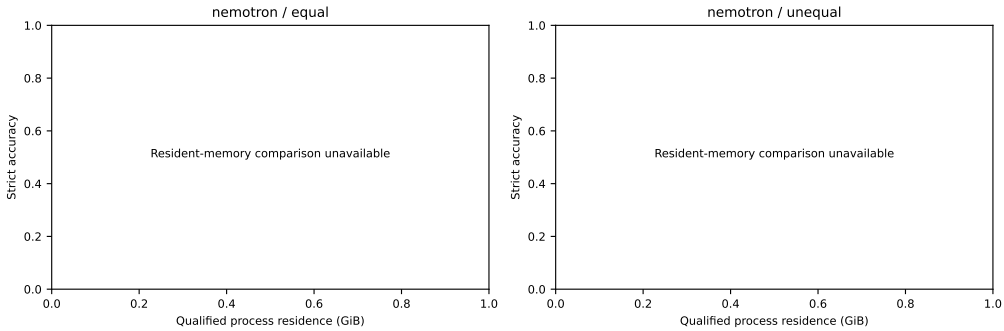


Figure 7
Accuracy versus measured resident memory; unavailable comparisons are explicit.

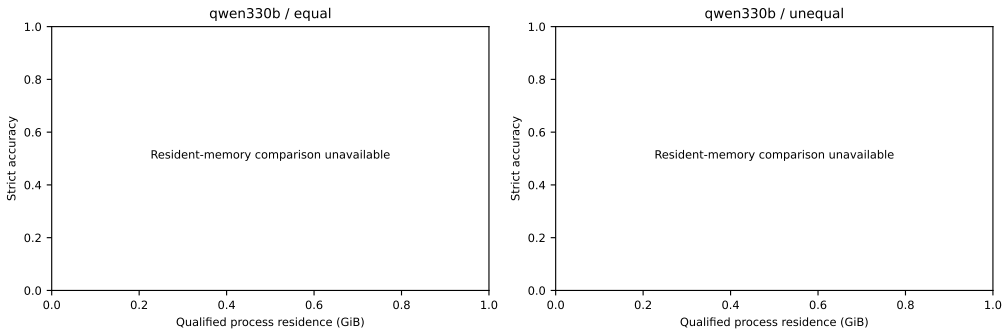


Figure 8
Accuracy versus measured resident memory; unavailable comparisons are explicit.

the complete author list, funding acknowledgments and material-availability statement before submission.

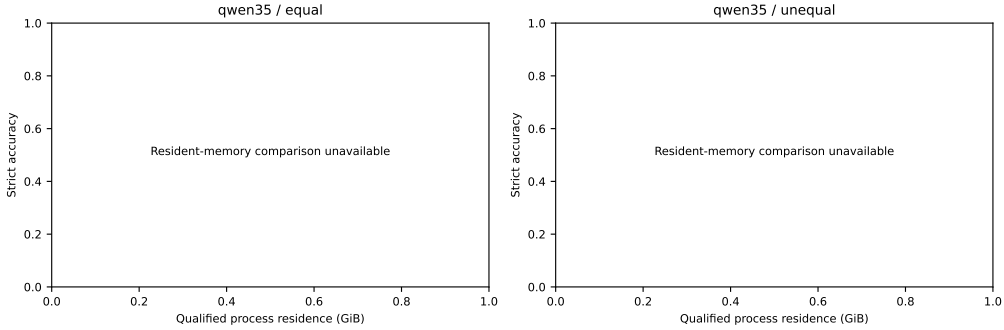


Figure 9
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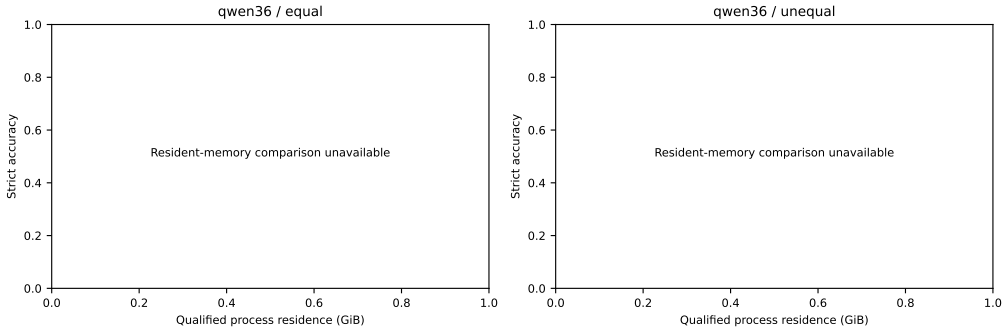


Figure 10
Accuracy versus measured resident memory; unavailable comparisons are explicit.

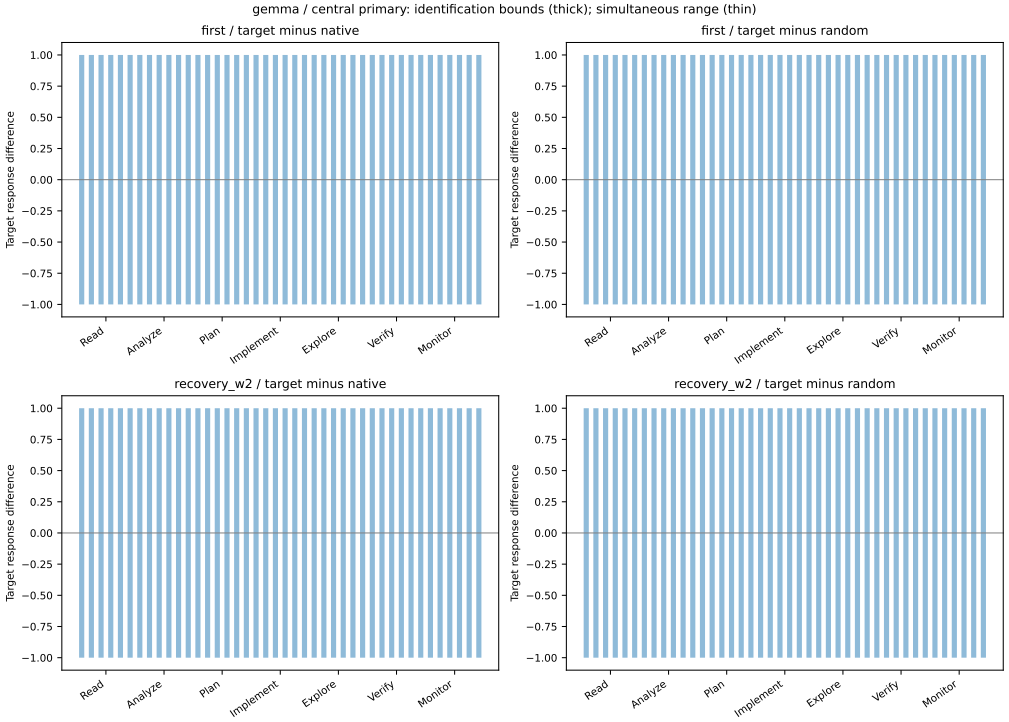


Figure 11
Central primary identification bounds and study-wide simultaneous uncertainty.

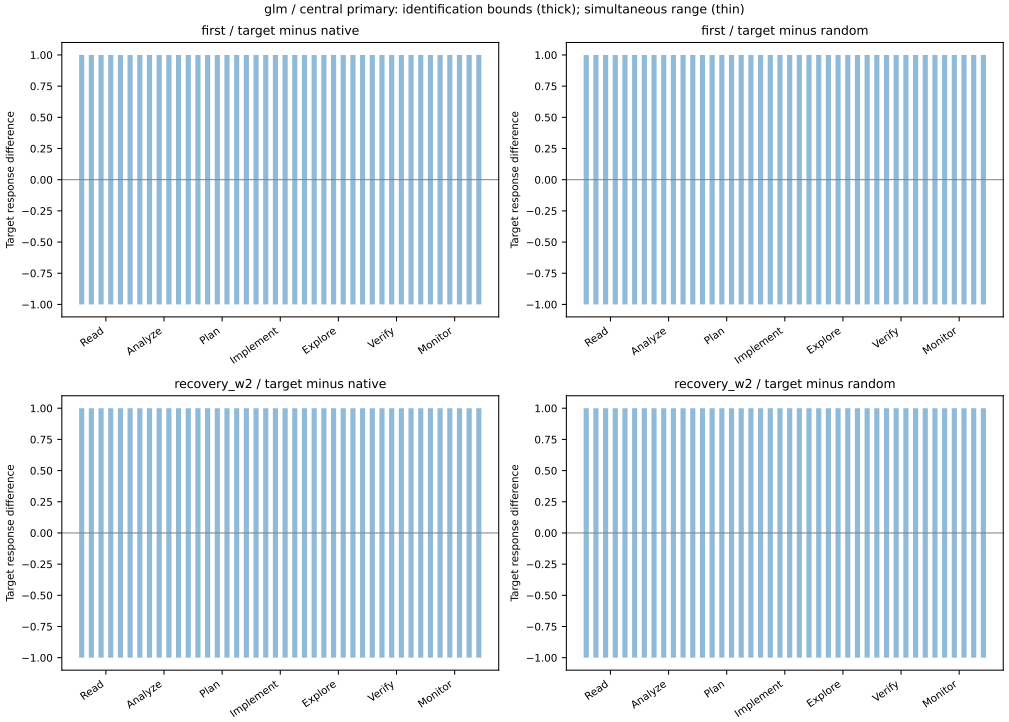


Figure 12
Central primary identification bounds and study-wide simultaneous uncertainty.

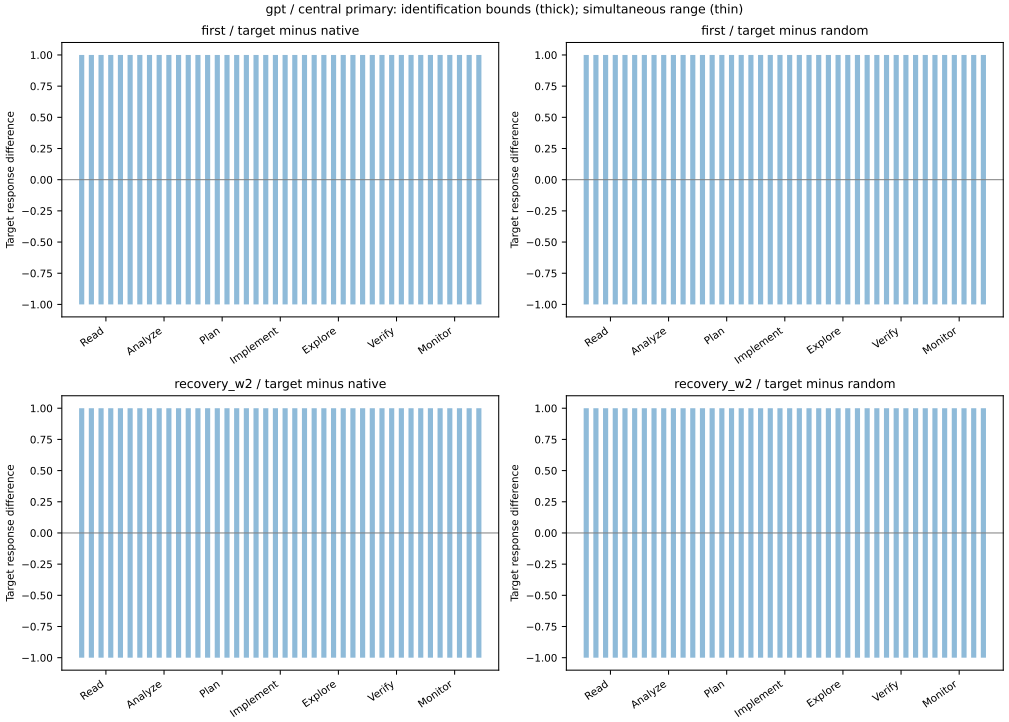


Figure 13
Central primary identification bounds and study-wide simultaneous uncertainty.

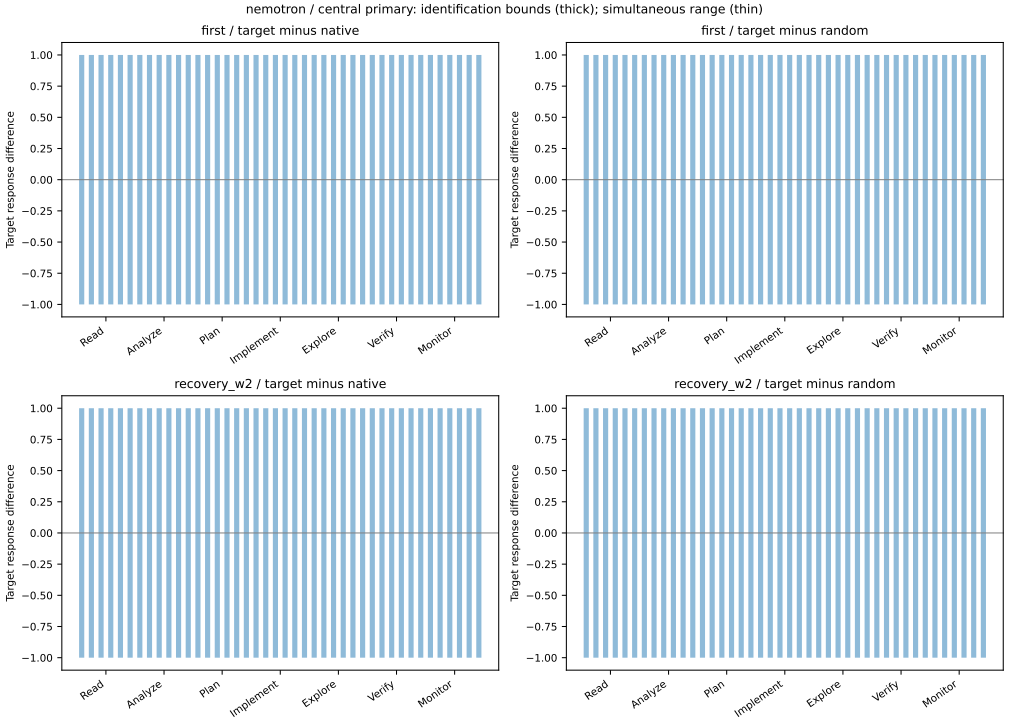


Figure 14
Central primary identification bounds and study-wide simultaneous uncertainty.

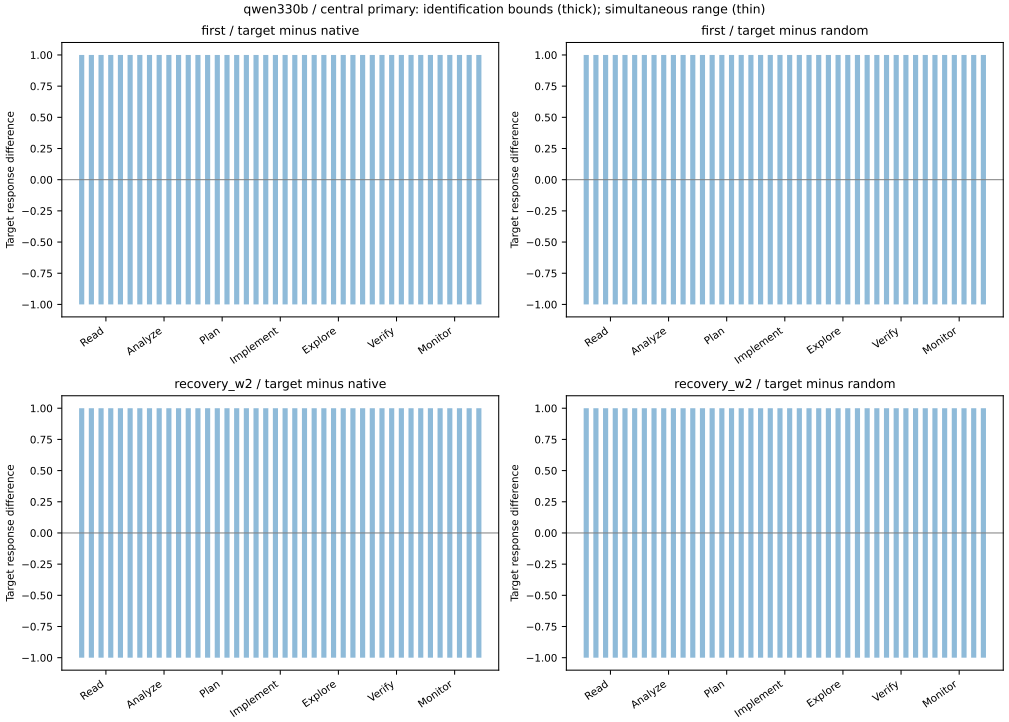


Figure 15
Central primary identification bounds and study-wide simultaneous uncertainty.

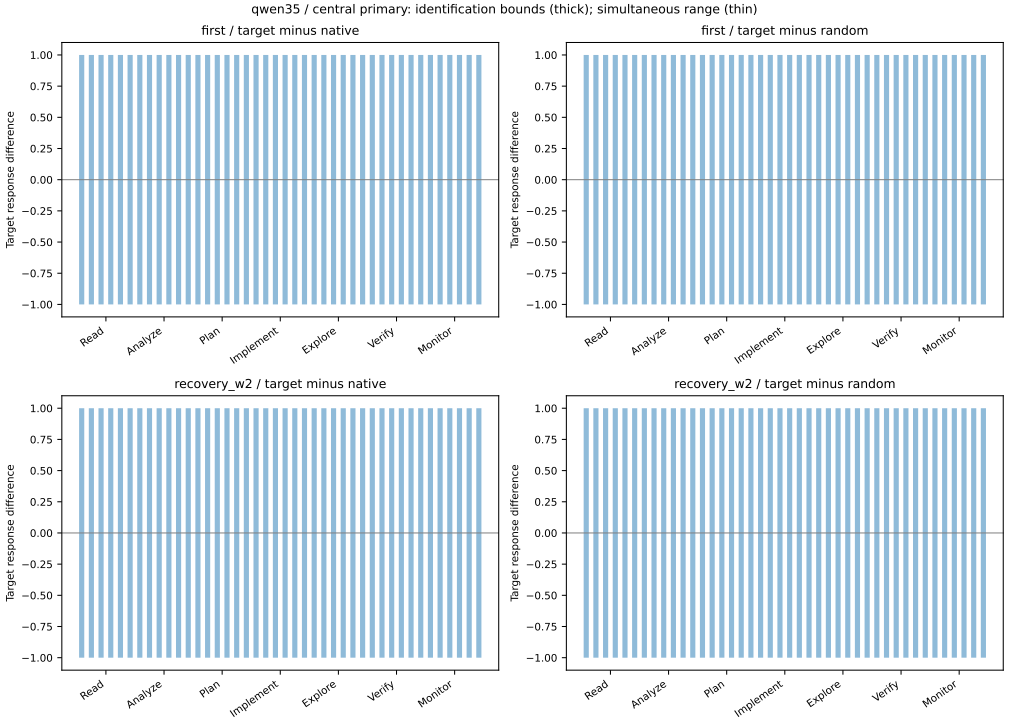


Figure 16
Central primary identification bounds and study-wide simultaneous uncertainty.

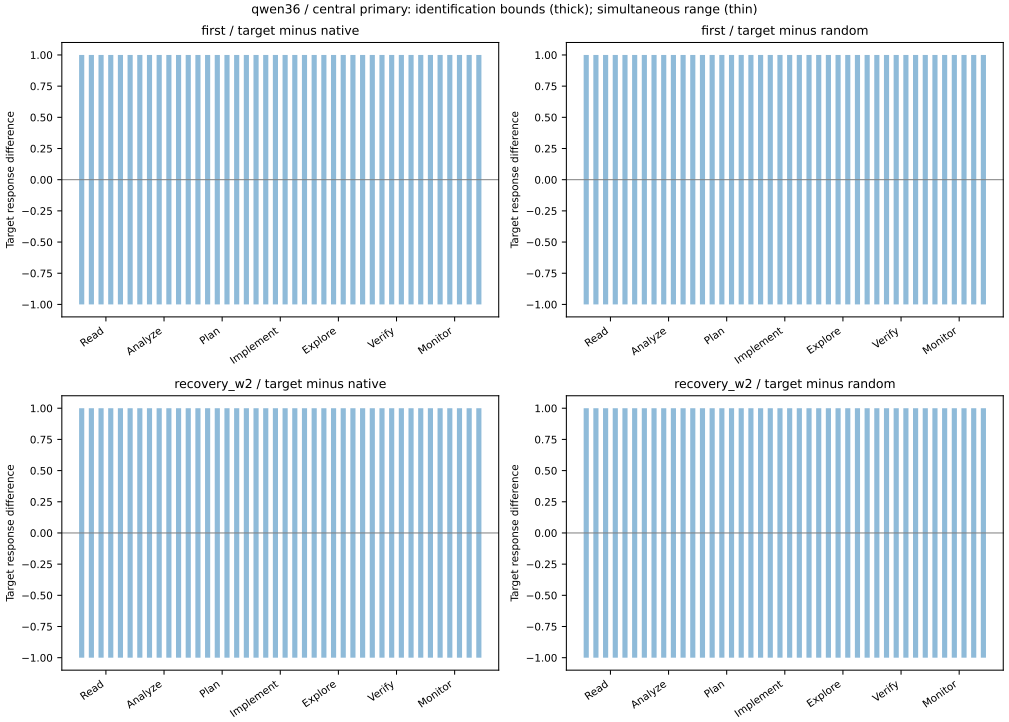


Figure 17
Central primary identification bounds and study-wide simultaneous uncertainty.

Appendix A: Experimental Setup

Historical experimental scope.. The generative judge assigns Schoenfeld episode labels. Boundary probes test prospective label decodability in Qwen3.5-35B-A3B, GPT-OSS-20B, and Gemma-4-26B-A4B. Correlation analysis relates each model's routing summaries to correctness and labelled reasoning episodes. Identity and margin guiding evaluate frozen policies under their original model-specific evaluation contracts. The current seven-model verification design is specified separately in Section H.

GEPA judge design.. The workflow uses GEPA prompt optimization (**?**) and the annotated DeepSeek-R1 SAT responses released by Li et al. (**?**): 3,125 units in 38 responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen predictions. Each request contains the SAT problem and the previous, current, and next response units. The original context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The Qwen3.8 alignment instead imports the 40-example prompt selected by the earlier final fit as its external seed, so it does not dynamically sample new demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the final-fit runs use a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design.. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Each model receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output. Qwen, GPT-OSS, and Gemma provide 41, 25, and 31 hidden-state indices, respectively. An intermediate index is the input to the equally numbered decoder block; the final output has no corresponding router.

At each index, seven separate target-versus-rest logistic regressions use raw activations (2,048 dimensions for Qwen, 2,880 for GPT-OSS, and 2,816 for Gemma), balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The layerwise binary fitting procedure and target-stratified 80/20 split follow Sun et al. (**?**), adapted here from step-position labels to Schoenfeld episodes, with seed 42. This protocol choice limits generalization assessment: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected Qwen probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation. Only boundary vectors are stored; the models do not regenerate the released or benchmark solutions.

Additional response-grouped probe evaluation.. A separate CPU-only mode reuses the saved activations and the same logistic regression settings. Five outer `StratifiedGroupKFold` folds (seed 42) hold out 8, 8, 7, 8, and 7 complete responses. The folds are identical across encoders and targets, verified against exact text, labels, and unit identities. Within each development set, a single grouped 20% validation holdout selects each target's layer by AUROC, breaking ties toward the lower index. This leaves

24 fit and six or seven validation responses; the selected classifier is refitted on all 30 or 31 development responses before outer testing. The inner seed is $42 + f + 1$ for zero-based fold f.

A position-only logistic baseline uses $\log(1 + j)$ and $\log(1 + t)$, where j and t count preceding annotated units and response tokens. Total length and future content are excluded. Tokenization makes the token-count feature model-specific. Metrics weight units equally within each fold, then average folds equally; macro results also average the seven targets equally. Confidence intervals resample complete responses within each outer test fold (5,000 percentile draws, seed 42), pairing probe and baseline predictions. They condition on saved folds, chosen layers, and fits, omitting refitting and split-seed variation; no multiple-comparison adjustment is applied. The grouped study retains fold-specific estimators and does not replace the legacy layers used by the correlation and intervention experiments.

Correlation-pipeline design.. Each of the three models has 4,647 attempts from 1,547 source problems: 32 attempts per AIME24, AIME25, and AMC23 problem, and one per MATH-500, OlympiadBench, and Minerva problem. The benchmark selection and repeat counts adapt the mathematical subset of SPIRAL's evaluation suite (**?**). There are 100 complete avg@32 groups. Generation uses temperature 0.6, top-p 0.95, a 32,768-token completion budget, and a 49,152-token context. Generation and replay quantizations are not identical: Qwen uses GPTQ generation and Unsloth runtime 4-bit replay; Gemma uses NVFP4 generation and NF4 replay; GPT-OSS uses its MXFP4 checkpoint. Native top-k is 8 of 256, 8 of 128, and 4 of 32, respectively. Probe-selected router indices are Qwen 27/34/35/36/38/39, Gemma 16/19/21/22/25/27/29, and GPT-OSS 15/19/21/23.

All token-limit completions remain included. Known-gold attempts with no extracted answer count as incorrect in correlation analysis. Whole-continuation features include final-answer tokens; full-reasoning and absolute-position views use all reasoning tokens independently of tagging. Class views use available sentence labels and omit missing features pairwise. Qwen uses a frozen Qwen3.8-NVFP4 judge at temperature zero, low effort, and a 4,096-token output budget, with a 100,000-unit sample-zero cap. Imported Gemma and GPT-OSS labels use temperature 1, top-p 0.95, top-k 20, low effort, and a 16,384-token budget. Gemma labels cover all six benchmarks; GPT-OSS labels cover MATH-500 and part of AIME24. These annotation protocols and populations are not matched across models. Source-integrity checks do not establish label accuracy.

Guiding design.. Identity and margin policies use a dataset-stratified 70/30 source-problem split (seed 42), problem-weighted calibration, and at least four supporting problems per target. Their completed stochastic evaluations pair 1,395 attempts on 465 held-out problems with condition-specific baselines. Sampling settings and prompts match within each pair; original attempt seeds are retained. Hooks act during prefill and decoding with native expert dispatch, eager execution, and no speculative decoding or prefix cache. Section F gives the calibration exclusions, policy definitions, and numerical comparisons. Intervention evaluation outcomes use the common offline answer-equivalence rule in Section F.1; calibration and observational labels retain their original values.

Table model and metric conventions.. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall

have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen's κ is chance-corrected nominal agreement, Kendall's τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight $[0, 1]$ rescalings. Routing entropy is Shannon entropy of the softmax router distribution; the correlation pipeline's switch rate is the fraction of adjacent tokens whose top-1 expert changes; its top-k overlap is the adjacent-set intersection divided by k (not Jaccard overlap); and router margin is the top-1 minus top-2 router probability. Top-k mass sums the selected probabilities, while the boundary margin is $p_{(k)} - p_{(k+1)}$. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance.. The five active packages are under `src/moe_exp`; their saved outputs are under the corresponding `results` subdirectories. Source hashes, comparison checks, and regeneration commands are listed in the reproducibility appendix. Correlation tables and guiding comparisons have different populations and must not be pooled.

Table B.1
Preliminary `gepaLLMAsJudge` agreement. The composite rescales Cohen's κ and Kendall's τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s):* Qwen3.6-27B `Q4_K_XL`, used as the episode-label judge. *Metrics:* exact-match accuracy, Cohen's κ, Kendall's τ_b, their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	**0.6994**	**0.6178**	**0.6667**	**0.8211**

Appendix B: gepaLLMAsJudge

The `gepaLLMAsJudge` workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by `llama.cpp` on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs.. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen's κ from 0.5710 to 0.6178, Kendall's τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation.. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward.. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA's optimization signal,

Table B.2
Aggregate selected-prompt outer-fold results over 2,789 units. "Balanced light" is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s):* Qwen3.6-27B `Q4_K_XL`, used as the episode-label judge. *Metrics:* exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	**0.6734**	**0.6796**	**0.6505**	**0.8063**
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table B.3
Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s):* Qwen3.6-27B `Q4_K_XL`, used as the episode-label judge. *Metrics:* class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables B.2 and B.3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold

Table B.4
Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s):* Qwen3.6-27B `Q4_K_XL`, used both for episode-label prediction and the same-model GEPA reward. *Metrics:* exact-match accuracy, balanced accuracy, macro-F1, Cohen's κ, Kendall's τ_b, absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	**0.7543**	**0.7769**	**0.7270**	**0.6968**	**0.6832**
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and "Final Answer." These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA's paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Completed LLM-judge final fit.. The Qwen3.6 final fit uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336 units). Seed and optimized prompts both return a valid label for all 407 validation units. Table B.4 reports exact gold-label metrics rather than the LLM judge's diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt's overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13 points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-

Table B.5
Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s):* Qwen3.6-27B `Q4_K_XL`, used both for episode-label prediction and the same-model GEPA reward. *Metrics:* class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

Table B.6
Final Qwen3.8 model-alignment results on the 407-unit development validation set. Both prompts have 100% valid-output coverage; the 336-unit locked test is not evaluated. *Model(s):* Qwen3.8-27B `UD-Q4_K_XL`, used for both classification and the same-model GEPA reward. *Metrics:* exact-match accuracy, balanced accuracy, macro-F1, Cohen's κ, and Kendall's τ_b.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
Imported 40-example seed	**0.7174**	**0.7400**	**0.6938**	**0.6523**	**0.6122**
GEPA optimized	0.7125	0.7288	0.6743	0.6475	0.6009
Absolute change	-0.0049	-0.0112	-0.0196	-0.0049	-0.0113

sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

Final Qwen3.8 model-alignment pass.. The final alignment uses Qwen3.8-27B `UD-Q4_K_XL`, seed 42, the heavy GEPA budget, and medium reasoning effort for the same-model judge. It keeps the same response-grouped 2,382/407/336 train/validation/locked-test split. Rather than drawing new few-shot examples, it imports the previous 40-example Qwen3.6 optimized prompt as its external seed. Gold-label agreement supplies 40% of the internal judge reward; functional fit, contextual coherence, and boundary precision supply the remaining 60%. Final selection uses exact balanced accuracy with a maximum allowed per-class recall drop of 0.10.

Table B.7
Per-class recall in the final Qwen3.8 alignment. The selection gate rejects any candidate whose recall drops by more than 0.10 for any class. *Model(s):* Qwen3.8-27B `UD-Q4_K_XL`. *Metrics:* class support and seed versus GEPA-optimized recall.

Class	Support	Seed recall	Opt. recall
Read	45	0.778	0.756
Analyze	122	0.549	0.533
Plan	32	0.750	0.563
Implement	103	0.874	0.874
Explore	17	0.529	0.647
Verify	70	0.700	0.786
Monitor	18	1.000	0.944

The optimized candidate changes 49 labels: it fixes 17 seed errors, breaks 19 seed-correct cases, and changes 13 errors to other errors. Its largest failure is Plan recall, which falls by 0.188 and therefore breaches the recall gate; the selected prompt is byte-identical to the imported seed. An unclustered exact McNemar test on the 17 versus 19 discordant correctness outcomes gives $p = 0.868$, offering no evidence of an aggregate improvement, although this unit-level calculation understates response clustering.

This negative exact-label result contrasts with the internal GEPA objective, whose best candidate reward rises from 0.7684 to 0.7762. Because only 40% of that reward is gold agreement, the discrepancy is consistent with optimizing the judge's broader functional criteria without improving corpus-label classification. The optimized instruction is shorter (2,591 versus 5,201 words) but internally inconsistent: an early rule treats "Let me compute that" as Monitor while a later rule treats "Let me calculate" as Plan. Plan-to-Monitor errors rise from 6 to 10, consistent with that boundary confusion. The appropriate conclusion is that the final alignment completed correctly and the safety gate worked: GEPA found no safe improvement over the existing seed. The complete artifact is stored at `results/gepaLLMAsJudge/qwen3.8-27b-medium-final-s42-v3`.

Frozen-prompt deployment and reasoning effort.. The NVFP4 deployment comparison evaluates the frozen 40-example prompt on the same 407 development-validation units, with temperature zero, a 4,096-token budget, seed 42, one worker, and one uncached pass. Table B.8 reports the saved outputs. Both settings have 100% valid-output coverage and no request errors. They disagree on 41/407 labels (10.1%): identical aggregate accuracy does not imply interchangeable annotations. This is exploratory evidence on reused validation responses, not independent validation of labels on generated benchmark traces.

Table B.8
Frozen Qwen3.8-27B NVFP4 judge on 407 reused validation units. Metrics are exact accuracy, balanced accuracy, and macro-F1.

Effort	Accuracy	Balanced accuracy	Macro-F1
Low	0.7101	0.7415	0.6863
Medium	0.7101	0.7444	0.6832

The 336-unit partition is excluded from these final fits. Because its responses have been inspected in preliminary work, it should not be described as genuinely untouched; an independent generalization estimate requires unseen responses.

Appendix C: probeTest

The `probeTest` workflow tests a different annotation mechanism from the generative judge in `gepaLLMAsJudge`. It asks whether each model's representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence or the latent dynamics of an encoder's own generated trace.

Three-model comparison.. All three encoders use the same 3,087 units and target-stratified sentence split: 2,469 training and 618 test units. Table C.1 reports each target's accuracy-selected hidden-state index, F1, and AUROC. All 38 response IDs occur in both partitions, and layer selection uses the evaluation split. These are descriptive within-corpus comparisons, not response-held-out generalization estimates. There are 287 Qwen, 175 GPT-OSS, and 217 Gemma binary fits. The detailed external inspection below applies only to Qwen.

Table C.1
Accuracy-selected boundary probes: Qwen3.5 GPTQ, GPT-OSS-20B, and Gemma-4 NF4. Each model reports hidden-state index, F1, and AUROC; selection and evaluation share the sentence-level test split.

	Qwen			GPT-OSS			Gemma		
Target	Index	F1	AUROC	Index	F1	AUROC	Index	F1	AUROC
Read	34	0.704	0.934	19	0.686	0.906	21	0.675	0.898
Analyze	36	0.647	0.835	21	0.540	0.767	27	0.611	0.816
Plan	39	0.485	0.855	15	0.434	0.852	25	0.500	0.907
Implement	38	0.771	0.930	21	0.745	0.905	29	0.735	0.922
Explore	40	0.489	0.888	23	0.405	0.842	16	0.447	0.835
Verify	27	0.726	0.912	21	0.667	0.880	19	0.603	0.872
Monitor	35	0.482	0.903	15	0.344	0.774	22	0.382	0.831

Qwen within-corpus results.. Table C.2 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence,

Table C.2
Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. ΔMaj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s):* `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics:* target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	ΔMaj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

Table C.3
Response-grouped prospective probes on 38 responses and 3,087 units. AUROC is averaged equally across five outer folds and seven targets. Brackets are 95% response-bootstrap intervals (5,000 draws), conditional on saved folds, selected layers, and fits. The paired gain subtracts the prefix-position baseline. Legacy scores select layers by sentence-test accuracy, so the two protocols differ in both splitting and selection.

Model	Legacy AUROC	Grouped AUROC	Position AUROC	Paired gain
Qwen	0.894	0.852 [0.838, 0.868]	0.566 [0.552, 0.594]	+0.286 [0.259, 0.302]
GPT-OSS	0.847	0.780 [0.761, 0.797]	0.571 [0.557, 0.600]	+0.209 [0.179, 0.227]
Gemma	0.869	0.788 [0.772, 0.805]	0.566 [0.553, 0.594]	+0.221 [0.195, 0.237]

and only 38 distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

C.1 Response-grouped validation and a position baseline

The additional grouped study tests the same representations on complete responses that are absent from both fitting and layer selection. Table C.3 reports macro-AUROC 0.852 (95% interval [0.838,0.868]) for Qwen, 0.780 ([0.761,0.797]) for GPT-OSS, and 0.788 ([0.772,0.805]) for Gemma. The corresponding legacy means are 0.894, 0.847, and 0.869. These differences change both the split and the layer-selection criterion, so they cannot be attributed solely to response overlap. The grouped results nevertheless establish a stronger claim than the original table: upcoming-label information remains accessible on held-out responses within the released SAT corpus.

The prefix-position baseline reaches macro-AUROC 0.566, 0.571, and 0.566, respectively. Each encoder exceeds its baseline, and all 21 per-target paired AUROC intervals exclude zero; these intervals are not adjusted for multiple comparisons (Table C.4). Figure C.1 shows both predictions with response-bootstrap uncertainty. Thus the signal exceeds this particular two-feature positional rule. The control does not exclude more flexible position effects, problem-type or formatting shortcuts, or correlations between episodes and lexical prefix content.

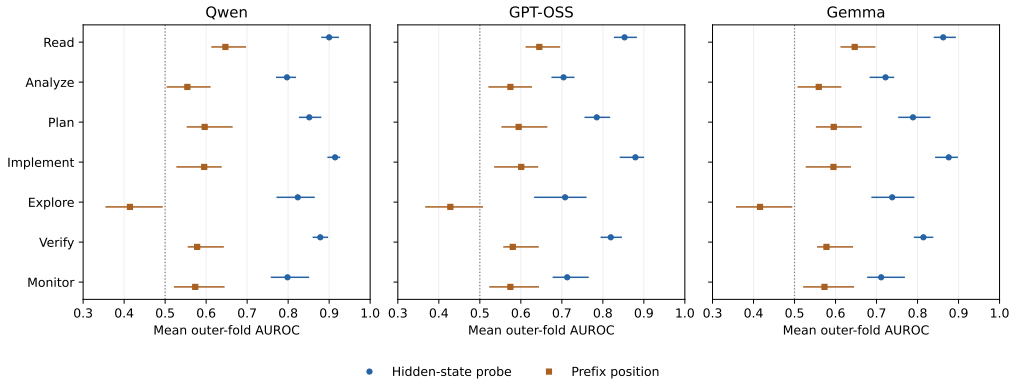


Figure C.1
Prospective AUROC on response-disjoint outer test folds. Points are means of five fold scores and horizontal bars are 95% conditional response-bootstrap intervals. Both the hidden-state probe and the simple prefix-position baseline are fitted without the test responses.

Table C.4
Response-grouped results by target. AUROC and the paired AUROC gain over the position baseline include 95% conditional response-bootstrap intervals. Balanced accuracy (BA) and binary F1 are mean-fold point estimates at the unchanged decision threshold; their intervals are retained in the result files. Indices list the validation-selected hidden states in outer-fold order, not a globally selected layer.

Model	Target	AUROC	Gain over position	BA	F1	Indices (folds 1–5)
Qwen	Read	0.900 [0.881, 0.924]	+0.253 [0.209, 0.288]	0.805	0.599	34, 29, 36, 32, 23
Qwen	Analyze	0.797 [0.770, 0.819]	+0.243 [0.178, 0.296]	0.725	0.586	22, 26, 35, 22, 22
Qwen	Plan	0.852 [0.826, 0.881]	+0.255 [0.196, 0.292]	0.754	0.440	22, 29, 29, 29, 33
Qwen	Implement	0.914 [0.896, 0.927]	+0.319 [0.277, 0.385]	0.825	0.696	32, 31, 30, 19, 18
Qwen	Explore	0.823 [0.772, 0.865]	+0.409 [0.323, 0.478]	0.674	0.341	35, 31, 29, 34, 24
Qwen	Verify	0.878 [0.860, 0.897]	+0.300 [0.232, 0.325]	0.798	0.634	24, 24, 24, 20, 15
Qwen	Monitor	0.799 [0.758, 0.851]	+0.225 [0.156, 0.292]	0.671	0.355	21, 36, 22, 34, 31
GPT-OSS	Read	0.853 [0.827, 0.883]	+0.208 [0.164, 0.241]	0.724	0.528	21, 18, 17, 16, 13
GPT-OSS	Analyze	0.704 [0.675, 0.731]	+0.130 [0.072, 0.185]	0.609	0.416	16, 20, 17, 20, 20
GPT-OSS	Plan	0.785 [0.755, 0.818]	+0.191 [0.118, 0.232]	0.619	0.286	20, 18, 20, 23, 7
GPT-OSS	Implement	0.879 [0.841, 0.901]	+0.278 [0.233, 0.335]	0.768	0.640	16, 20, 18, 21, 18
GPT-OSS	Explore	0.708 [0.632, 0.760]	+0.280 [0.155, 0.369]	0.561	0.178	23, 22, 19, 11, 22
GPT-OSS	Verify	0.819 [0.794, 0.847]	+0.239 [0.174, 0.264]	0.699	0.517	16, 18, 18, 17, 16
GPT-OSS	Monitor	0.713 [0.678, 0.766]	+0.139 [0.078, 0.211]	0.589	0.233	16, 9, 21, 13, 22
Gemma	Read	0.863 [0.840, 0.893]	+0.216 [0.174, 0.253]	0.747	0.547	16, 20, 16, 19, 21
Gemma	Analyze	0.722 [0.683, 0.743]	+0.163 [0.088, 0.215]	0.653	0.488	6, 20, 20, 23, 22
Gemma	Plan	0.789 [0.753, 0.832]	+0.193 [0.129, 0.241]	0.655	0.337	19, 29, 16, 23, 16
Gemma	Implement	0.876 [0.843, 0.898]	+0.281 [0.244, 0.337]	0.779	0.644	24, 29, 20, 23, 25
Gemma	Explore	0.738 [0.687, 0.792]	+0.322 [0.229, 0.402]	0.572	0.185	17, 19, 17, 29, 19
Gemma	Verify	0.814 [0.791, 0.838]	+0.236 [0.167, 0.262]	0.725	0.543	25, 15, 28, 16, 15
Gemma	Monitor	0.711 [0.677, 0.769]	+0.138 [0.070, 0.217]	0.593	0.240	28, 0, 29, 25, 25

Ranking performance still does not imply reliable seven-class annotation. Macro binary F1 is 0.521 for Qwen, 0.400 for GPT-OSS, and 0.426 for Gemma at the unchanged decision rule. In Qwen, Explore reaches AUROC 0.823 but F1 only 0.341; Monitor reaches 0.799 and 0.355. The selected layers also vary across folds: Qwen Read uses indices 23–36 and Implement 18–32. This variability argues against interpreting one selected index as a unique reasoning location. The study intentionally produces fold-

Table C.5
Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table C.6 compares them with the subsequent provisional pragmatic review. *Model(s):* `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics:* unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

specific probes; the existing routing experiments retain their original measurement layers.

Across the five folds there are 1,505 Qwen, 945 GPT-OSS, and 1,155 Gemma fits, including 35 selected-layer refits and 35 position baselines per model. The numbers of inner candidate fits reaching the fixed iteration limit are 0, 2, and 2, respectively; the selected-layer refits have 0, 0, and 0 convergence warnings. All position fits converge, and all 5,000 bootstrap draws retain binary support for every reported target. Diagnostics and indices are retained rather than used to change the solver budget after evaluation.

The remaining inferential limits are the small number of responses and the scope of the test. The intervals condition on this single split plan, selected layers, and fitted classifiers; they omit refitting and split-seed variation. Same-corpus response generalisation does not establish transfer to new benchmarks, annotation validity on the encoders' own generations, or a causal role for the probed directions. The legacy external Qwen inspection below addresses a different population and decision rule.

C.2 External benchmark inspection

All 80 benchmark examples complete without omitted units, yielding 581 sentence boundaries. Table C.5 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication.. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its

Table C.6
Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s):* `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics:* unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table C.6 reports the complete dataset-level results.

The class-level comparison in Table C.7 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit's semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark's annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early,

Table C.7
Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s):* `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics:* accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution (Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

Table D.1
Prespecified measures for the expanded correlation analysis.

Analysis	Unit	Measure	Reason
Continuous feature versus correctness or a binary reasoning-event indicator	Generation trace	Point-biserial r_{pb}	The predictor is continuous and the outcome is binary; the signed coefficient is exactly Pearson's r after coding the outcome as $0/1$.
Mean feature versus avg@N correctness	Source problem	Spearman's ρ	Avg@N is bounded, often tied, and not assumed to have a linear interval relationship with a feature. Ranking tests monotone association and one row per problem avoids treating repeated attempts as independent.
Continuous feature–feature and continuous expert-rate associations	Trace or source problem	Spearman's ρ	These exploratory scans ask whether quantities vary monotonically without assuming comparable scales or a linear relationship.
Correct-versus-incorrect feature difference within a repeated problem	Mixed-outcome source problem	Mean paired contrast	This directly controls for the problem by forming one correct-minus-incorrect contrast per problem; uncertainty is obtained by resampling problems.

Appendix D: Correlation Pipeline

D.1 Statistical measures and literature basis

The analysis selects the association measure from the outcome scale and unit of analysis, as summarized in Table D.1. The detailed Qwen results below use the six-benchmark run: 4,647 attempts, including token-limit completions, with full-reasoning, class, and position views. The 100 complete avg@32 groups support repeated-attempt analysis. Classes use sampled labelled attempts; full and position views use all reasoning tokens. Section E adds the completed Gemma and GPT-OSS runs, with model-specific class coverage and annotation protocols.

For a continuous feature X and binary target $Y \in \{0, 1\}$, the first coefficient is

$$r_{\mathrm{pb}} = \mathrm{Corr}(X, Y). \tag{D.1}$$

Thus positive values mean that the feature tends to be higher for correct (or event-present) traces, and negative values mean that it tends to be lower. Liu et al. directly use point-biserial correlation for continuous language-model quantities against binary answer correctness, making it the closest precedent for this case (**?**).

For repeated-attempt datasets, correctness and each predictor are first aggregated once per source problem. Only groups containing exactly the declared avg@N unique sample identifiers, no duplicate generations detected by the saved fingerprints, and a complete set of binary outcomes enter the primary Spearman analysis. Spearman correlation is appropriate here because its target is monotone rank association rather than a specifically linear change in avg@N. Hong et al. correlate a scalar projection of residual-stream states with graded reasoning scores using Spearman correlation (**?**); Xu et al.

use Spearman correlation between language-model-derived conceptual quantities and human ratings (**?**); and Hewitt and Manning use it to compare geometry derived from contextual representations with gold syntactic structure (**?**). These studies are analogous precedents for rank correlation with representation-derived quantities; they do not by themselves establish Spearman as a unique choice for avg@32. Zhang et al. additionally report Pearson and Spearman correlations together when relating model uncertainty, including entropy-based measures, to graded human hallucination judgments (**?**).

Uncertainty is reported with 95% percentile bootstrap intervals that resample whole source problems, keeping attempts from the same problem together. Xu et al. provide relevant precedent for bootstrap intervals and use 10,000 resamples in one category–feature analysis (**?**); the resample count in this pipeline remains explicit and configurable rather than being justified by that number alone. Benjamini–Hochberg adjusted p-values (stored as q-values) (**?**) are computed separately within each declared dataset/scope, target, aggregation, or cross-feature family. Naive trace-level p-values are retained only as diagnostics when a dataset contains repeated attempts, and layerwise and expert-identity screens remain exploratory.

D.2 Completed six-benchmark run and analysis population

The Qwen numerical results use the run under `results/correlation_pipeline/reasoning-vllm-v1`. Forward replay and the saved aggregate analyses are complete for 4,647 attempts from 1,547 source problems: 500 MATH-500, 960 AIME24, 960 AIME25, 675 OlympiadBench, 1,280 AMC23, and 272 Minerva attempts. AIME and AMC retain 32 attempts per problem; the other benchmarks retain one. GPQA-Diamond and MMLU-Pro are absent.

Generation, replay, and annotation.. Generation serves `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` through vLLM with temperature 0.6 and top-p 0.95. Replay uses `unsloth/Qwen3.5-35B-A3B` with runtime 4-bit quantization at the probe-selected decoder indices 27, 34, 35, 36, 38, and 39. Hidden-state features describe block inputs, not exact post-attention router inputs. Router confidence is $1 - H(p)/\log 256$. The continuation analysis has 83 scalar features; across-layer means are emphasized and layerwise screens remain exploratory. Saved bootstrap intervals use 500 replicates and seed 42, resampling whole source problems.

The sentence judge is `unsloth/Qwen3.8-27B-NVFP4`, temperature zero, thinking enabled with low reasoning effort, and a 4,096-token completion budget. Inputs contain the question and previous/current/next sentences. The frozen program SHA-256 is `69706a02c0484f345cf57775729ce7551feb5932954b9ad0b8f5bb63585c6fd9`. Its accuracy on these generated math traces has not been independently measured; the earlier GGUF alignment evaluation uses a different deployment.

Population and token scopes.. All supplied generations enter whole-continuation, full-reasoning, position, and expert analyses, including token-limit completions. Whole-continuation metrics include final-answer tokens. The `full/reasoning` view uses every reasoning token, independently of annotation coverage. Position views also use all reasoning tokens within their absolute windows. Class views alone use available validated sentence labels: the sampling protocol selects sample zero, with error-rate-weighted benchmark quotas, seed 42, and a 100,000-sentence cap. Missing labels never

exclude a generation from the other views. Missing class/window features are pairwise missing, not zero.

Table D.2: Completed-run coverage and accuracy, Qwen3.5-35B-A3B. All supplied attempts are retained, including limit hits. AIME/AMC use 32 attempts per problem; other benchmarks use one. Unparseable answers with known gold are incorrect. Pooled accuracy is descriptive.

Dataset	Input	Limit hits	Analyzed	Correct	Errors	Input acc.	Analyzed acc.
MATH-500	500	17	500	345	155	0.690	0.690
AIME24	960	105	960	817	143	0.851	0.851
AIME25	960	165	960	773	187	0.805	0.805
Olympiad	675	89	675	322	353	0.477	0.477
AMC23	1280	17	1280	1238	42	0.967	0.967
Minerva	272	19	272	73	199	0.268	0.268
Pooled	4647	412	4647	3568	1079	0.768	0.768

Table D.3: Whole-continuation aggregate features versus correctness: point-biserial r (finite traces), Qwen3.5-35B-A3B. Token-limit completions are included. A dash denotes an undefined coefficient.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.291 (500)	−0.726 (960)	−0.727 (960)	−0.387 (675)	−0.238 (1,280)	−0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	−0.035 (960)	0.222 (960)	0.118 (675)	−0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Hidden-state norm	0.203 (500)	−0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	−0.152 (500)	−0.058 (960)	−0.077 (960)	−0.102 (675)	−0.229 (1,280)	−0.096 (272)
Top-1 switch rate	−0.071 (500)	−0.297 (960)	0.141 (960)	0.005 (675)	−0.160 (1,280)	−0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	−0.155 (960)	0.016 (675)	−0.062 (1,280)	−0.010 (272)

Termination and accuracy denominators.. There are 3,568 correct answers among 4,647 attempts (0.768). All 412 completion-limit hits (8.9%) remain in the analysis; only two are correct. Among the 4,235 non-limited attempts, 3,566 are correct (0.842). There are 388 unparseable answers counted as incorrect under the known-gold scoring contract. These denominators differ from accuracy computed only among parseable answers in generation summaries. Termination and length are strong potential confounders: these results do not estimate natural-length performance or a length-adjusted routing effect.

D.3 Correctness associations and uncertainty

Whole-continuation router confidence has positive point-biserial coefficients on all six benchmarks: MATH-500 $r = 0.199$, AIME24 0.533, AIME25 0.493, Olympiad 0.243, AMC23 0.175, and Minerva 0.112. Saved pointwise 95% problem-bootstrap intervals exclude zero except for AMC23 ($[-0.116, 0.678]$). MATH-500's interval is $[0.099, 0.276]$; AIME24's is $[0.382, 0.678]$ and AIME25's is $[0.302, 0.608]$. These are marginal associations, without length or dataset adjustment.

Leading-expert margin ($M = p_{(1)} - p_{(2)}$) is positive throughout; Minerva's interval includes zero. Hidden-state step distance is also positive throughout ($r = 0.100$–0.557), while token count is negative (-0.727–-0.238). Attempts within AIME/AMC problems are dependent; naive trace-level p-values and their BH adjustments are diagnostic only.

Full-reasoning confidence gives similar coefficients: 0.191, 0.527, 0.486, 0.238, 0.164, and 0.124 in the same dataset order. All 4,647 traces have reasoning tokens. Similarity between these two scopes is descriptive; it does not remove length or termination effects.

D.4 Class-conditioned routing and hidden states

Table D.4: Class coverage in the completed run. Qwen3.8-27B labels on Qwen3.5-35B-A3B generations. Units are tagged sentences; dataset columns count traces with selected class tokens. Classes overlap across traces; missing classes are not zero-valued features.

Class	Units	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva	Total traces
Read	6,358	378	30	26	514	27	270	1245
Analyze	22,378	424	30	30	619	38	271	1412
Plan	5,654	407	28	28	507	28	271	1269
Implement	23,844	477	30	30	667	39	271	1514
Explore	7,151	260	28	27	474	22	251	1062
Verify	10,745	400	29	30	617	38	270	1384
Monitor	23,870	451	30	29	583	29	272	1394

The 100,000 sampled sentences have complete labels across 1,546 nonempty attempt samples; one of the 1,547 sample-zero attempts has no labelled sentence. All seven class analyses are complete, with coverage reported separately from the full population. Explore has tokens in 1,062 traces; Verify in 1,384. AIME/AMC class views contain at most the 30/30/40 sample-zero attempts rather than all 32 attempts per problem. Annotation counts describe sampled sentences, not class prevalence in the complete corpus. Class transition metrics require adjacent original tokens within the same sentence.

Class conditioning remains heterogeneous. Verify confidence is 0.066 on MATH-500, 0.123 on Olympiad, and -0.028 on Minerva; Plan confidence is -0.067, 0.039, and -0.121, respectively. These separately selected cohorts do not test a difference between classes. The smaller AIME/AMC class cohorts remain especially sensitive to individual errors. Annotation errors, length, difficulty, and coverage are uncontrolled, so these screens do not identify beneficial or harmful reasoning states.

D.5 Class counts, transitions, and outcome associations

Class-conditioned router features and the proportion of reasoning assigned to each class describe different properties. The following plots use *sampled sentence shares*: the number of labelled sentences of a class divided by the number of labelled sentences in that attempt. An observed zero share is retained; unlabelled sentences are never assigned a class. These are estimates from the sampling protocol, not complete-sequence class fractions. Each source problem contributes only its sample-zero attempt; benchmark comparisons retain their separate denominators. Attempts with no labelled sentences are excluded.

Table D.5: Mean sampled sentence share (percent), with equal weight per labelled attempt. Empty-label attempts are excluded; absent classes in an observed sample have zero share. These are not full-corpus class proportions.

Dataset	Attempts	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
MATH-500	499	8.9	19.5	9.3	29.3	4.1	11.8	17.1
AIME24	30	6.7	33.9	2.9	28.3	7.0	14.4	6.8
AIME25	30	5.7	33.2	4.8	27.9	6.7	13.4	8.3
Olympiad	675	6.9	26.8	6.0	29.7	6.4	13.2	11.0
AMC23	40	6.6	25.9	5.8	32.0	3.8	15.2	10.6
Minerva	272	9.4	19.8	8.4	21.8	7.7	10.3	22.6

Class-state dynamics remain descriptive. Relating complete transition paths to correctness and interventions conditioned on those paths is future work. The currently sampled labels do not justify reconstructing missing transitions.

D.6 Position-dependent estimates and coverage

The position reference uses all 4,647 supplied generations and mean reasoning length 12,498.062 tokens. Ten non-cumulative windows have boundaries $\lceil i\bar{L}/10 \rceil$, approximately 1,249.8 tokens apart; overflow begins after the final boundary at 12,499. Every reasoning token contributes to its window regardless of tagging. Token count is omitted as a position predictor. Coverage falls from 4,647 traces in the first window to 1,789 in overflow, so later-window coefficients describe different cohorts.

Table D.6: Reasoning length in tokens, all attempts including limit hits. This replaces the position-coverage count plot; lengths are observed under the generation budget.

Dataset	Attempts	Mean	Median	90th percentile
MATH-500	500	6,226.5	4,250	12,772
AIME24	960	15,908.4	14,011	32,766
AIME25	960	17,874.9	16,620	32,768
Olympiad	675	13,402.5	9,150	32,768
AMC23	1280	8,452.8	7,822	14,929
Minerva	272	9,805.0	7,465	19,533

D.7 Metric-pair dependence

Token count and confidence are negatively associated on all six datasets: $\rho = -0.681$, -0.627, -0.836, -0.870, -0.845, and -0.280 in MATH/AIME24/AIME25/Olympiad/AMC/Minerva order. This dependence is one reason confidence–correctness correlations need length-aware follow-up. Top-eight mass S and the selected/unselected mean probability gap G are algebraically redundant for $k = 8$ and $E = 256$:

$$G = \frac{S}{k} - \frac{1-S}{E-k} = \frac{E}{k(E-k)}S - \frac{1}{E-k}. \tag{D.2}$$

The affine relation survives token/layer averaging. These metrics should not count as independent corroboration or enter an unregularized model twice. The appendix presents correlation heatmaps for all 20 scopes, including class-conditioned metrics

versus correctness and token count. The numeric tables remain available as a separate generated artifact.

The legacy regex flags (backtracking, contradiction, and self-correction) are removed from the report's outcome panels, to keep the outcome panels focused on model-assigned classes. They are not the seven predicted reasoning classes. Detector outputs remain in the source artifacts.

D.8 Repeated-attempt findings

There are 100 eligible complete avg@32 groups: 30 AIME24, 30 AIME25, and 40 AMC23 problems. Problem means use all declared attempts, including limit hits, and binary outcomes are complete. The run supports both between-problem rank associations and within-problem correct/incorrect contrasts.

Table D.7: Whole-continuation mean feature versus avg@32 correctness, Spearman ρ (source problems). All 32 attempts, including token-limit completions, enter each eligible group.

Feature	AIME24	AIME25	AMC23
Token count	-0.861 (30)	-0.909 (30)	-0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	-0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	-0.112 (30)	0.388 (30)	-0.215 (40)
Selection-boundary margin	-0.367 (30)	0.135 (30)	-0.145 (40)
Selected/rest mean gap	-0.112 (30)	0.388 (30)	-0.215 (40)
Hidden-state norm	-0.384 (30)	-0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	-0.331 (40)
Top-1 switch rate	-0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	-0.047 (30)	-0.196 (30)	-0.327 (40)

Table D.8: Within-problem confidence differences, Qwen3.5: mean correct minus incorrect, equally weighted over mixed-outcome problems; saved 95% whole-problem bootstrap intervals.

Dataset	Mixed problems	Difference	95% interval
AIME24	14	0.00268	[0.00152, 0.00412]
AIME25	15	0.00123	[0.00065, 0.00176]
AMC23	6	0.01102	[0.00455, 0.01709]

Mean whole-continuation confidence correlates positively with avg@32 on AIME24 ($\rho = 0.632$, CI $[0.383, 0.818]$) and AIME25 (0.705, CI $[0.460, 0.854]$). AMC23 is negative but uncertain (-0.200, CI $[-0.565, 0.243]$), despite its positive attempt-level coefficient. Thus attempt-level and problem-level associations answer different questions. AIME24 and AIME25 mean token count have strong negative avg@32 associations (-0.861 and -0.909).

Within-problem confidence differences are positive over 14, 15, and 6 mixed-outcome problems, respectively; Table D.8 reports magnitudes and bootstrap intervals. Conditioning on a shared question reduces between-question variation but does not control attempt length or termination. Class views still lack repeated labelled attempts, so these repeated-attempt findings must not be generalized to class-specific effects.

D.9 Expert screen, storage, and implications

Table D.9: Post-hoc largest absolute correctness coefficient per dataset in the expert-rate screen, Qwen3.5-35B-A3B. Point-biserial r and BH q within dataset/target; no expert bootstrap intervals. Numeric IDs pool six distinct layer-local modules. Repeated-attempt trace-level q-values are diagnostic, not cluster-adjusted inference.

Dataset	Pooled-ID rate	Traces	Errors	r	q
MATH-500	Top-1 61 / top-2 7	500	155	0.328	$3.36e-11$
AIME24	Top-1 ID 123	960	143	−0.568	$1.87e-80$
AIME25	Top-1 133 / top-2 85	960	187	0.556	$4.72e-76$
Olympiad	Top-1 70 / top-2 181	675	353	0.312	$6.25e-14$
AMC23	Top-8 ID 217	1280	42	−0.665	$2.83e-161$
Minerva	Top-1 162 / top-2 165	272	199	0.409	$1.31e-09$

The expert screen includes top-1 rates, any-top-eight rates, and up to 128 ordered top-1/top-2 pairs selected by routing support (minimum four traces). The largest absolute correctness coefficient per dataset is selected post hoc. Figure D.11 shows all selected-pool IDs against correctness and total token count, without selecting a maximum. Numeric IDs pool six different layer-local modules; they do not identify a portable expert or establish an intervention target. Trace-level BH values on repeated-attempt datasets do not account for clustering.

Metric values associated with accuracy.. Correlation signs alone do not specify the observed metric ranges associated with higher accuracy. Figure D.12 therefore reports accuracy directly against router confidence, the selection-boundary margin, and the selected/rest mean probability gap. Bins are computed separately per benchmark; this is an exploratory whole-continuation analysis, not an online threshold-selection experiment.

Common router notation and unavailable extensions.. Let $p_{(j)}$ denote the jth largest router probability, E the number of experts, and k the selected count ($E = 256$, $k = 8$ here). Router confidence is $C = 1 - H(p)/\log E$; leading-expert margin is $M = p_{(1)} - p_{(2)}$; selected probability mass is $S = \sum_{j \le k} p_{(j)}$; selection-boundary margin is $B = p_{(k)} - p_{(k+1)}$; and selected/rest mean gap is $G = S/k - (1 - S)/(E - k)$. Thus M is not a selected-versus-rest statistic. These definitions apply across models with different native selection counts.

Rank-prefix curves (top 2, 4, 6, then all selected versus the rest) require additional probability summaries. Only the all-selected endpoint G is saved. Separate entropy of the selected and unselected distributions also cannot be recovered from global entropy, selected mass, or expert IDs. These extensions require new feature extraction and replay.

Replay covers 62,088,209 continuation tokens. Compact tensor artifacts occupy 5.970 GB versus a projected 1,907.350 GB of raw selected-layer payload; dataset files total 9.259 GB (decimal units). Storage and source-fingerprint checks are recorded in `report/correlation_run_audit.json`. No raw router/hidden tensors or expert weights are saved. The completed run supplies six-benchmark, full-reasoning, class, position, lexical, and repeated-attempt analyses. Next steps are independent label assessment, length-/termination-aware models, common-cohort class/window comparisons, and validation holding out whole problems. Separate held-out guiding results are reported in Section F.

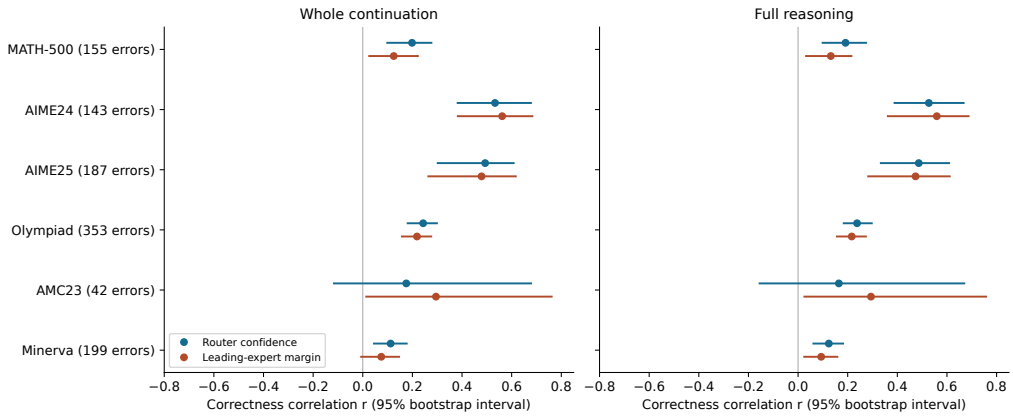


Figure D.1
Qwen3.5 router confidence and leading-expert margin versus correctness, with saved pointwise 95% problem-bootstrap intervals across all six benchmarks. Both views retain all attempts, including limit hits; full reasoning excludes non-reasoning continuation tokens. Error counts refer to attempts, not independent source problems.

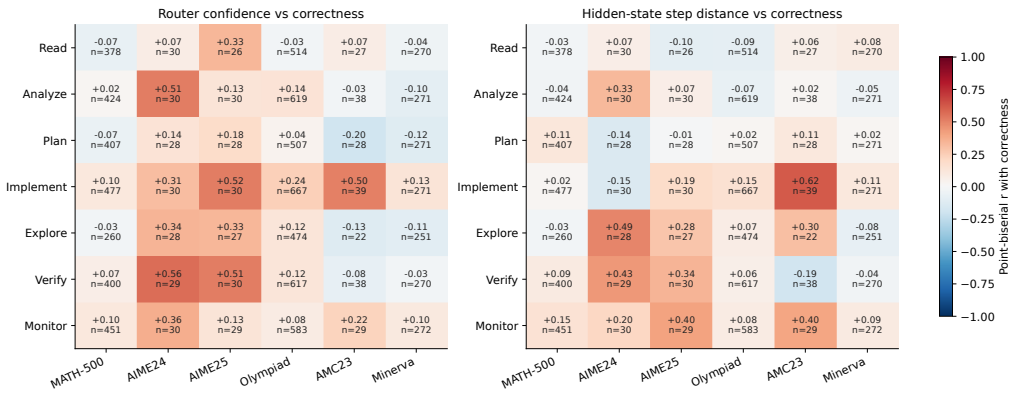


Figure D.2
Class-conditioned confidence and hidden-state step distance versus whole-attempt correctness. Cells show point-biserial r and finite trace count. Classes use sampled Qwen3.8 labels on Qwen3.5 generations, with different token-bearing cohorts. No significance markings are implied.

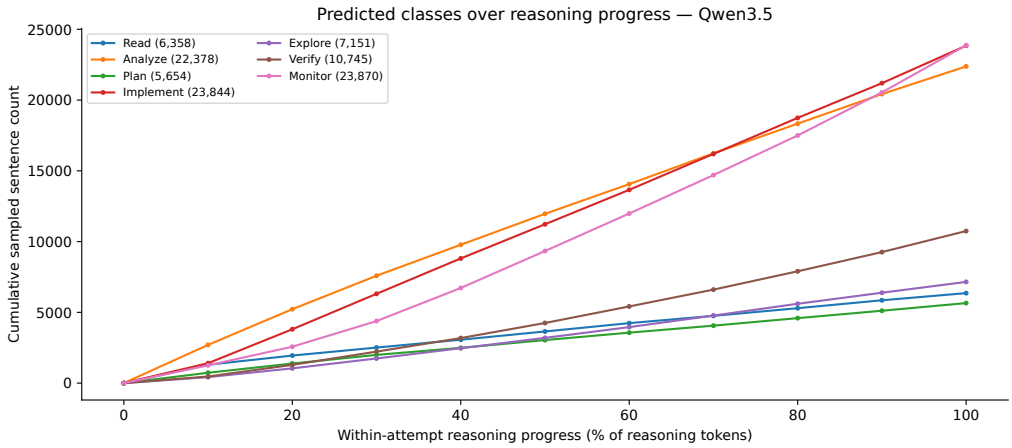


Figure D.3
Cumulative counts of sampled sentences by predicted class, pooled across benchmarks. Counts start at zero; a sentence is counted when its last reasoning token is reached. Progress is normalized by each attempt's full reasoning-token length, unlike the absolute windows in Figure D.8. Legend totals equal the rightmost endpoints. These are raw sampled counts, not cumulative percentages or estimates of full-corpus counts; longer traces and benchmark sampling quotas affect them.

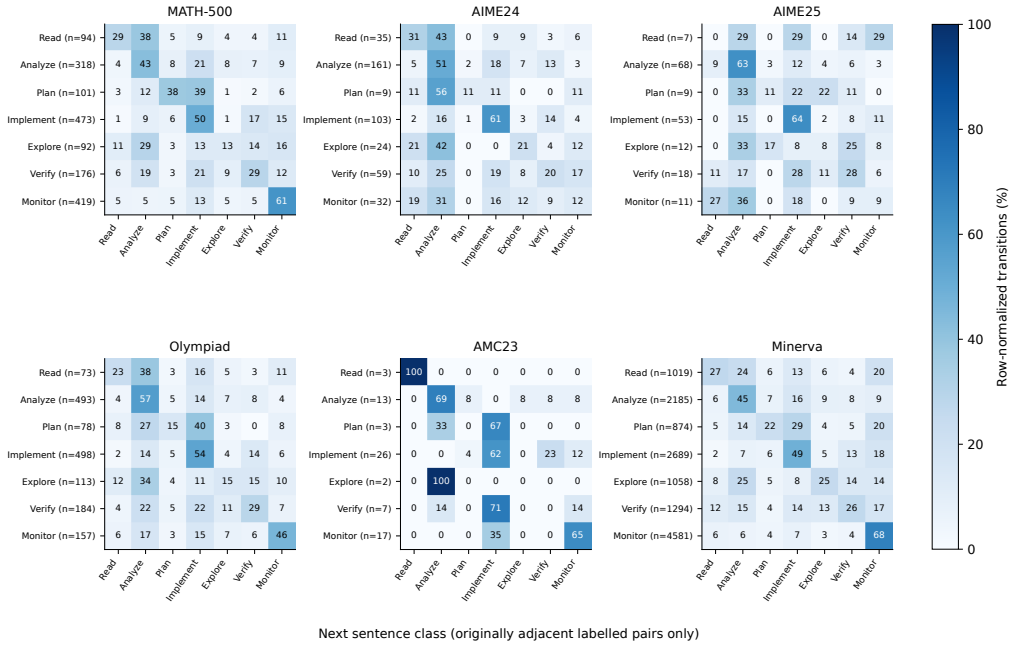


Figure D.4
Observed class-state transitions by benchmark. Rows are the current class, columns the next class; cells are row percentages and row labels give pair counts. Only originally consecutive sentence indices with both labels are counted. Gaps in sampling are never bridged. Self-transitions are retained; empty rows are undefined. This describes the observed adjacent-pair sample, not the complete transition process or an effect of changing state.



Figure D.5
Sampled class proportions versus whole-attempt correctness (point-biserial r) and total continuation-token count (Spearman ρ). All labelled attempts enter each class, including observed zero proportions. These differ from correlations of router features within class-selected tokens. No significance claim is made; class shares sum to one and cannot be interpreted as independent effects.

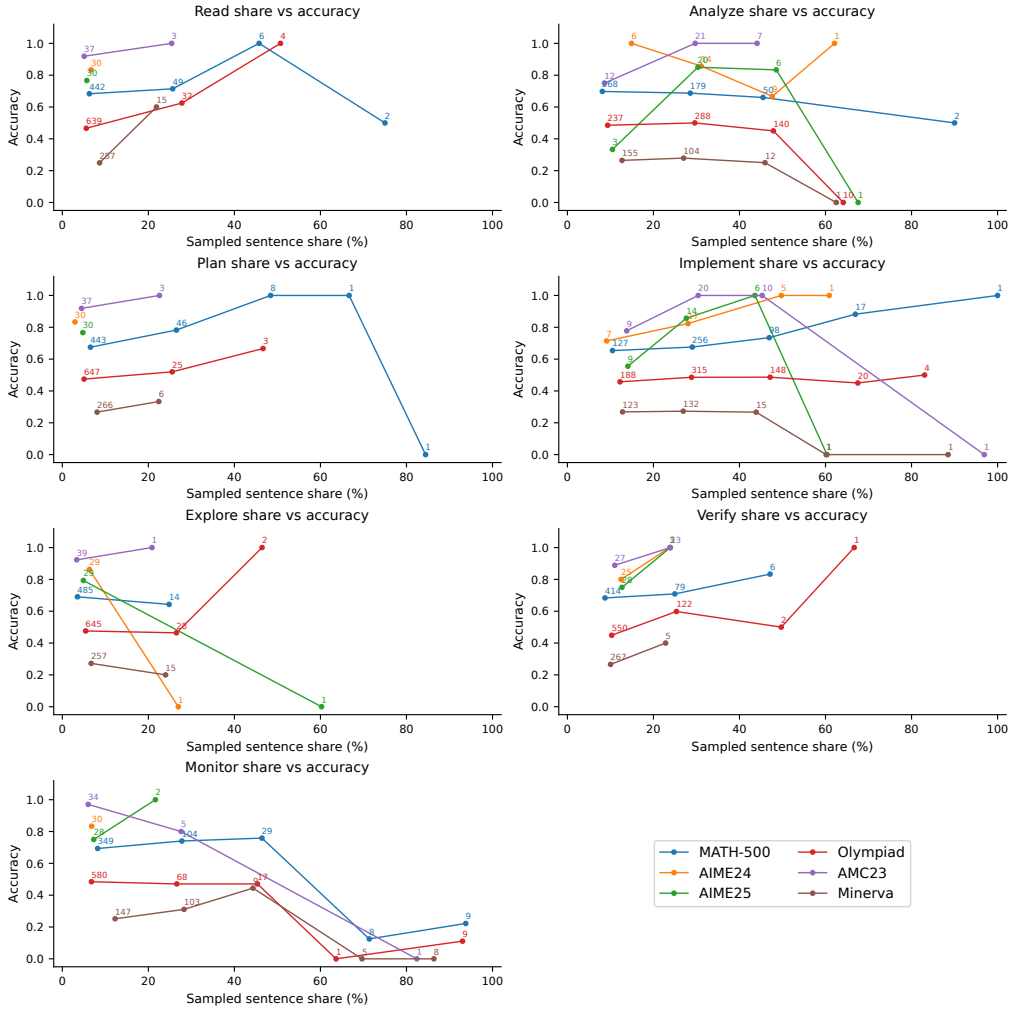


Figure D.6
Actual outcome levels versus sampled class proportions. Fixed share bins are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ percent. Each point is at the mean observed share and shows accuracy; small labels give attempt counts. Empty bins are omitted. Colours distinguish benchmarks; lines only connect occupied bins. Sparse bins, sentence sampling, length, and difficulty can explain apparent trends; these descriptive plots do not identify optimal class proportions or a validated accuracy–length Pareto frontier.

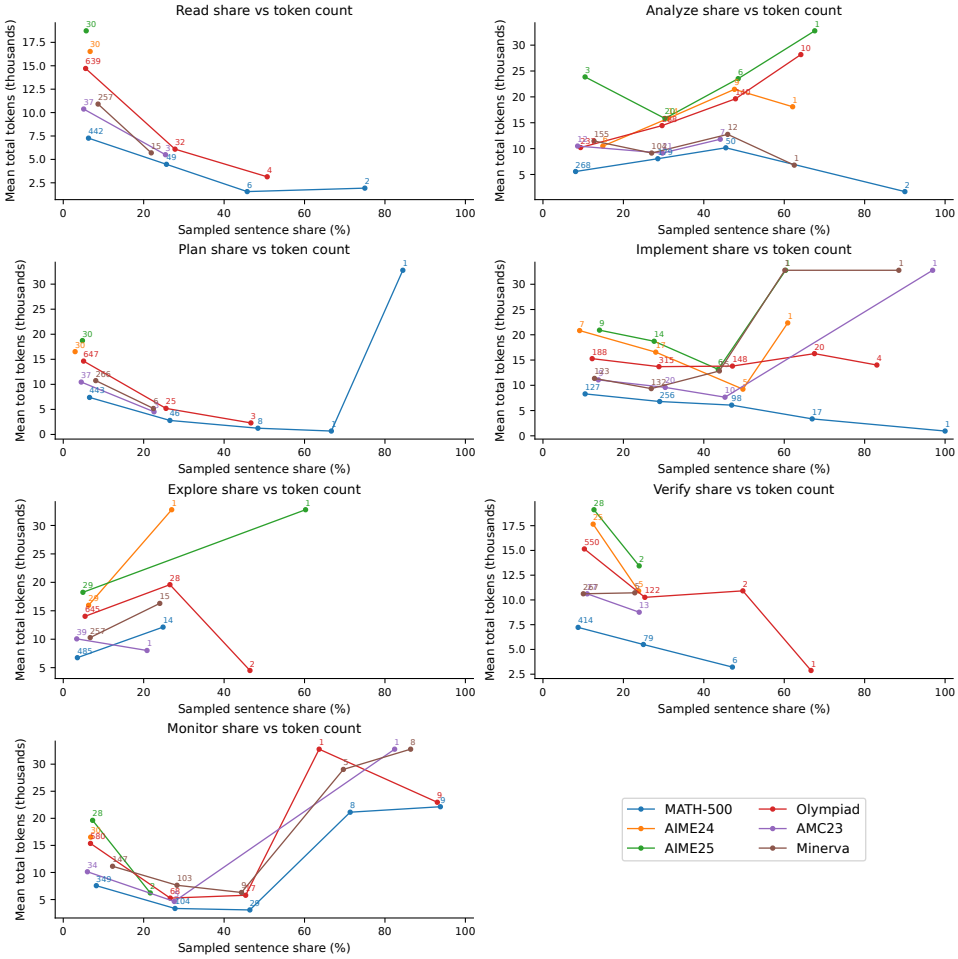


Figure D.7
Mean total continuation tokens versus sampled class share. Bin boundaries, colours, sample-count labels, and sampling limitations are as in Figure D.6. Token count includes the final answer and token-limit completions; lines are descriptive, not intervention effects.

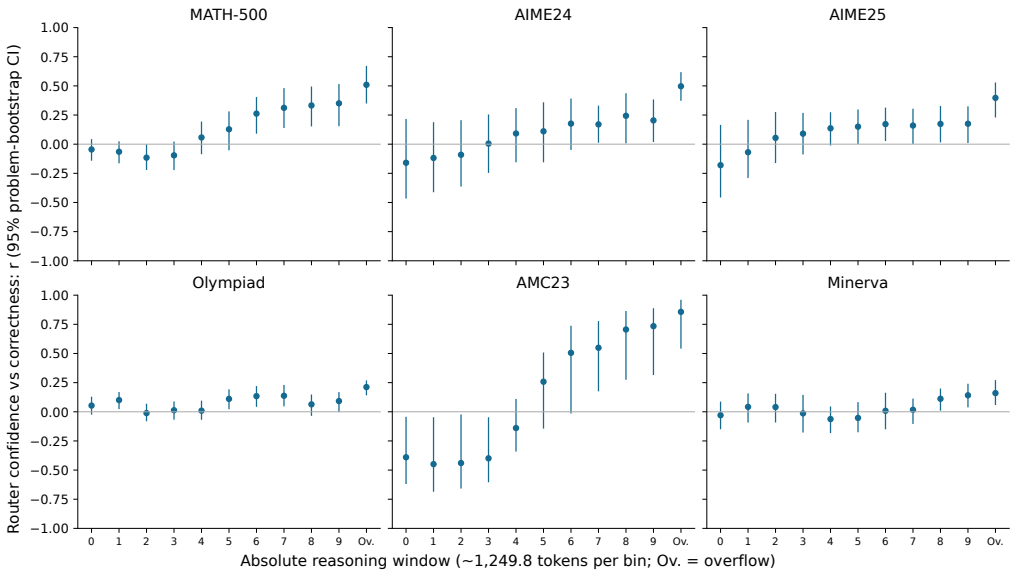


Figure D.8
Router confidence versus whole-attempt correctness in absolute reasoning windows, with saved pointwise 95% source-problem bootstrap intervals. Each benchmark has its own panel. Missing estimates remain blank. Later windows use changing token-bearing cohorts; these are not within-trace causal effects.

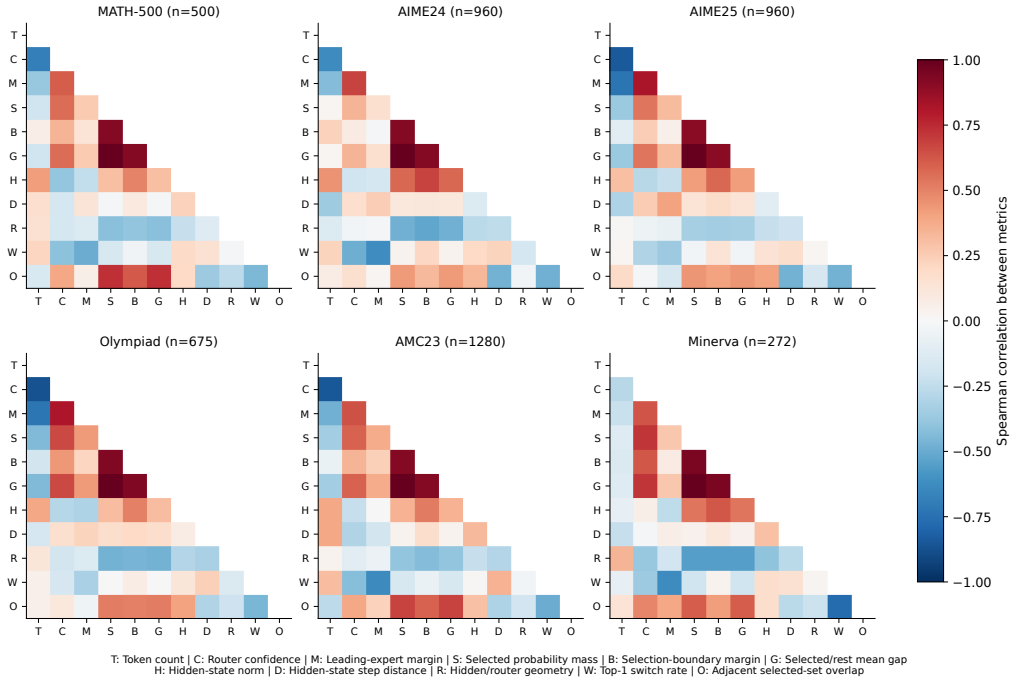


Figure D.9
Whole-continuation metric–metric Spearman correlations across all six benchmarks. Lower triangles display each association once; diagonals and upper triangles are omitted. Repeated attempts and limit hits remain included. These are descriptive metric relationships, not correctness effects.

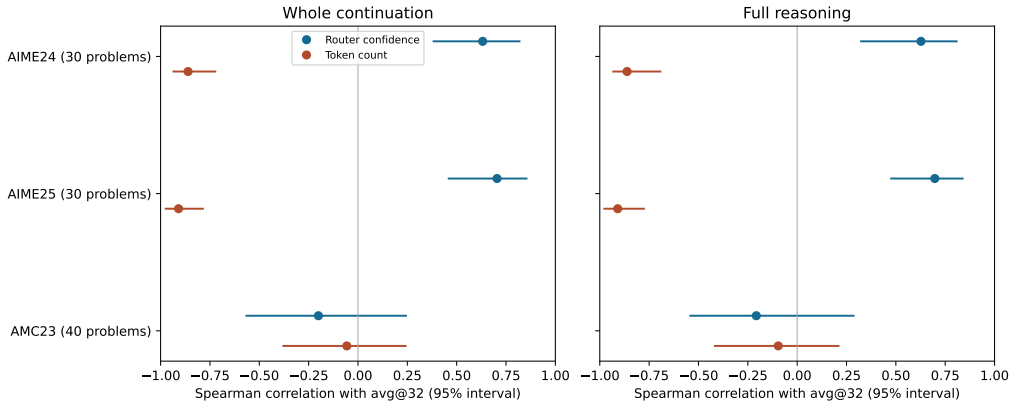


Figure D.10
Mean confidence and token count versus problem-level avg@32, with saved 95% whole-problem bootstrap intervals. Whole-continuation and full-reasoning views use the same complete 30/30/40 problem groups. These estimates remain unadjusted for difficulty, length, and termination.

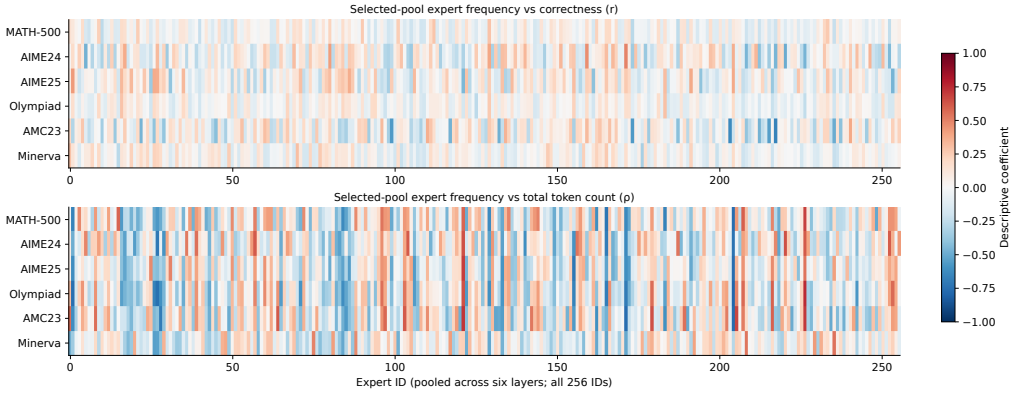


Figure D.11
Selected-pool membership frequency for every expert ID versus correctness (r, top) and total continuation tokens (ρ, bottom). All 256 IDs are shown in their original order for each benchmark. Rates pool tokens and six layers; the same ID refers to distinct layer-local modules. These associations do not establish that activating an expert will improve accuracy or reduce length.

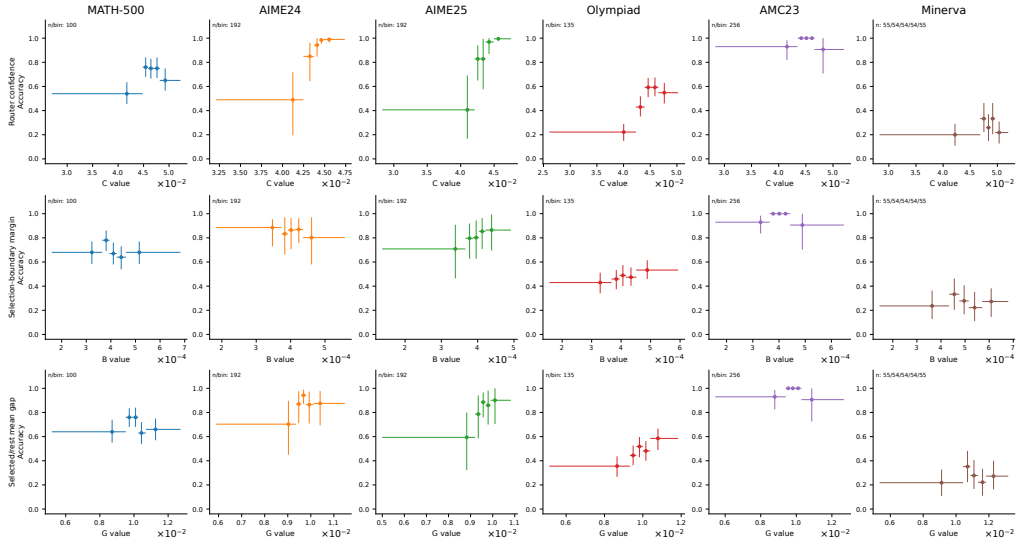


Figure D.12
Accuracy at observed router-metric values. Each benchmark uses up to five quantile bins; horizontal bars show actual value ranges, points show mean values and accuracy, and panel labels give attempt counts per bin (left to right, or a shared count when equal). Vertical bars are pointwise 95% percentile intervals from 500 whole-source-problem bootstrap resamples within each fixed bin (seed 42). Degenerate intervals can occur in all-correct/all-incorrect bins and do not imply certainty. All attempts and limit hits are retained. Bins can share source problems, so their estimates are not independent; neither bin selection nor length and termination confounding is addressed by these intervals.

Appendix E: Three-model correlation results

Forward replay and saved correlation analyses are complete for Qwen3.5-35B-A3B, Gemma-4-26B-A4B, and GPT-OSS-20B. Each includes 4,647 attempts from 1,547 source problems on the same six benchmarks, with 100 complete avg@32 groups. All token-limit completions are retained. Completion of class analysis means analysis of available labels, not full annotation coverage. The refreshed Gemma and GPT-OSS class analyses use the imported stratified labels and cover all six benchmarks; individual class estimates still depend on token-bearing and finite-outcome support. Qwen uses the local deterministic NVFP4 judge described above. Gemma and GPT-OSS use imported Qwen3.8-27B labels, with temperature 1, top-p 0.95, top-k 20, low reasoning effort and a 16,384-token judge budget. Gemma and GPT-OSS share the stratified sampling procedure and judge settings, but have model-specific selected problems and sentence quotas. Qwen retains a different annotation protocol. Generation uses Qwen GPTQ-Int4, Gemma NVFP4, and GPT-OSS checkpoints respectively; replay settings and selected decoder indices are below. Token counts and absolute position windows are model-specific. Correlations are marginal descriptive associations, not controlled model effects.

Table E.1: Replay configuration and mean reasoning length used for position windows.

Model	Replay	Decoder indices	Mean tokens
Qwen3.5-35B-A3B	unsloth-4bit	27, 34, 35, 36, 38, 39	12,498.1
Gemma-4-26B-A4B	bnb-4bit	16, 19, 21, 22, 25, 27, 29	10,512.6
GPT-OSS-20B	mxfp4-bf16	15, 19, 21, 23	5,977.6

Table E.2: All-attempt accuracy and termination by model. Invalid known-gold answers count as incorrect; repeated attempts are not independent problems.

Model	Dataset	Attempts	Correct	Accuracy	Limit hits
Qwen3.5-35B-A3B	MATH-500	500	345	0.690	17
Qwen3.5-35B-A3B	AIME24	960	817	0.851	105
Qwen3.5-35B-A3B	AIME25	960	773	0.805	165
Qwen3.5-35B-A3B	Olympiad	675	322	0.477	89
Qwen3.5-35B-A3B	AMC23	1280	1238	0.967	17
Qwen3.5-35B-A3B	Minerva	272	73	0.268	19
Gemma-4-26B-A4B	MATH-500	500	353	0.706	6
Gemma-4-26B-A4B	AIME24	960	852	0.887	54
Gemma-4-26B-A4B	AIME25	960	791	0.824	103
Gemma-4-26B-A4B	Olympiad	675	349	0.517	36
Gemma-4-26B-A4B	AMC23	1280	1249	0.976	3
Gemma-4-26B-A4B	Minerva	272	82	0.301	2
GPT-OSS-20B	MATH-500	500	332	0.664	4
GPT-OSS-20B	AIME24	960	681	0.709	99
GPT-OSS-20B	AIME25	960	668	0.696	118
GPT-OSS-20B	Olympiad	675	285	0.422	34
GPT-OSS-20B	AMC23	1280	1152	0.900	6
GPT-OSS-20B	Minerva	272	31	0.114	0

Table E.3: Class token-bearing attempts; classes overlap. Zero coverage implies unavailable estimates.

Model	Class	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Qwen3.5-35B-A3B	Read	378	30	26	514	27	270

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Model	Class	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Qwen3.5-35B-A3B	Analyze	424	30	30	619	38	271
Qwen3.5-35B-A3B	Plan	407	28	28	507	28	271
Qwen3.5-35B-A3B	Implement	477	30	30	667	39	271
Qwen3.5-35B-A3B	Explore	260	28	27	474	22	251
Qwen3.5-35B-A3B	Verify	400	29	30	617	38	270
Qwen3.5-35B-A3B	Monitor	451	30	29	583	29	272
Gemma-4-26B-A4B	Read	434	28	30	601	37	265
Gemma-4-26B-A4B	Analyze	433	30	30	628	40	265
Gemma-4-26B-A4B	Plan	313	26	28	501	27	194
Gemma-4-26B-A4B	Implement	490	30	30	639	40	266
Gemma-4-26B-A4B	Explore	215	24	23	459	20	200
Gemma-4-26B-A4B	Verify	430	30	30	629	39	251
Gemma-4-26B-A4B	Monitor	329	30	29	542	34	217
GPT-OSS-20B	Read	314	28	28	517	32	177
GPT-OSS-20B	Analyze	341	30	30	587	34	216
GPT-OSS-20B	Plan	216	28	30	450	26	114
GPT-OSS-20B	Implement	470	30	30	634	40	248
GPT-OSS-20B	Explore	212	27	28	441	19	128
GPT-OSS-20B	Verify	346	30	30	583	32	169
GPT-OSS-20B	Monitor	368	30	30	577	29	221

E.1 Stratified annotation coverage

The September 21 export replaces the earlier Gemma error-weighted sample and GPT-OSS corpus-prefix labels. Each model has 100,000 selected sentences, drawn with seed 42 from one completion per source problem, with quotas proportional to dataset $\times$ correctness $\times$ reasoning-length-quartile supply. The four 25,000-sentence parts are merged, not independent experimental replicates. GPT-OSS uses the corrected length-based sampling plan, which falls back to replay tokens when server reasoning-token counts are zero. GPT-OSS contributes 100,000 valid labels on 1,510 problems; Gemma contributes 99,999 on 1,503 problems. One Gemma sentence has a missing judge response and is excluded; its trace remains partially labelled. Both models now have class-token coverage on all six benchmarks. Sampled class populations remain smaller than the full 4,647-attempt corpus and need not contain the same problems across models. Sampling eligibility uses the saved scored-trace population; subsequently retained invalid answers in the full-corpus accuracy analysis do not imply annotation eligibility.

Table E.4: Stratified annotation population. Problems each contribute one labelled completion; valid labels are sentence counts, not token counts. Missing labels are retained in the sampling record.

Model	Dataset	Problems	Valid labels	Missing
Gemma	MATH-500	494	20,825	0
Gemma	AIME24	30	2,887	0
Gemma	AIME25	30	3,861	0
Gemma	Olympiad	639	54,556	1
Gemma	AMC23	40	2,573	0
Gemma	Minerva	270	15,297	0
GPT-OSS	MATH-500	496	14,050	0
GPT-OSS	AIME24	30	4,128	0
GPT-OSS	AIME25	30	4,987	0
GPT-OSS	Olympiad	642	70,215	0
GPT-OSS	AMC23	40	1,373	0
GPT-OSS	Minerva	272	5,247	0

The common annotation procedure improves benchmark coverage but does not match model cohorts. Full class/metric matrices are in the appended atlases. Qwen

retains its earlier sampling and judge deployment, so the three-model class comparisons are not a controlled model comparison.

Across models, confidence–correctness associations do not have a uniform sign. Qwen is positive on all six benchmarks; Gemma is negative on AMC23 and Minerva, and GPT-OSS is negative on MATH-500, AIME25, AMC23, and Minerva. The pointwise intervals in the comparison figures show substantial uncertainty for several of these estimates. Token-count associations are negative except for Gemma on AMC23. These differences motivate model-specific validation rather than assuming that a Qwen-derived routing rule transfers to other models.

E.2 Qwen3.5-35B-A3B

Table E.5: Qwen3.5-35B-A3B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.291 (500)	−0.726 (960)	−0.727 (960)	−0.387 (675)	−0.238 (1,280)	−0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	−0.035 (960)	0.222 (960)	0.118 (675)	−0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Hidden-state norm	0.203 (500)	−0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	−0.152 (500)	−0.058 (960)	−0.077 (960)	−0.102 (675)	−0.229 (1,280)	−0.096 (272)
Top-1 switch rate	−0.071 (500)	−0.297 (960)	0.141 (960)	0.005 (675)	−0.160 (1,280)	−0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	−0.155 (960)	0.016 (675)	−0.062 (1,280)	−0.010 (272)

Confidence coefficients in benchmark order are 0.199, 0.533, 0.493, 0.243, 0.175, 0.112.

Table E.6: Qwen3.5-35B-A3B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.861 (30)	−0.909 (30)	−0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	−0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	−0.112 (30)	0.388 (30)	−0.215 (40)
Selection-boundary margin	−0.367 (30)	0.135 (30)	−0.145 (40)
Selected/rest mean gap	−0.112 (30)	0.388 (30)	−0.215 (40)
Hidden-state norm	−0.384 (30)	−0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	−0.331 (40)
Top-1 switch rate	−0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	−0.047 (30)	−0.196 (30)	−0.327 (40)

E.3 Gemma-4-26B-A4B

Table E.7: Gemma-4-26B-A4B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.155 (500)	−0.672 (960)	−0.709 (960)	−0.240 (675)	0.080 (1,280)	−0.205 (272)
Router confidence	0.080 (500)	0.392 (960)	0.313 (960)	0.027 (675)	−0.256 (1,280)	−0.037 (272)
Leading-expert margin	0.108 (500)	0.297 (960)	0.456 (960)	0.036 (675)	−0.243 (1,280)	0.010 (272)

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Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Selected probability mass	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	−0.217 (1,280)	−0.061 (272)
Selection-boundary margin	−0.027 (500)	0.261 (960)	0.068 (960)	−0.043 (675)	−0.128 (1,280)	−0.028 (272)
Selected/rest mean gap	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	−0.217 (1,280)	−0.061 (272)
Hidden-state norm	0.036 (500)	0.161 (960)	0.007 (960)	−0.005 (675)	0.104 (1,280)	0.003 (272)
Hidden-state step distance	0.010 (500)	0.272 (960)	0.440 (960)	0.001 (675)	−0.141 (1,280)	0.059 (272)
Hidden/router geometry	−0.068 (500)	0.020 (960)	0.000 (960)	−0.019 (675)	−0.043 (1,280)	−0.076 (272)
Top-1 switch rate	0.052 (500)	−0.133 (960)	0.222 (960)	0.048 (675)	−0.116 (1,280)	0.100 (272)
Adjacent selected-set overlap	−0.021 (500)	0.129 (960)	−0.244 (960)	−0.049 (675)	0.038 (1,280)	−0.117 (272)

Confidence coefficients in benchmark order are 0.080, 0.392, 0.313, 0.027, -0.256, -0.037.

Table E.8: Gemma-4-26B-A4B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.744 (30)	−0.917 (30)	−0.099 (40)
Router confidence	0.473 (30)	0.468 (30)	0.071 (40)
Leading-expert margin	0.586 (30)	0.777 (30)	0.191 (40)
Selected probability mass	0.416 (30)	0.403 (30)	0.027 (40)
Selection-boundary margin	0.345 (30)	0.182 (30)	−0.063 (40)
Selected/rest mean gap	0.416 (30)	0.403 (30)	0.027 (40)
Hidden-state norm	0.026 (30)	−0.042 (30)	−0.060 (40)
Hidden-state step distance	0.410 (30)	0.547 (30)	−0.063 (40)
Hidden/router geometry	0.300 (30)	0.042 (30)	−0.004 (40)
Top-1 switch rate	−0.367 (30)	0.285 (30)	−0.215 (40)
Adjacent selected-set overlap	0.213 (30)	−0.321 (30)	0.059 (40)

E.4 GPT-OSS-20B

Table E.9: GPT-OSS-20B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.221 (500)	−0.606 (960)	−0.646 (960)	−0.362 (675)	−0.169 (1,280)	−0.094 (272)
Router confidence	−0.061 (500)	0.210 (960)	−0.075 (960)	0.051 (675)	−0.077 (1,280)	−0.160 (272)
Leading-expert margin	0.041 (500)	0.137 (960)	−0.176 (960)	0.154 (675)	−0.018 (1,280)	−0.082 (272)
Selected probability mass	−0.052 (500)	0.128 (960)	−0.102 (960)	0.089 (675)	−0.033 (1,280)	−0.166 (272)
Selection-boundary margin	−0.036 (500)	−0.218 (960)	−0.091 (960)	−0.064 (675)	−0.005 (1,280)	−0.032 (272)
Selected/rest mean gap	−0.052 (500)	0.128 (960)	−0.102 (960)	0.089 (675)	−0.033 (1,280)	−0.166 (272)
Hidden-state norm	0.192 (500)	0.494 (960)	0.366 (960)	0.329 (675)	0.049 (1,280)	−0.006 (272)
Hidden-state step distance	−0.150 (500)	−0.186 (960)	−0.404 (960)	−0.210 (675)	−0.094 (1,280)	0.045 (272)
Hidden/router geometry	0.002 (500)	−0.239 (960)	−0.311 (960)	−0.076 (675)	−0.027 (1,280)	0.053 (272)
Top-1 switch rate	0.181 (500)	0.149 (960)	0.376 (960)	0.110 (675)	−0.006 (1,280)	0.331 (272)
Adjacent selected-set overlap	−0.198 (500)	−0.068 (960)	−0.278 (960)	−0.143 (675)	−0.052 (1,280)	−0.317 (272)

Confidence coefficients in benchmark order are -0.061, 0.210, -0.075, 0.051, -0.077, -0.160.

Table E.10: GPT-OSS-20B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.889 (30)	−0.908 (30)	−0.338 (40)
Router confidence	0.337 (30)	0.104 (30)	0.186 (40)
Leading-expert margin	0.248 (30)	−0.028 (30)	0.262 (40)

Continued from previous page

Metric	AIME24	AIME25	AMC23
Selected probability mass	0.186 (30)	0.046 (30)	0.272 (40)
Selection-boundary margin	−0.356 (30)	−0.273 (30)	0.060 (40)
Selected/rest mean gap	0.186 (30)	0.046 (30)	0.272 (40)
Hidden-state norm	0.733 (30)	0.655 (30)	0.338 (40)
Hidden-state step distance	−0.341 (30)	−0.655 (30)	−0.289 (40)
Hidden/router geometry	−0.658 (30)	−0.474 (30)	−0.177 (40)
Top-1 switch rate	0.162 (30)	0.546 (30)	−0.151 (40)
Adjacent selected-set overlap	−0.029 (30)	−0.340 (30)	−0.089 (40)

Aggregate correctness and metric-pair coefficients were independently recomputed from saved CSVs in all 60 model/scope combinations; repeated-problem mean correlations and within-problem contrasts were also checked. This validates numerical consistency, not judge accuracy or replay fidelity. Audit counts and numerical errors are saved in `report/multimodel_correlation_audit.json`; input SHA-256 hashes are in `report/multimodel_correlation_sources.json`. The appended atlases include every Gemma and GPT-OSS scope, with unavailable coefficients explicitly marked. Full correctness tables are available in `report/multimodel_correlation_tables.tex`.

E.5 Directed class transitions by answer correctness

For each model, we separate labelled attempts by whole-answer correctness and count directed transitions between originally adjacent labelled sentences. Rows are source classes and columns are destination classes. Self-transitions are retained; missing labels and sampling gaps are never bridged. Each cell is the total observed count divided by the number of labelled attempts in that outcome group: equivalently, the mean per-attempt count, including zero-pair attempts. Empty-label attempts are excluded. These are counts per attempt, not row-normalized probabilities or full-trace transition estimates.

The two panels per model pool the six benchmarks and all six panels share one colour scale. Benchmark-specific counts and denominators are saved in the accompanying statistics JSON. Outcome groups differ in benchmark mix, reasoning length, and label coverage; Qwen also retains a different annotation protocol from Gemma and GPT-OSS. Differences are descriptive associations, not evidence that a transition causes correctness or a controlled comparison between models.

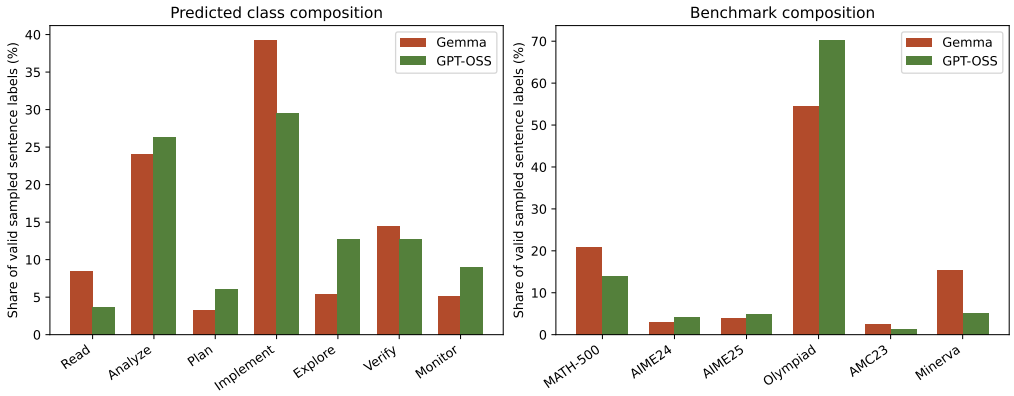


Figure E.1
Gemma and GPT-OSS stratified label composition. Denominators are 99,999 and 100,000 valid sampled sentences. Equal sample budgets do not imply equal class or benchmark proportions. These sentence shares are neither complete-corpus token shares nor adjusted model effects.

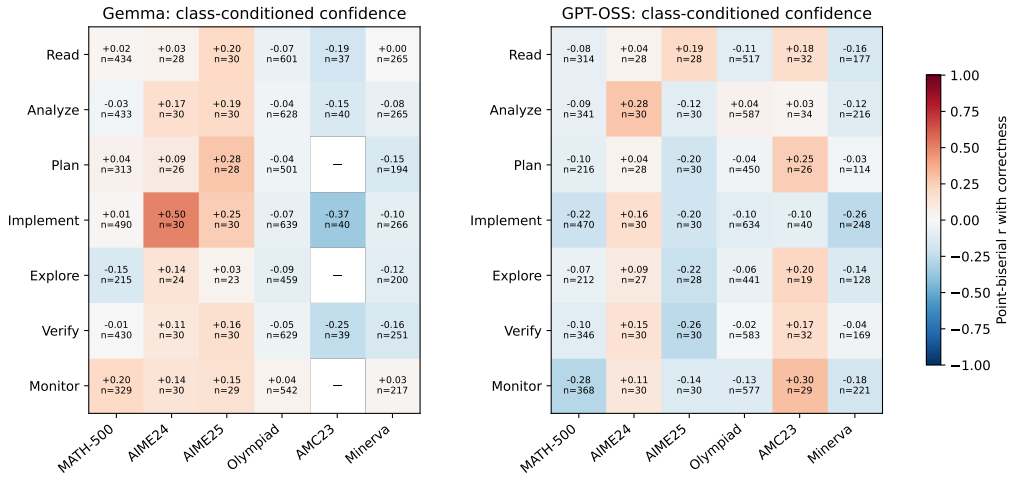


Figure E.2
Class-conditioned router confidence versus whole-attempt correctness after stratified relabelling. Each cell reports the saved point-biserial coefficient and its finite-attempt count. Cohorts include only attempts with tokens assigned to that class; missing coefficients are undefined, not zero. Sparse AIME/AMC samples, near-ceiling outcomes, and model-specific problem selection limit comparisons. This descriptive screen does not adjust for length or difficulty or establish a causal class effect.

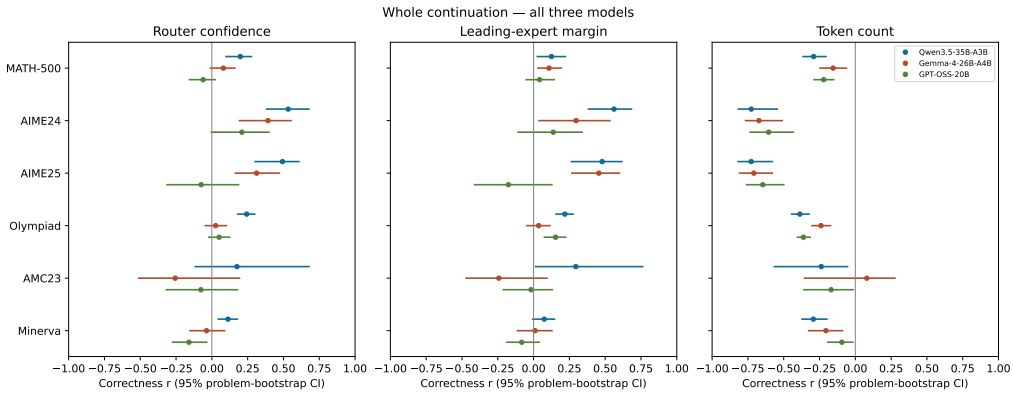


Figure E.3
Whole continuation: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

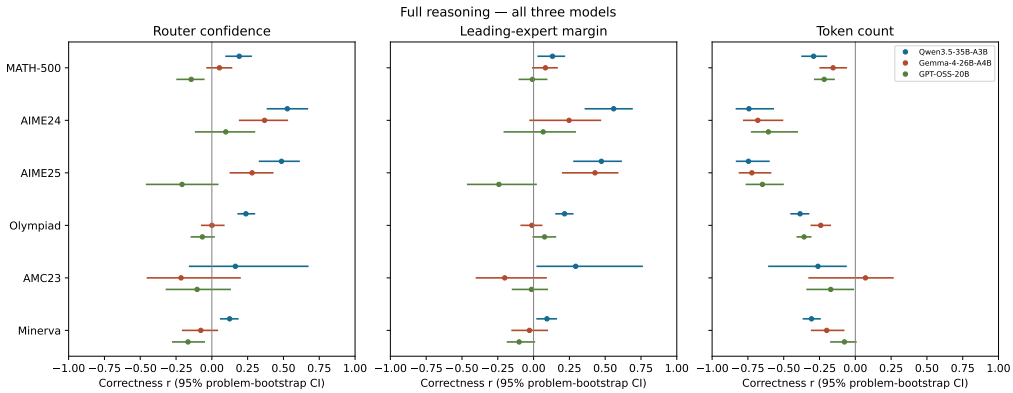


Figure E.4
Full reasoning: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

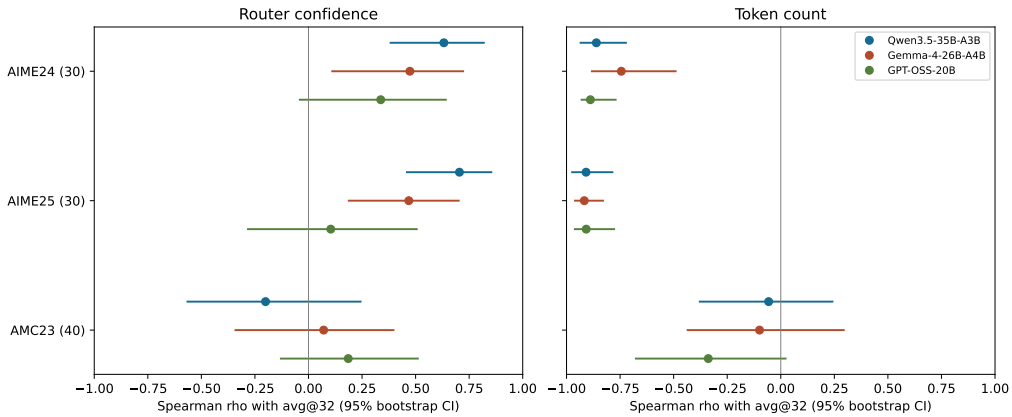


Figure E.5
All models: whole-continuation problem means versus avg@32, with saved pointwise problem-bootstrap intervals. All 32 attempts enter each problem mean.

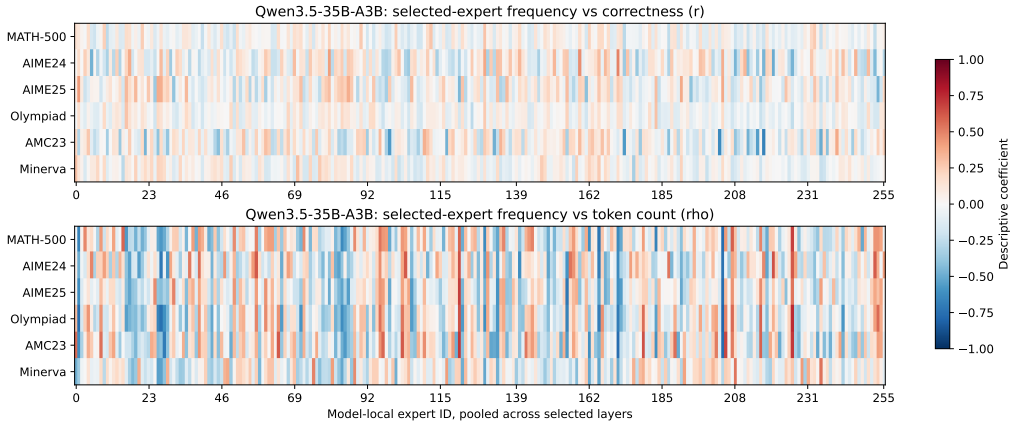


Figure E.6
Qwen3.5-35B-A3B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

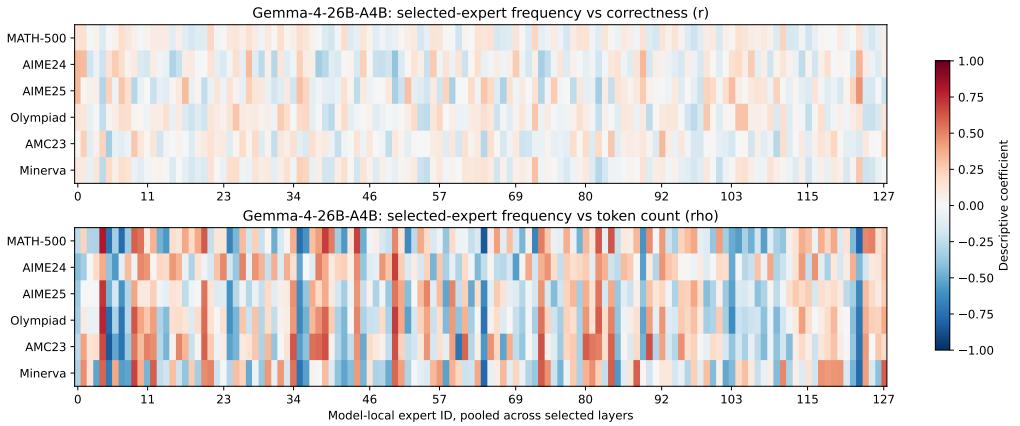


Figure E.7
Gemma-4-26B-A4B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

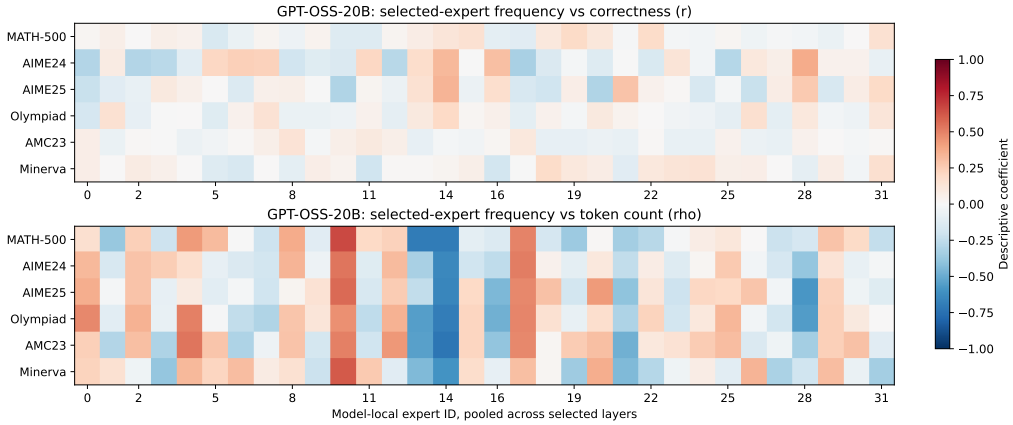


Figure E.8
GPT-OSS-20B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

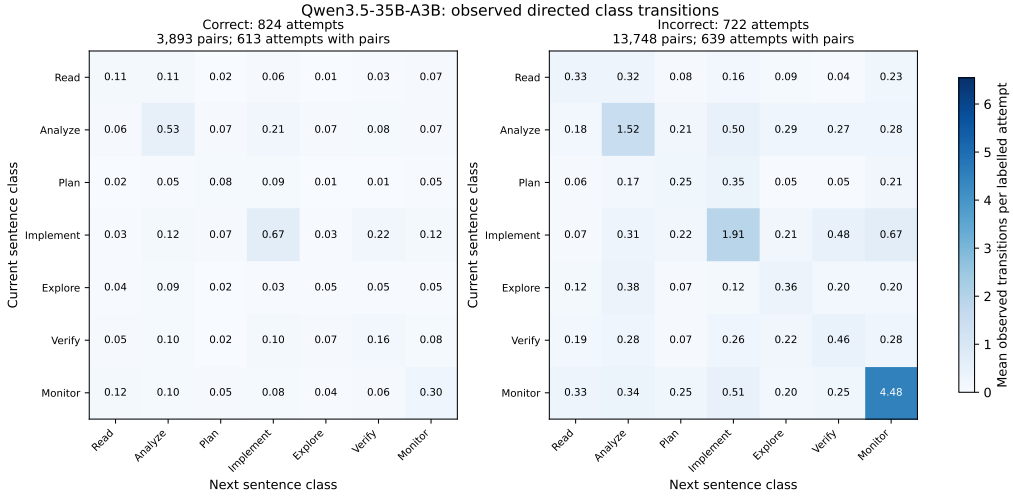


Figure E.9
Qwen3.5-35B-A3B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

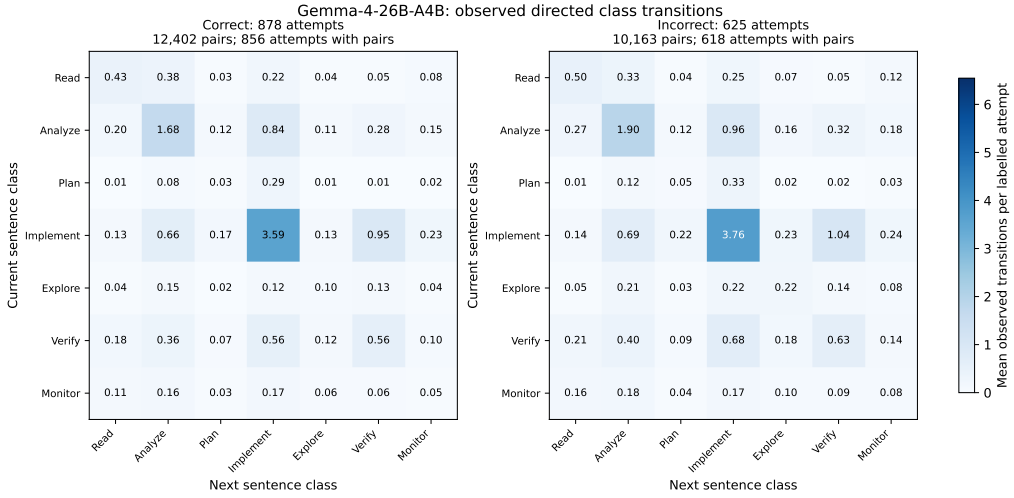


Figure E.10
Gemma-4-26B-A4B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

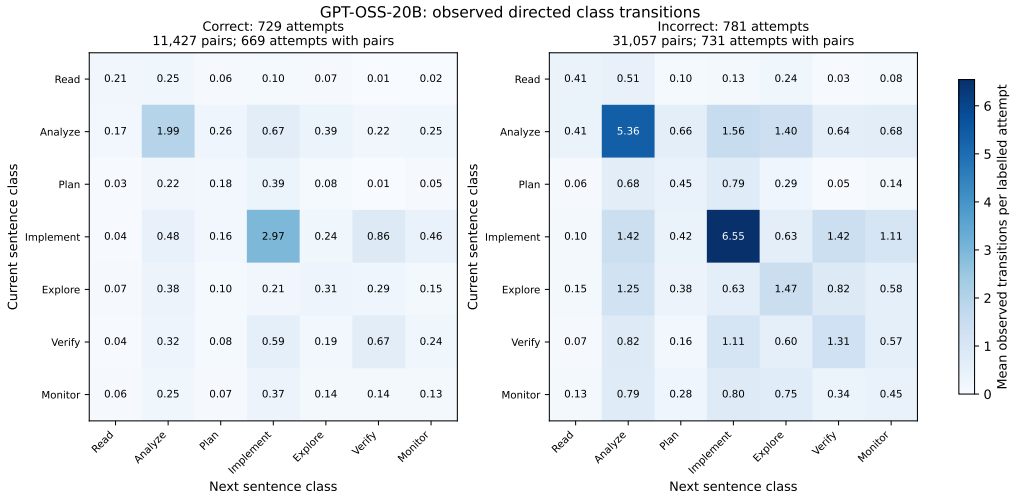


Figure E.11
GPT-OSS-20B: directed class transitions for correct and incorrect answers. Cells show mean observed counts per labelled attempt, including self-transitions on the diagonal. Panel headings give denominators and adjacent-pair coverage; zero-pair attempts remain in the mean. The shared colour scale is identical across all three models. Sampling and population differences limit interpretation.

Appendix F: Held-out Routing Interventions

Calibration and evaluation population.. The original Qwen identity and margin policies split source problems within each benchmark, allocating 1,082 problems to calibration and 465 to evaluation (70/30, seed 42). Calibration uses 3,014 scored traces from 1,001 problems; 238 traces with missing saved correctness are excluded. This differs from the correlation analyzer, which counts known-gold missing answers as incorrect. The problem-weighted calibration accuracy is 0.5817 on this scored subset. Consequently, calibration targets may be sensitive to answer-extraction failures and must not be interpreted as estimates over all allocated traces. Each problem contributes total weight one, divided across its usable attempts. Evaluation retains 150 MATH-500, 9 AIME24, 9 AIME25, 203 OlympiadBench, 12 AMC23, and 82 Minerva problems. With 32 attempts on each AIME/AMC problem, this gives 1,395 paired attempts. No evaluation problem is used in policy calibration.

Identity policy.. At each of the six selected Qwen layers, expert inclusion is weighted by its routing fraction and problem weight. Experts require support from at least four distinct problems and positive accuracy lift over the calibration baseline. At most eight identities per layer are selected; positive lifts are normalized to the largest retained lift. A signed strength scales the additive router-logit bias. Strength $+1$ promotes these identities; strength -1 suppresses the same identities. This negative-strength control suppresses positive-lift experts; the additional negative-polarity conditions below select a separate set of negative-lift experts. Unlike the pooled-ID observational screen, policy targets are layer-qualified.

Margin policy.. The policy calibrates the trace-mean selection-boundary margin $p_{(k)} - p_{(k+1)}$, with $k = 8$, using five problem-weighted quantile bins. Among bins with at least four supporting problems, it selects the highest-accuracy positive-lift bin, breaking ties by support and then lower margin. At inference, probability mixing moves the token-level gap toward the selected interval: uniform mixing reduces the gap, whereas mixing with a distribution uniform over the selected top-k increases it. Trace-level calibration and token-level application have different units. Both policies transfer routing targets from Unsloth 4-bit replay to GPTQ generation, which is an additional unisolated deployment difference.

Matched evaluation.. Each policy is compared with its own baseline using identical problem IDs, prompts, prompt token IDs, per-attempt seeds, sampling settings, engine configuration, and recorded software versions. Generation uses temperature 0.6, top-p 0.95, and a 32,768-token limit. The original Qwen identity $+1$ and margin comparisons use maximum concurrency 16; Qwen identity -1 and $+2$ use 25. The seven OSS identity conditions share concurrency 25 and the same recorded generation contract. Baselines are interchangeable only after compatibility and output-hash checks. Primary summary tables include complete matched stochastic pairs. All ten complete identity comparisons appear in Table F.4; the Qwen paper run was interrupted and its completed subset is reported separately in Section F.2. Separate greedy and restricted-benchmark outputs are not pooled here. The engine writes completed records incrementally and can resume without continuing a partially generated sequence. Baseline reuse requires matching contracts and a verified output hash; the Qwen and GPT-OSS margin runs at concurrency 25 reuse compatible saved baselines. The evaluated controllers use calibrated identities or current margins, without online episode predictions. Annotation

and probing support measurement and target exploration. Accuracy tables use the uniform offline equivalence rule in Section F.1; original judgments remain archived. Calibration labels and fitted policies are held fixed.

F.1 Consistent offline answer-equivalence audit

The qualitative inspection identified equivalent final answers rejected by the historical normalized-string/numeric fallback. A uniform offline pass therefore rescores both conditions in every saved, manifested identity and margin evaluation: 13 complete full stochastic comparisons, two complete greedy comparisons, the 39-problem Minerva comparison, and the interrupted Qwen subset. It reads 40,097 stored condition records, including reused baseline copies; this is not a count of independent generated solutions. The original text, tokens, manifests, policies, and judgments are preserved. Revised decisions and all changed answers are recorded separately in `report/guiding_rescoring.json`.

The rule first extracts the last nonempty balanced boxed candidate, then uses the existing answer-marker and last-number fallbacks when no such box is present. Only that candidate is compared with the gold answer. It normalizes display notation, plain-text labels, common units, currency, and explicit list separators. Mathematical comparison uses Math-Verify 0.9.0 with latex2sympy2_extended 1.11.0, ANTLR 4.13.2, and SymPy 1.14.0[1]. Variables and structured objects use strict comparison; exact-number answers require equality, while decimal/scientific scalar answers allow relative tolerance 10^{-6} with no absolute floor. Thus zero does not match a small nonzero answer merely because its magnitude is below 10^{-6}. Scientific notation is parsed numerically, rather than as a symbolic expression involving e. Text denoting multiple answers is not silently reduced to its first number. Standard unit annotations follow the parser's normalization conventions; this is not a dimensional-analysis check.

The pass changes 4,054 stored judgments from incorrect to correct and one from correct to incorrect. It also changes the extracted candidate in 2,065 records. The single downward change concerns a capped Qwen baseline whose extracted box contains the literal placeholder "answer" rather than its earlier saved numeric fallback. These counts include repeated baseline copies. Changes occur on every benchmark, mainly OlympiadBench, MATH-500, and Minerva. The analysis reruns the same 5,000 paired, dataset-stratified source-problem bootstrap resamples (seed 42), including the equal-problem sensitivity statistic. The 494-ID interrupted comparison remains separate and its accuracy counts are unchanged.

No revised positive accuracy interval excludes zero in the complete comparisons under either reported weighting. In particular, the earlier positive secondary OSS findings and greedy Qwen margin interval do not survive the common scoring rule. The Qwen concurrency-25 margin lower endpoint is exactly zero. Strong negative effects under paper-style promotion remain. Generation costs and termination flags are unchanged.

The checker is tested on equivalent and distinct fractions, text labels, scientific notation, ordered answers, units, currency, multiple-answer text, and very small numerical values, as well as final-candidate extraction. It does not provide independent human adjudication. There are 699 records with an unparsed candidate or reference, 332 with unsupported embedded text, and 30 parser/comparison errors; these receive

1 `https://github.com/huggingface/Math-Verify`

Table F.1
Selected changes after applying the same revised scoring rule to both conditions. Changes and 95% intervals are percentage points. Attempt weighting is primary; problem weighting is secondary. Historical values are retained here only to document the audit and are not mixed into current result tables.

Condition	Weighting	Historical change	Revised change	Revised 95% CI
Qwen margin (c=25)	Attempt	+1.08	+0.93	[0.00, +1.86]
GPT-OSS margin	Attempt	-3.15	-0.86	[-2.87, +1.15]
GPT-OSS fixed positive +2	Attempt	-1.43	-3.30	[-6.16, -0.93]
GPT-OSS fixed positive +2	Problem	+4.87	-0.10	[-3.12, +3.08]
GPT-OSS fixed negative +1	Problem	+4.94	+2.35	[-0.58, +5.44]
Qwen greedy margin	Attempt	+2.15	+0.86	[-1.29, +3.01]

an incorrect decision unless normalized display equality already succeeds. Error categories and changed judgments remain available for further review. A two-second parser/comparison timeout and a 2,048-character limit for nonnumeric expressions bound degenerate inputs. Calibration and observational correlation labels retain their original values: the revised evaluations test the existing frozen policies, not policies refitted with revised labels.

Table F.2
Complete matched Qwen3.5 GPTQ stochastic evaluations, 1,395 attempts on 465 held-out problems per comparison. B/G denotes baseline/guided; changes are percentage points. Flips are wrong-to-right/right-to-wrong. Intervals are paired, dataset-stratified source-problem bootstrap percentiles (5,000 resamples, seed 42).

Policy	B accuracy	G accuracy	Change	95% CI	Flips	Limit hits B/G	Mean tokens B/G
Identity $+1$	0.8495	0.8466	-0.29	[-1.15, +0.65]	38/42	140/146	13863/13706
Identity -1	0.8409	0.8516	+1.08	[-0.14, +2.37]	46/31	154/138	13830/13919
Margin 1	0.8430	0.8437	+0.07	[-0.86, +1.01]	40/39	144/143	13848/13840

Observed effects and limits.. Qwen identity $+1$ changes 80 recorded outcomes and loses four net correct answers; margin guiding changes 79 and gains one. Identity -1 changes 77 and gains 15. The corresponding rescored changes are -0.29, $+0.07$, and $+1.08$ points. These original three comparisons do not establish a beneficial direction. The negative-strength point estimate is positive, but differences across strengths are not a controlled ranking because their baselines and concurrency differ. These are exploratory comparisons without a separate strength-selection set. Paired, dataset-stratified source-problem bootstrap intervals use 5,000 resamples with seed 42. All attempts and both conditions remain together when a problem is sampled; pooled accuracy retains its attempt weighting. The 95% intervals for identity $+1$, identity -1, and margin guiding are respectively $[-1.15, +0.65]$, $[-0.14, +2.37]$, and $[-0.86, +1.01]$ percentage points. All include zero, so these comparisons do not establish the direction of the accuracy effect. They do not establish equivalence either. Each AIME evaluation has only nine source problems, which limits the precision of benchmark-specific bootstrap estimates. Intervals are conditional on the frozen policies and saved attempts, and are not adjusted for multiple comparisons. Pooled attempt accuracy also weights each AIME/AMC problem 32 times as heavily as each single-attempt problem; benchmark counts are reported to make this weighting explicit.

Table F.3
Qwen3.5 GPTQ guiding results by benchmark. Correct counts are baseline/guided; accuracy changes are percentage points. Intervals resample whole problems (5,000 resamples, seed 42); each AIME benchmark contains only nine evaluation problems.

Policy	Dataset	Attempts	Correct B/G	Change	95% CI
Identity +1	aime24	288	185/177	-2.78	[-4.17, -1.39]
Identity +1	aime25	288	246/246	+0.00	[-1.74, +2.78]
Identity +1	amc23	384	382/384	+0.52	[+0.00, +1.56]
Identity +1	math500	150	148/148	+0.00	[+0.00, +0.00]
Identity +1	minerva	82	51/54	+3.66	[-2.44, +9.76]
Identity +1	olympiad	203	173/172	-0.49	[-3.94, +2.96]
Identity −1	aime24	288	178/182	+1.39	[-2.08, +4.86]
Identity −1	aime25	288	242/248	+2.08	[-1.39, +6.25]
Identity −1	amc23	384	384/384	+0.00	[+0.00, +0.00]
Identity −1	math500	150	148/146	-1.33	[-3.33, +0.00]
Identity −1	minerva	82	50/53	+3.66	[-1.22, +8.54]
Identity −1	olympiad	203	171/175	+1.97	[-1.48, +5.42]
Margin 1	aime24	288	180/184	+1.39	[-0.69, +3.82]
Margin 1	aime25	288	243/247	+1.39	[-0.35, +3.47]
Margin 1	amc23	384	381/382	+0.26	[-0.52, +1.04]
Margin 1	math500	150	147/148	+0.67	[+0.00, +2.00]
Margin 1	minerva	82	51/49	-2.44	[-9.76, +3.66]
Margin 1	olympiad	203	174/167	-3.45	[-7.39, +0.00]

Completed identity comparisons.. For Qwen fixed positive-polarity guiding, strength $+1$ changes accuracy by -0.29 points with interval $[-1.15, +0.65]$, and strength -1 gives $+1.08$ with $[-0.14, +2.37]$. Increasing promotion to strength $+2$ gives -2.80 points with $[-5.16, -0.72]$. Thus stronger promotion reduces primary attempt-weighted accuracy under the revised scorer.

All seven GPT-OSS identity conditions use four targets at each of layers 15, 19, 21, and 23, matching native routing top-$k = 4$. Each evaluates 1,395 matched attempts from 465 held-out problems at concurrency 25, with a frozen chat-template date of 22 September 2026. Rescored baseline accuracy is 77.85% (1,086/1,395) in every condition. Fixed positive strengths $+1$, $+2$, and -1 give changes of 0.00, -3.30, and -1.00 points. Only the strength-$+2$ interval excludes zero: $[-6.16, -0.93]$. Fixed negative-polarity promotion gives $+0.57$ points with $[-1.58, +2.65]$ and does not establish a benefit.

Paper-style promotion at strength $+1$ reduces accuracy to 16.20% (226/1,395) for positive targets and 18.28% (255/1,395) for negative targets. Changes are -61.65 points with $[-69.32, -53.41]$ and -59.57 with $[-67.24, -51.40]$, respectively. Mean generated length rises from 6,497 to 22,489 and 23,639 tokens; token-limit hits increase from 74 to 670 and 815. Suppressing the positive set with paper-style strength -1 gives 75.63% accuracy and a smaller but below-zero primary change of -2.22 points ($[-4.16, -0.43]$); mean length is 6,698 tokens with 106 limit hits.

Paper-style steering adapts the extreme-score intervention with $\epsilon = 0.01$ while retaining our accuracy-lift selector (**?**); it is not a full reproduction of SteerMoE. The four-target budget does not prevent severe degradation from promotion. Promoting a full top-k set fixes membership and equalises its hook scores; this is a possible mechanism, not a verified explanation of termination.

Table F.4
Complete identity-guiding stochastic comparisons. Accuracy is percent; changes and 95% intervals are percentage points relative to each matched baseline. Fixed denotes the lift-weighted additive bias; Paper denotes extreme-score steering with epsilon 0.01. Polarity identifies the selected expert set, independently of signed strength. Each comparison has 1,395 attempts on 465 problems; intervals use 5,000 paired, dataset-stratified source-problem bootstrap resamples (seed 42).

Model	Method	Polarity	Strength	B accuracy	G accuracy	Change	95% CI
GPT-OSS	Fixed	Negative	+1	77.85	78.42	+0.57	[-1.58, +2.65]
GPT-OSS	Paper	Negative	+1	77.85	18.28	-59.57	[-67.24, -51.40]
GPT-OSS	Paper	Positive	-1	77.85	75.63	-2.22	[-4.16, -0.43]
GPT-OSS	Paper	Positive	+1	77.85	16.20	-61.65	[-69.32, -53.41]
GPT-OSS	Fixed	Positive	-1	77.85	76.85	-1.00	[-3.44, +1.22]
GPT-OSS	Fixed	Positive	+1	77.85	77.85	+0.00	[-2.15, +2.01]
GPT-OSS	Fixed	Positive	+2	77.85	74.55	-3.30	[-6.16, -0.93]
Qwen	Fixed	Positive	+1	84.95	84.66	-0.29	[-1.15, +0.65]
Qwen	Fixed	Positive	-1	84.09	85.16	+1.08	[-0.14, +2.37]
Qwen	Fixed	Positive	+2	84.09	81.29	-2.80	[-5.16, -0.72]

F.2 Interrupted Qwen paper-style intervention: partial results

The Qwen positive-expert paper-style intervention at strength $+1$ was stopped by user request after 494 of 1,395 planned attempts (35.4%). The manifest records a KeyboardInterrupt; 901 attempts have no saved guided result. This policy targets eight experts at each of six layers, matching the model routing top-$k = 8$; it is unaffected by the OSS target-count correction. The intervention uses epsilon 0.01, concurrency 25, and a 32,768-token completion budget. The saved outputs remain resumable; the run is not marked complete.

On the 494 exact matched baseline/guided IDs (16 source problems), baseline accuracy is 70.45% (348/494), and guided accuracy is 0.81% (4/494), a change of -69.64 percentage points. There are 4 wrong-to-right and 348 right-to-wrong flips. Mean generated length changes from 19,352 to 32,768 tokens (medians 17,252 and 32,768); token-limit hits change from 110/494 to 494/494.

Table F.5
Partial Qwen paper-style intervention. Accuracy and token counts compare only the same completed IDs in each condition. B/G denotes baseline/guided. A dash indicates no completed guided attempts, not zero accuracy.

Benchmark	Completed/planned	Problems	Accuracy B/G (%)	Change (pp)	Mean tokens B/G	Limit hits B/G
AIME24	288/288	9	61.81/0.00	-61.81	20,947/32,768	74/288
AIME25	206/288	7	82.52/1.94	-80.58	17,121/32,768	36/206
AMC23	0/384	0	—	—	—	—
MATH-500	0/150	0	—	—	—	—
Minerva	0/82	0	—	—	—	—
Olympiad	0/203	0	—	—	—	—

These are descriptive partial results, excluded from the complete-run comparison and weighting tables. The completed subset is not a random sample: benchmark submission order, scheduling, and generation length can affect which requests have finished. Multiple attempts from one problem are also dependent. Consequently the pooled difference is not an estimate for the full 1,395-attempt evaluation, and no population confidence interval or significance claim is reported. Missing attempts are not

counted as wrong or filled from the baseline. Increased token-limit incidence does not by itself establish repetitive looping.

All saved guided records have unique IDs, scored boolean outcomes, matching baseline inputs, prompt token IDs, and per-attempt sampling settings. Policy, engine, software, scoring, and template-date contracts also match; hook diagnostics confirm activity at all selected layers. Source hashes and benchmark denominators are saved in `report/partial_guiding_statistics.json`.

F.3 Qualitative inspection of paper-style failures

A reproducible, assistant-assisted excerpt inspection examines 22 saved historically baseline-correct/guided-incorrect pairs from strength-$+1$ paper-style promotion. For each complete GPT-OSS polarity, four cases are sampled without replacement from outputs ending at the token limit and four from outputs ending normally. Six additional capped cases are sampled from the interrupted Qwen positive run. Sampling uses sorted attempt IDs and seed 20260924, with separate streams per condition and termination stratum. The OSS samples cover eight and seven distinct problems; the Qwen sample covers four. Selection conditions on recorded correctness loss and termination, and repeated attempts can share a problem. The resulting counts describe this sample, not the prevalence of failure modes.

The review examines fixed beginning, middle, and ending excerpts, together with the most frequent exact 16-word span, against the corresponding baseline answer. Repetition is described from the text, not inferred from output length alone. The whole-text diagnostic counts case-sensitive, whitespace-delimited spans, allowing overlap; it does not capture every form of repetition, particularly unspaced symbol chains. This is neither independent human annotation nor exhaustive verification of the derivations.

Table F.6
Descriptions of 22 cases sampled as correctness losses under the original scorer. Capped/stopped gives the sampled termination strata. Repetition can coexist with mathematical errors; the arithmetic-error column describes the remaining case without an observed long cycle. Formatting disagreement denotes an equivalent final answer rejected by the saved scorer. Counts are conditional on the sampling design.

Condition	Capped/stopped	Repetition	Arithmetic error	Formatting	No box
GPT-OSS positive	4/4	6	0	2	4
GPT-OSS negative	4/4	7	1	0	4
Qwen positive (partial)	6/0	6	0	0	6

Repetition and termination.. All 14 sampled capped outputs contain repeated arithmetic or verbal cycles and lack a boxed-answer marker. Five of the eight normally ended OSS outputs also contain cycles before returning an incorrect boxed answer. Across these 19 repetitive outputs, the most frequent exact 16-word span occurs 62–9,906 times; the matched baseline maxima range from one to four. For example, positive OSS case P03 repeats `390?` while its baseline returns 49. Qwen case Q06 repeatedly emits "Wait, wait." while its baseline returns 82. Negative OSS case N06 repeats an unresolved sum and eventually returns 112 instead of 144. Thus repetition can accompany both failure to finish and a normally terminated wrong answer. The separate non-cyclic arithmetic example N05 correctly enumerates three, two, and two choices but evaluates $3 \times 2 \times 2 = 6$, returning 6 instead of 12.

Scoring disagreement.. Two normally ended positive OSS cases expose formatting sensitivity in the saved fallback scorer. Case P05 returns `14/15`, equivalent to the gold $\frac{14}{15}$; case P08 returns `East`, equivalent to the gold `\text{east}`. Both are stored as incorrect, whereas their baselines use the gold LaTeX representation and are stored as correct. The recorded method is `normalized_exact_numeric_fallback`; its normalized-string or floating-point comparison rejects these pairs. The classification concerns final-answer equivalence, not the validity of every intermediate step. The subsequent offline audit in Section F.1 corrects both examples and applies the same rule to all saved baseline and guided evaluations. Current accuracy and bootstrap tables use those revised decisions. This qualitative sample retains its original IDs and selection based on historical judgments; P05 and P08 are consequently no longer correctness losses under the revised rule. Resampling after correction would answer a different qualitative question.

Scope of interpretation.. The review supplies direct text examples of repetition, incorrect arithmetic, and scoring artefacts. It does not isolate why the routing intervention causes a particular trajectory, quantify population-level looping, or extend the interrupted Qwen result to its missing attempts. Full-population termination counts remain those in the generation-diagnostic tables. The sampled IDs, source hashes, excerpts, text diagnostics, and case-level notes are stored in `report/paper_failure_inspection.json` and `report/paper_failure_review.json`.

F.4 Expert-selection comparison with SteerMoE

The comparison separates expert detection from the intervention applied to those experts. SteerMoE uses paired contrastive inputs and counts top-k expert inclusions separately in the two populations. Its score is the token-pooled activation-rate difference

$$\Delta_{\ell e}^{\text{paper}} = \frac{A_{\ell e}^{+}}{N^{+}} - \frac{A_{\ell e}^{-}}{N^{-}}, \tag{F.1}$$

where A counts tokens selecting the layer–expert identity and N counts all tokens in the corresponding population. Detection therefore uses routing frequencies, not hidden-state similarity. Positive and negative differences identify opposite behavioral associations; steering changes the selected experts' log-softmax scores to the original maximum plus ϵ or minimum minus ϵ (**?**), Sections 3.1–3.2. The authors' released custom-steering notebook ranks layer–expert identities together by absolute risk difference[2]. Our policy instead assigns a separate budget to each of a preselected set of layers.

Implemented estimator.. For calibration attempt r, let $y_r \in \{0, 1\}$ be correctness, $f_{r\ell e}$ the fraction of saved routing tokens selecting expert e at layer ℓ, and $w_r = 1/m_{p(r)}$, where $m_{p(r)}$ is the number of usable attempts for its source problem. With $W = \sum_r w_r$, the implemented score is

$$\pi = \frac{\sum_r w_r y_r}{W}, \qquad L_{\ell e} = \frac{\sum_r w_r f_{r\ell e} y_r}{\sum_r w_r f_{r\ell e}} - \pi. \tag{F.2}$$

2 `https://github.com/adobe-research/SteerMoE/blob/main/custom_steering.ipynb`

Thus calibration gives every problem total weight one and first normalizes routing counts within each trace. It contrasts observed correct and incorrect completions; it does not construct paired counterfactual behavioral inputs. Experts need at least four supporting problems in the audited policies. Within each selected layer, eligible experts are ordered by signed lift, then support, then expert ID, and capped at native top-k (eight for Qwen and four for GPT-OSS). Fixed bias scales each retained lift by the largest retained magnitude; paper-style steering uses only membership in the selected set, not those relative magnitudes.

Why activation-based scores need not select the same experts.. Define $\mu_1 = E_w[f_{r\ell e} \mid y_r = 1]$, $\mu_0 = E_w[f_{r\ell e} \mid y_r = 0]$, and $\bar{f} = E_w[f_{r\ell e}]$. Algebra applied to our estimator gives

$$L_{\ell e} = \frac{\pi(1-\pi)}{\bar{f}}(\mu_1 - \mu_0), \qquad \bar{f} > 0. \tag{F.3}$$

Under these same weights, lift and activation-rate difference have the same sign but can have different rankings: lift additionally divides by the expert's mean activation frequency. This identity does not equate either score to the paper's token-pooled estimate on different, paired inputs.

The saved frequency masses, activation-weighted accuracies, and calibration baseline suffice to reconstruct $\mu_1 - \mu_0$ without replaying a model. Table F.7 changes only the ranking score and retains our weights, sign filter, support threshold, layers, and target budget. The audit also reconstructs every saved selected set and normalized score. Only 14 of 48 Qwen positive layer–expert identities remain selected; the four-target GPT-OSS policies retain 6 of 16 positive identities and 7 of 16 negative identities. This demonstrates selection sensitivity even before changing the data construction or token weighting. It does not measure intervention performance for these alternative sets.

Table F.7
Selection sensitivity using saved calibration sufficient statistics. RD here uses the same problem-balanced trace weights, support filter, polarity, layer set, and per-layer budget as lift; it is not a reproduction of the paper's token-pooled contrastive estimator. Overlap counts layer–expert identities.

Model	Polarity	Layers	Overlap	Overlap per layer
GPT-OSS	negative	4	7/16	15: 1/4, 19: 2/4, 21: 1/4, 23: 3/4
GPT-OSS	positive	4	6/16	15: 2/4, 19: 1/4, 21: 1/4, 23: 2/4
Qwen	positive	6	14/48	27: 4/8, 34: 3/8, 35: 2/8, 36: 1/8, 38: 3/8, 39: 1/8

Interpretation of the intervention comparison.. The completed Qwen fixed runs share selection statistics and target sets. The completed GPT-OSS fixed and paper-style conditions use four targets per layer, with the same selected identities within each polarity. Their results compare intervention rules using our selector; they do not evaluate the full SteerMoE detection and steering pipeline.

The completed OSS comparison already holds the selected identities fixed within each polarity while changing the intervention rule. A selector comparison should instead fix the split, generation contract, layers, budget, and intervention, changing only the ranking estimator. The present OSS budget is four targets per layer, matching native dispatch; smaller budgets of one or two targets would test less extensive promotion.

Matched random sets would help separate selector information from generic perturbation. A faithful detection replication additionally requires paired contrastive inputs and token-pooled activation counts. The present audit changes no policies and launches no generation runs.

Additional completed margin comparisons.. All three complete stochastic margin comparisons have intervals containing zero after consistent rescoring. Qwen at concurrency 16 gives $+0.07$ points ($[-0.86, +1.01]$); Qwen at concurrency 25 gains 13 net correct attempts, or $+0.93$ points ($[0.00, +1.86]$). GPT-OSS loses 12 net correct attempts, or -0.86 points ($[-2.87, +1.15]$). The Qwen concurrency-25 lower endpoint is zero, so its positive point estimate does not establish a direction under this interval criterion.

Table F.8
Complete stochastic margin comparisons at strength 1, with 1,395 attempts on 465 problems each. B/G denotes baseline/guided accuracy in percent; changes and intervals are percentage points. Concurrency is the maximum active-sequence count. Each row uses its own matched baseline; this is not a controlled concurrency comparison.

Model	Concurrency	B	G	Change	95% CI
Qwen	16	84.30	84.37	+0.07	[-0.86, +1.01]
Qwen	25	84.09	85.02	+0.93	[+0.00, +1.86]
GPT-OSS	25	77.85	76.99	-0.86	[-2.87, +1.15]

Table F.9
Generated-token means and token-limit hits for all 13 complete stochastic comparisons. Each row has 1,395 matched attempts; B/G denotes baseline/guided. Token statistics are descriptive.

Condition	Mean tokens B	Mean tokens G	Limit hits B	Limit hits G
Qwen fixed positive +1	13863	13706	140	146
Qwen fixed positive +2	13830	14092	154	169
Qwen fixed positive -1	13830	13919	154	138
GPT-OSS fixed positive +1	6497	6580	74	72
GPT-OSS fixed positive +2	6497	8629	74	164
GPT-OSS fixed positive -1	6497	6371	74	60
GPT-OSS fixed negative +1	6497	6900	74	87
GPT-OSS paper positive +1	6497	22489	74	670
GPT-OSS paper positive -1	6497	6698	74	106
GPT-OSS paper negative +1	6497	23639	74	815
Qwen margin (c=16)	13848	13840	144	143
Qwen margin (c=25)	13830	13810	154	146
GPT-OSS margin (c=25)	6497	6409	74	83

Benchmark contributions and evaluation weighting.. Table F.10 shows the benchmark-level point estimates. The primary statistic weights all 1,395 attempts equally. Table F.11 adds a secondary statistic that first averages correctness across each problem's attempts and then averages the 465 problems. A benchmark's weight remains proportional to its problem count; this is neither a benchmark macro-average nor pass@32. Both statistics use the same 5,000 paired, dataset-stratified problem resamples (seed 42), retaining all attempts when a problem is drawn. Full benchmark counts and intervals are saved in `report/guiding_sensitivity.json`.

Table F.10
Benchmark-level accuracy changes in percentage points for all completed stochastic interventions. Positive/negative denotes expert polarity; c denotes concurrency. Entries are point estimates, not significance decisions. Benchmark counts (problems/attempts) are shown in the header. Full benchmark intervals and correct counts are saved in guiding_sensitivity.json.

Condition	MATH-500 150/150	AIME24 9/288	AIME25 9/288	Olympiad 203/203	AMC23 12/384	Minerva 82/82
Qwen fixed positive +1	+0.00	-2.78	+0.00	-0.49	+0.52	+3.66
Qwen fixed positive +2	+0.00	-5.90	-1.74	-0.99	-3.91	+0.00
Qwen fixed positive -1	-1.33	+1.39	+2.08	+1.97	+0.00	+3.66
GPT-OSS fixed positive +1	+5.33	+1.39	-0.35	-2.46	-1.30	-1.22
GPT-OSS fixed positive +2	+2.00	-8.33	-7.64	-2.46	-0.26	+3.66
GPT-OSS fixed positive -1	+3.33	-2.08	-3.12	-2.96	+0.78	-1.22
GPT-OSS fixed negative +1	+4.67	-1.04	-1.39	+0.00	+1.04	+4.88
GPT-OSS paper positive +1	-47.33	-56.60	-76.39	-60.10	-68.75	-24.39
GPT-OSS paper positive -1	+2.67	-3.47	-5.21	-3.45	-0.78	+0.00
GPT-OSS paper negative +1	-47.33	-56.60	-74.31	-56.16	-63.80	-29.27
Qwen margin (c=16)	+0.67	+1.39	+1.39	-3.45	+0.26	-2.44
Qwen margin (c=25)	-0.67	+2.43	+2.43	+0.00	+0.00	+0.00
GPT-OSS margin (c=25)	+2.67	-1.04	-0.69	-3.45	-1.30	+1.22

Table F.11
Sensitivity to the evaluation unit: accuracy changes and 95% intervals in percentage points. Attempt weighting is the primary statistic; equal-problem weighting first averages correctness over each problem's attempts and then averages the 465 problems. Both use the same 5,000 paired, dataset-stratified problem resamples (seed 42). Equal problem weights are not equal benchmark weights. This is a secondary exploratory analysis without multiplicity adjustment.

Condition	Attempt change	95% CI	Problem change	95% CI
Qwen fixed positive +1	-0.29	[-1.15, +0.65]	+0.39	[-1.51, +2.31]
Qwen fixed positive +2	-2.80	[-5.16, -0.72]	-0.68	[-2.76, +1.33]
Qwen fixed positive -1	+1.08	[-0.14, +2.37]	+1.14	[-0.75, +2.94]
GPT-OSS fixed positive +1	+0.00	[-2.15, +2.01]	+0.42	[-2.37, +3.12]
GPT-OSS fixed positive +2	-3.30	[-6.16, -0.93]	-0.10	[-3.12, +3.08]
GPT-OSS fixed positive -1	-1.00	[-3.44, +1.22]	-0.51	[-3.41, +2.49]
GPT-OSS fixed negative +1	+0.57	[-1.58, +2.65]	+2.35	[-0.58, +5.44]
GPT-OSS paper positive +1	-61.65	[-69.32, -53.41]	-50.15	[-54.88, -45.45]
GPT-OSS paper positive -1	-2.22	[-4.16, -0.43]	-0.83	[-3.76, +2.23]
GPT-OSS paper negative +1	-59.57	[-67.24, -51.40]	-49.13	[-53.80, -44.56]
Qwen margin (c=16)	+0.07	[-0.86, +1.01]	-1.66	[-3.67, +0.45]
Qwen margin (c=25)	+0.93	[+0.00, +1.86]	-0.12	[-2.07, +1.80]
GPT-OSS margin (c=25)	-0.86	[-2.87, +1.15]	-0.50	[-3.49, +2.46]

Qwen margin guiding at concurrency 25 gains seven correct attempts on AIME24 and seven on AIME25, with zero net change on AMC23. MATH-500 loses one correct attempt; Minerva and OlympiadBench each have zero net change. The AIME improvements involve nine improved problems and one worsened problem. Equal-problem weighting gives -0.12 points with $[-2.07, +1.80]$, compared with the primary $+0.93$ and $[0.00, +1.86]$. Neither weighting establishes a beneficial direction.

For GPT-OSS fixed positive strength $+2$, the revised equal-problem estimate is -0.10 points ($[-3.12, +3.08]$), while fixed negative-polarity promotion gives $+2.35$ ($[-0.58, +5.44]$). The previous secondary positive intervals do not survive rescoring. For positive strength $+2$, MATH-500 and Minerva gain three correct attempts each, OlympiadBench loses five, and AIME24, AIME25, and AMC23 lose 24, 22, and one. Qwen fixed strength $+2$ gives -0.68 equal-problem points ($[-2.76, +1.33]$), and GPT-

Table F.12
Benchmark contributions to Qwen margin guiding at concurrency 25. B/G denotes baseline/guided correct counts; net is guided minus baseline correct attempts. Changes and pointwise 95% intervals are percentage points, with 5,000 paired problem resamples (seed 42).

Benchmark	Problems	Attempts	Correct B/G	Net	Change	95% CI
MATH-500	150	150	148/147	-1	-0.67	[-2.00, +0.00]
AIME24	9	288	178/185	+7	+2.43	[+0.00, +5.56]
AIME25	9	288	242/249	+7	+2.43	[+1.04, +4.17]
Olympiad	203	203	171/171	+0	+0.00	[-3.45, +3.45]
AMC23	12	384	384/384	+0	+0.00	[+0.00, +0.00]
Minerva	82	82	50/50	+0	+0.00	[-6.10, +6.10]

OSS paper-style suppression gives -0.83 ($[-3.76, +2.23]$); their below-zero primary intervals do not persist under this secondary weighting. GPT-OSS paper-style positive and negative promotion remain strongly negative: -50.15 ($[-54.88, -45.45]$) and -49.13 ($[-53.80, -44.56]$) points. No complete stochastic comparison has a positive interval excluding zero under either weighting.

These intervals do not account for configuration search, independent decoding-seed variation, or multiple comparisons.

What the margin hook actually changes.. Full diagnostics for the completed margin comparison record 20,657,039 hook token evaluations per selected layer. The top-eight identity set changes on 32.7–41.8% of these evaluations. Mean distance to the target interval falls from 2.19–4.71×10^{-4} to 0.285–0.902×10^{-4} across layers. Thus the numerical operation moves the intended gap, but it does not isolate a membership-preserving intervention after finite-precision log-probability conversion and casting. These diagnostics describe the hooked logits, not an independent recording of the kernel's dispatched expert IDs. Their denominator can include prefill, padding, and recomputation and is not a generated-token count. The identity -1 comparison uses minimal diagnostics, so detailed hook statistics are not available on a common basis across all conditions.

Separate generation protocols.. The separate greedy comparisons use 465 attempts on 465 problems. Qwen identity guiding gives -0.22 points ($[-2.58, +2.15]$), and margin guiding gives $+0.86$ ($[-1.29, +3.01]$). The 39-problem Minerva-only identity comparison changes from 24 to 25 correct answers, or $+2.56$ points ($[-5.13, +12.82]$). All three intervals contain zero. These distinct decoding protocols and populations remain separate from the full stochastic comparisons.

Appendix G: Synthesis and Limitations

The frozen Qwen3.8 judge retains the 40-example seed because the optimized candidate fails the per-class recall gate. Its NVFP4 deployment reaches 0.710 accuracy on 407 reused validation units under both low and medium effort, while changing 10.1% of labels between settings. Neither prompt selection nor this deployment check establishes annotation accuracy on the generated benchmark traces. Gemma and GPT-OSS now share a stratified 100,000-sentence sampling protocol and judge configuration, with 99,999 and 100,000 valid labels respectively. Their selected cohorts remain model-specific; Qwen uses a different sampling protocol and judge deployment.

All three models retain prospective episode information on held-out responses from the released SAT corpus. Five response-disjoint outer folds with inner-validation layer selection give macro-AUROC 0.852 ([0.838,0.868]) for Qwen, 0.780 ([0.761,0.797]) for GPT-OSS, and 0.788 ([0.772,0.805]) for Gemma. Each exceeds its two-feature prefix-position baseline. These 95% intervals resample responses conditional on the saved folds and fits. The study strengthens same-corpus generalization evidence but leaves external transfer, richer shortcut controls, and online detector validity unresolved. The original sentence-split scores and routing-layer choices retain their overlap and test-selection limitations. Qwen's external 581-unit inspection has 50.3% primary and 60.8% compatible agreement under a provisional hybrid review; it is neither independent human validation nor evidence for the other encoders.

The three correlation corpora each cover 4,647 attempts, 1,547 problems, and 100 complete avg@32 groups. Associations depend on model, benchmark, and aggregation level. For Qwen, whole-continuation confidence correlates with avg@32 at $\rho = 0.632$ on AIME24 and 0.705 on AIME25, whereas AMC23 gives -0.200 with an interval containing zero. These estimates do not control length, termination, or difficulty. Problem-respecting bootstrap intervals address repeated attempts for the reported primary features; naive attempt-level significance screens remain exploratory. Class coverage now spans all six benchmarks for Gemma and GPT-OSS. Sparse class/outcome cells and unmatched problem cohorts still limit comparisons; equal sentence budgets do not balance benchmark or class shares. No common causal interpretation of router confidence follows from these data.

The stratified class analysis also shows benchmark-dependent signs. For Gemma's Implement-labelled tokens, confidence correlates with correctness at $r = 0.498$ on AIME24 (30 attempts) and -0.367 on AMC23 (40 attempts). For GPT-OSS, the corresponding coefficients are 0.164 and -0.098. These are descriptive estimates on different model-specific cohorts, not paired tests of model differences. Gemma's AMC23 Plan, Explore, and Monitor cohorts contain only correct attempts, so their correctness correlations are undefined despite nonzero label coverage.

After consistent offline rescoring, no complete intervention establishes a beneficial accuracy direction using the reported 95% interval criterion. Qwen fixed positive strength $+2$ loses 2.80 attempt-weighted points ($[-5.16, -0.72]$); GPT-OSS fixed positive strength $+2$ loses 3.30 ($[-6.16, -0.93]$). GPT-OSS paper-style suppression loses 2.22 points ($[-4.16, -0.43]$), and positive/negative paper-style promotion loses 61.65 and 59.57 points with intervals well below zero. All stochastic margin intervals contain zero: Qwen concurrency 25 gives $+0.93$ points ($[0.00, +1.86]$), while GPT-OSS gives -0.86 ($[-2.87, +1.15]$). Equal-problem weighting removes the below-zero intervals for fixed strength $+2$ and paper-style suppression, but paper-style promotion remains strongly negative. The earlier positive OSS secondary intervals and greedy Qwen margin inter-

val do not survive rescoring. These exploratory intervals remain conditional on saved attempts and frozen policies and are not adjusted for multiple comparisons.

The Qwen paper-style positive intervention was interrupted after 494 of 1,395 attempts. On those same completed IDs, baseline accuracy is 70.45% and guided accuracy is 0.81%; all 494 guided outputs hit the 32,768-token limit. This subset covers all AIME24 attempts and part of AIME25 only, so it does not estimate the full evaluation effect (Section F.2). Each effect uses its own matched baseline, generated or reused after compatibility and output-hash checks. Calibration excludes unscored traces and selects targets from observational associations; Qwen also transfers across quantizations. Margin manipulation changes expert membership in finite precision. Strong paper steering retains our selection estimator and is not a full replication of the paper. The evaluated controllers do not use online episode predictions: annotations and probes support measurement and target exploration. The qualitative sample in Section F.3 contains direct repetition examples in capped and normally ended outputs, an arithmetic error, and two equivalent final answers rejected by the saved scorer. These descriptions are conditional on a small stratified sample; they establish neither population failure-mode frequencies nor a causal mechanism. The documented formatting discrepancies motivated the completed offline rescoring pass; the 22-case inspection retains its original sampling frame. These findings test the actionability of routing associations, without establishing a portable policy or isolating identity from mixture-weight changes. Intervals are conditional on saved runs, omit independent seed variation, and are not adjusted for multiple comparisons. Objective and strength selection require an independent evaluation design.

Model	Routing loss change	Simultaneous 95% interval
GPT-OSS	-0.0786	[-0.0957, -0.0618]
Nemotron	-0.1372	[-0.1563, -0.1182]
Qwen3.5	-0.1409	[-0.1536, -0.1278]
Qwen3.6	-0.1390	[-0.1528, -0.1260]
Gemma	-0.1487	[-0.1614, -0.1369]
GLM	-0.1368	[-0.1530, -0.1204]
Qwen3-30B	-0.0943	[-0.1093, -0.0810]

Table H.1
CLEAN routing readout over the retrospective judge-context baseline. All seven intervals exclude zero under the saved seven-comparison correction. These are label-readout results, not causal intervention effects.

Appendix H: Preserved Verification Evidence and Protocol

This appendix retains the verification study's original observation snapshot and frozen protocol. Pending-stage statements below describe that snapshot, not current scheduler ownership. The later seven-model campaign has not established a qualified joint accuracy–memory result; engineering completion and failed allocations are not scientific effects.

The paper separates information in native routing, the ability to change routing, the effect on reasoning operations, and the resulting utility. The saved CLEAN readout improves label-prediction cross-entropy over a retrospective judge-context baseline in all seven checkpoints (Table H.1). Its simultaneous intervals concern label readout; they do not establish that promoting a correlated expert causes the corresponding reasoning operation.

Historical next-class anticipation is already measured for GPT-OSS and Qwen3.6. Their routing loss changes are -0.02073 ($[-0.02548, -0.01601]$) and -0.05570 ($[-0.06944, -0.04154]$), respectively, under the saved past-information controls and family-grouped analysis. Equivalent prepared token-window caches are unavailable for the other five checkpoints. Their positive native readout results remain usable; an anticipation claim for all seven would need the missing cells. The historical FRESH verification contrast against native is $+5.5$ percentage points, with simultaneous 95% interval $[-4.5, +15.5]$. It establishes neither improvement nor a precise absence of control. The completed dense ledger contains 187,524 identities: 187,136 valid, 388 unusable and none unresolved.

The prospective method uses one discovery-nominated singleton expert per checkpoint. Family/source-trace matched checked-versus-unchecked profiles use only the last 256 already-emitted prediction rows. This nomination is a destination-state association heuristic; checked windows can contain the verification itself. It does not establish anticipation or exact position matching. The causal comparison determines actionability.

The action adds a positive bias to the expert's raw routing logit and calls the architecture's own selector. Native top-k, shared experts, grouped/sigmoid rules, normalization and learned scales are preserved. One pulse lasts at most 256 prediction rows and stops at operational native reasoning closure. Discovery-only same-state calibration targets an inclusion increment of 0.25 ± 0.025. Random controls exclude the nominee

Model	Backend	Recognizer	Dose/control
GPT-OSS	PASS	FAIL	Unmeasured
Nemotron	PASS	FAIL	Unmeasured
Qwen3.5	PASS	Unmeasured	Unmeasured
Qwen3.6	PASS	FAIL	Unmeasured
Gemma	PASS	FAIL	Unmeasured
GLM	PASS	Unmeasured	Unmeasured
Qwen3-30B	PASS	FAIL	Unmeasured

Table H.2
Prospective verification study status. A passing backend qualifies execution only. Unmeasured or failed qualification cells do not estimate a causal null.

and match native exposure, calibrated dose and ID-aligned relative weight displacement. No admissible pool is a qualification failure.

Candidate recognition includes intermediate/complete numeric or symbolic values checkable against an original problem constraint. It distinguishes a candidate-specific substantive check from intention or checking another value. Two draws from a fixed reader audit eligible and already-checked proposals. Qualification requires at least ten apparent discovery families and 70% apparent family precision. Disjoint held-out uncertainty, coverage and symbolic cases are separate; the reader is not human ground truth. Before the first recognizer or target was selected, we amended decoding to accept either a plain JSON object or one complete outer `json` code fence around the final response. The amendment applies to all seven models and preserves the original draws; truncated or ambiguous outputs remain unconfirmed.

The experiments have different populations. The conditional assay freezes the first qualified native seed 0 prefix on each of 96 independent families, without seed 1 fallback, before any treatment. Fresh native, targeted and matched-random continuations use two seeds and a 1,024-token cap. The event is verification before native closure and within that horizon. Closure competes with verification; all assigned branches remain in the analysis. The original-prompt experiment uses all 96 families, two seeds, three arms and a 16,384-token cap. It retains no-fire/failed assignments and reuses only exact compatible original native sources. Maximum final enrollment is 4,032 long runs plus 4,032 short continuations across seven checkpoints; discovery and validation costs are separate.

All seven checkpoints passed the 14 backend qualification checks under the current executor. One retained native discovery trace per checkpoint has also completed with validated routing captures. These traces belong to the frozen 48-family discovery manifests and are reused there. In the current execution snapshot, GPT-OSS, Nemotron, Gemma, Qwen3-30B and Qwen3.6 have completed all 48 discovery traces, each producing a 128-prefix audit. Qwen3.5 and GLM remain in discovery; Qwen3.6 has begun its retained largest-input reader measurement before the full semantic audit. The status table reports recognizer and calibrated-control qualification separately from causal and utility outcomes.

GPT-OSS has completed all 256 fixed eligibility-reader draws. Under the declared decoding amendment, 196 provide usable stopped judgments, 36 are truncated and 24 fail the original content validator; 84 of the 128 prefixes have two usable draws. Apparent family-weighted confirmation precision is 22.41% for eligibility across 29

families and 18.75% for already-checked proposals across eight families. Both fall below the 70% gate, and already-checked support also falls below ten families. GPT therefore stops before target nomination or treatment. This is a qualification limit of the current controller; GPT's causal effect and utility remain unmeasured. The other checkpoints retain their own gates.

Nemotron's complete 256-draw audit also fails recognition qualification. It provides 191 usable stopped judgments, 37 truncated draws and 28 content-invalid draws; 81 prefixes have two usable judgments. Apparent family-weighted confirmation is 25% for eligibility across 34 families and 10% for already-checked proposals across ten families. Both decisions meet the apparent support requirement but fail the 70% precision gate. Normally stopped agreeing readers also reject proposals, so the failure is not attributable solely to output truncation. Nemotron stops before nomination or treatment; its causal and utility outcomes remain unmeasured. Qwen3-30B's full audit likewise fails recognition qualification. Of its 256 draws, 193 provide usable stopped judgments, 39 are truncated and 24 fail content validation; 80 prefixes have two usable judgments. Apparent family-weighted confirmation is 10.10% for eligibility across 33 families and 33.33% for already-checked proposals across 12 families. Both support minima pass and both precision gates fail. Agreeing stopped readers reject 22 apparent eligibility proposals and five already-checked proposals. This model also stops before nomination or treatment, leaving its causal and utility outcomes unmeasured.

Gemma's complete 256-draw audit also fails recognition qualification. It provides 209 usable stopped judgments, 30 truncated draws and 17 content-invalid draws; 91 prefixes have two usable judgments. Apparent family-weighted confirmation is 25.52% for eligibility across 32 families and 41.67% for already-checked proposals across 12. Both apparent support minima pass and both precision gates fail. Agreeing stopped readers reject 14 apparent eligibility proposals and two already-checked proposals. Gemma stops before nomination or treatment, with causal and utility outcomes unmeasured. Qwen3.5, Qwen3.6 and GLM continue through their own gates.

Accuracy is graded offline, with gold absent from the generator and semantic reader. The token endpoint is generator plus online-controller output tokens. Input processing, runtime, annotation/grading, capture and research recovery are itemized separately. Unknown outcomes retain worst-case bounds. Seeds are averaged within family before shared family resampling across checkpoints; 42 contrasts receive simultaneous intervals. The shared equivalence grader matched 97 of 100 frozen agent-consensus labels and made no false equivalence judgments on clear-wrong controls. It failed qualification because all three disagreements occurred in the Minerva stratum, exceeding the prespecified limit of two. All nine draws for those cases stopped normally and produced valid JSON; the disagreements concerned contextual qualifications or interpretation of the reference. These consensus labels are not independent human ground truth. The failed grader is excluded from production adjudication. Strictly accepted answers and known capped or absent-final-answer failures retain their grades; uncapped strict rejects remain unknown and enter the accuracy bounds. The separate semantic reader and token endpoint retain their own contracts. No accuracy-loss margin has been chosen, so nonsignificant loss cannot establish retention. Backend PASS qualifies execution, not a semantic effect. A valid scoped limit, harm or tradeoff is a possible paper result.

New conditional verification and original-prompt utility outcomes remain unmeasured. The status table reports qualification separately.

Model	Sentences	Loss change	Simultaneous 95% CI
GPT-OSS	37,549	-0.07858	$[-0.09567, -0.06179]$
Nemotron	35,334	-0.13721	$[-0.15628, -0.11819]$
Qwen3.5	34,740	-0.14090	$[-0.15359, -0.12781]$
Qwen3.6	35,484	-0.13899	$[-0.15276, -0.12601]$
Gemma	31,902	-0.14866	$[-0.16136, -0.13692]$
GLM	38,931	-0.13679	$[-0.15302, -0.12035]$
Qwen3-30B	36,445	-0.09427	$[-0.10929, -0.08100]$

Table I.1
Routing-augmented minus retrospective judge-context cross-entropy in the saved CLEAN readout. Each row has 522 questions and 509 families. Intervals retain the model-specific saved seven-comparison procedure; this is not a fresh joint test across models. Sentences are measurements within questions.

Appendix I: Saved-Data Details and Exploratory Associations

The following analyses preserve their original populations, estimators and uncertainty procedures. The exact readout table accompanies the main result. Conditional candidate identities, same-cohort correctness associations and in-sample concentration are exploratory or descriptive evidence. They do not provide independent confirmation of expert usefulness. No outcome or significance filter was applied to the saved correlation figure.

I.1 Exact present-class readout estimates

I.2 Selection frequency, motion and candidate experts

For a window W, selection frequency is

$$f_{\ell,e}(W) = \frac{1}{|W|} \sum_{t \in W} \mathbf{1}\{e \in S_{\ell,t}\}. \tag{I.1}$$

It sums to k_ℓ, rather than one. Entropy or divergence of the mean selection distribution therefore uses $f_{\ell,e}/k_\ell$. If normalized executed weights are captured, their occupancy is a different statistic; IDs alone cannot recover it.

Temporal continuity can be measured by $|S_{\ell,t} \cap S_{\ell,t-d}|/k_\ell$ at declared lag d. A temporal claim requires comparison with an order-shuffled control that preserves the same allocation composition. Entropy of mean allocation minus mean per-token entropy is invariant to token ordering and cannot establish a state transition. Formulas, units, lag, support and required captured fields belong beside every reported metric.

Conditional expert association is estimated within comparable source conditions before averaging across problem families. A preserved Qwen3.6 analysis nominates layer 28 experts 189 and 9: matched selection-membership differences before a next-sentence Verify label are $+0.08118$ and $+0.06296$, positive in 40/48 and 44/48 discovery families. The analysis has 483 matched positive windows across 48 families, and fold-selected proposals retain positive held-out directions. This is conditional nomination under an LLM-assigned target; it does not make these experts semantic modules or independently validate the final fixed pair.

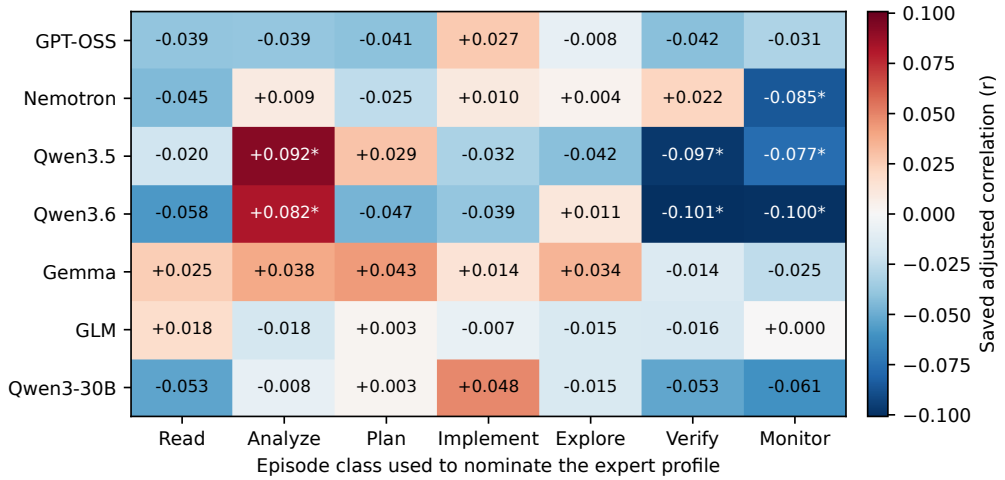


Figure I.1
Saved nuisance-adjusted correlations between whole-reasoning use of a class-nominated expert set and adjudicated correctness. All seven models and seven classes are shown. Asterisks preserve the original $q < 0.05$ threshold from Benjamini–Hochberg correction within each model's 21 mean-feature comparisons (seven classes by all/in/out usage); they do not represent a new joint test across the 49 displayed cells. Correlations use the original same-cohort nomination and adjustment rules.

A candidate's rank is also estimator-dependent. In the legacy selector audit, replacing accuracy-lift ranking with activation-rate differences retains only 14/48 Qwen targets, 6/16 GPT positive targets and 7/16 GPT negative targets under the same budgets. Analyses should report that sensitivity, support and held-out stability. A low-frequency expert is not thereby dispensable, and an expert correlated with an error is not thereby harmful: difficult questions can cause both expert use and errors.

I.3 Adjusted expert-profile associations with correctness

The saved A-corpus analysis also relates whole-reasoning use of class-enriched expert sets to adjudicated answer correctness. For each class, the archived rule nominates up to three enriched experts per selected layer from the full A corpus without using correctness labels. The feature is their selected-slot count over all reasoning tokens, divided by native k and reasoning-token count, then averaged equally across layers with a nominated set. It is not restricted to tokens labelled with the nominating class, and different class sets can overlap.

Figure I.1 presents all 49 model–class cells. Each is a saved partial Pearson correlation after the original adjustment for benchmark, log completion length, retrospective problem difficulty, the class's share of labelled tokens and a cap indicator when present. The difficulty estimate uses the other 31 same-model attempts for AIME/AMC and the other six models on single-attempt datasets. It is a retrospective analysis covariate, not an available deployment feature. Question support ranges from 1,545 to 1,547 per model.

The largest corrected associations occur for Qwen3.5 and Qwen3.6: Analyze-profile use correlates positively with correctness ($r = 0.0918$ and 0.0818), while Verify-profile

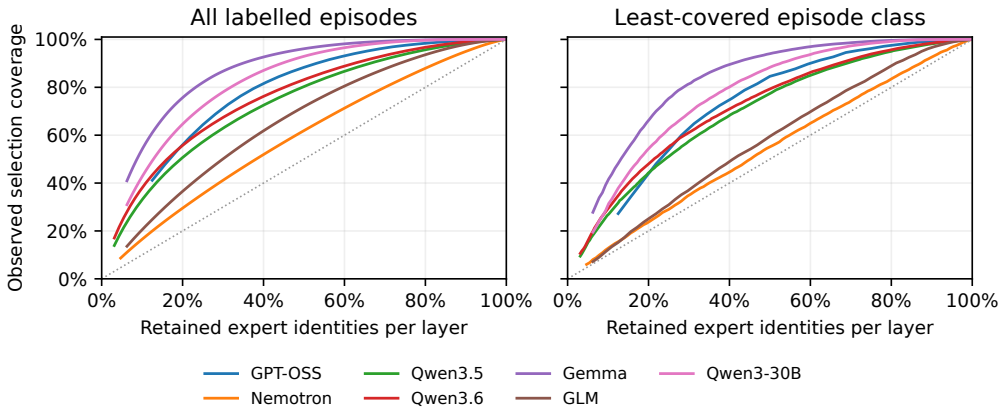


Figure I.2
In-sample coverage of original labelled expert selections when retaining the most frequently selected identities within each layer. Left: pooled labelled selections. Right: the least-covered episode class at each retained count. The dotted line is the retained identity fraction, not an evaluated random policy. The rank and coverage use the same data; no model is pruned or rerun. Native selection coverage measures neither accuracy nor memory saved.

use correlates negatively ($r = -0.0972$ and -0.1008), as does Monitor-profile use ($r = -0.0770$ and -0.0995). Nemotron's Monitor-profile association is also below zero ($r = -0.0853$) at its original corrected threshold. The remaining cells, including all cells for GPT-OSS, Gemma, GLM and Qwen3-30B, do not pass that threshold. Differences in threshold crossings are not tests of model differences.

The preserved confidence intervals use 1,000 question-ID bootstrap draws, refitting nuisance regressions while holding the nominated sets and difficulty estimates fixed. They do not cluster duplicate problem families. Those intervals, support and all original q values remain in the accompanying source CSV. This is an exploratory association screen on the nomination cohort. It supplies neither held-out expert usefulness nor the effect of removing a Verify- or Monitor-associated expert.

I.4 Labelled-state coverage under frequency-based retention

The saved seven-model A summaries allow a descriptive retention analysis without another model run. Let $n_{\ell,e,c}$ count original selected expert slots associated with labelled state c. For each layer, rank experts by $\sum_c n_{\ell,e,c}$, breaking ties by expert ID, and retain the first b identities as $K_\ell(b)$. The state-specific coverage is

$$R_c(b) = \frac{\sum_\ell \sum_{e \in K_\ell(b)} n_{\ell,e,c}}{\sum_\ell \sum_e n_{\ell,e,c}}. \tag{I.2}$$

The pooled curve uses all seven classes in numerator and denominator; the least-covered curve takes $\min_c R_c(b)$. Counts are weighted by original token-layer selections, without equal-family or equal-benchmark reweighting. The source A cohort contains 1,547 questions per model, but labels cover only a subset of their reasoning tokens.

Concentration differs substantially across these captured corpora (Figure I.2). At the common plotting landmark of half the expert identities per layer, pooled coverage

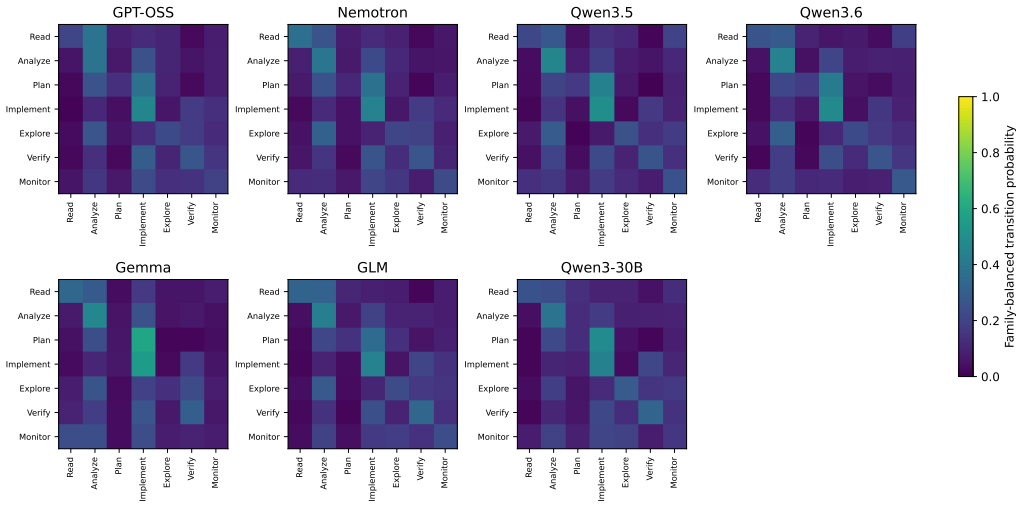


Figure I.3
Family-balanced next-class probabilities for consecutive observed sentences, using the common 509-family population. Rows are current classes and columns are next classes. Missing labels and sentence gaps break adjacency; unsupported rows are missing.

is 88.34% for GPT-OSS, 61.90% for Nemotron, 80.32% for Qwen3.5, 83.20% for Qwen3.6, 96.11% for Gemma, 71.92% for GLM and 92.94% for Qwen3-30B. The least-covered class retains respectively 84.58%, 55.13%, 77.39%, 79.30%, 94.04%, 59.52% and 88.35%. Monitor is least covered in six models; Read is least covered for Gemma. A pooled-frequency summary can therefore conceal a larger coverage loss for a particular labelled state.

Half retention is a plotting landmark, not a selected deployment budget. These deterministic summaries have no held-out stability estimate or inferential interval. Their model differences also reflect generated text and annotation coverage. They measure selected slots on native labelled traces, not complete top-k containment, executed-weight mass or coverage of unlabelled tokens. The keep-sets have not been checked for grouped-router feasibility. Changing a route can alter every subsequent token, so these curves nominate analysis questions rather than predict post-pruning accuracy or resident memory.

I.5 Observed adjacent-state transitions

The consecutive-sentence analysis also now covers all seven models on the common 522-question/509-family population (Figure I.3). An edge requires consecutive original sentence indices in the same reasoning segment; gaps and missing labels break adjacency. Labelled-sentence coverage ranges from 7.0% for Qwen3.6 to 28.1% for GPT-OSS. The matrices therefore describe observed labelled adjacencies rather than complete reasoning trajectories. Cell/source family support, coverage denominators and endpoint-censored dwell durations accompany the saved transition receipt.

I.6 Reproducible main-result figures

The main-result figures are regenerated from three compact saved CSVs in `generated/best_results_20261009/`. The bundle includes all seven readout rows, all 105 model/fold/arm selection cells and the complete saved anticipation export, including endpoints not promoted in the main text. A standalone plotting script and SHA-256 manifests bind the inputs and outputs. Confidence limits are copied unchanged; fold differences are computed by pairing the original common folds, without new resampling or scientific refitting. Historical uncertain and harmful intervention outcomes remain in their respective appendices.

Appendix J: Reproducibility Checklist

Current artifact inventory.. The judge includes grouped prompt-selection studies, final-fit validation, and a frozen-prompt NVFP4 effort comparison. Boundary probes are complete for Qwen, GPT-OSS, and Gemma; external probe inspection covers Qwen only. Each correlation model has a complete six-benchmark corpus, with differing class-label coverage. Guiding tables use complete Qwen fixed identity $+1, +2, -1$, both Qwen margin-1 pairs (concurrency 16 and 25), and GPT-OSS margin-1, plus GPT-OSS fixed positive-polarity $+1, +2, -1$, fixed negative-polarity $+1$, and paper positive-polarity $+1, -1$ and negative-polarity $+1$ pairs. The interrupted Qwen paper-style condition is reported separately on its 494 completed IDs and excluded from complete-run summaries. Independent annotation assessment and independent intervention-strength selection remain outstanding. Response-grouped probe evaluation is complete for all three encoders.

Grouped probe artifacts and regeneration.. The original `probes` directories remain the source of the routing layer choices. Each encoder additionally has a `probes_grouped_s42` directory under `results/probeTest`, with a run configuration, shared split plan, unit index, fold/target validation metrics and classifiers, outer-test predictions, final summaries, and completion status. The shared plan is `results/probeTest/grouped_s42/split_plan.json`. Source and code hashes and package versions identify the fitting inputs; incompatible checkpoints are rejected. The README in `src/moe_exp/probeTest` gives the CPU-only commands for all three encoders.

Run `python3 report/generate_grouped_probe_results.py` to rebuild the grouped tables and figure. The generator checks shared folds, one outer-test prediction per unit and target, validation-based selection, and mean-fold summaries. It records input and output fingerprints plus convergence and bootstrap-support diagnostics in `report/grouped_probe_sources.json`. All runs use four CPU workers with one BLAS thread each and expose no GPU to the fitting container.

Regenerating probe, effort, and guiding tables.. Run `python3 report/generate_current_results.py` from the repository root. It reads saved probe summaries and the judge-effort CSV, checks that all reported guiding manifests are complete and share comparison contracts, and verifies unique matching IDs, prompts, prompt tokens, per-attempt sampling settings, and Boolean outcomes. It then recomputes pooled and benchmark-specific correctness, flips, token counts, and truncation counts. `report/current_results_sources.json` records input SHA-256 hashes and the derived guiding statistics. The generator does not run inference or rejudge answers itself. It reads the uniform corrected decisions and bootstrap comparisons from `report/guiding_rescoring.json`, checking that raw source hashes and historical outcomes match the preserved inputs. The primary intervals use 5,000 paired source-problem resamples with seed 42. Margin and weighting tables read `report/guiding_sensitivity.json`, which verifies the rescoring sidecar hash and reproduces its primary intervals before computing the equal-problem statistic. The complete rescoring and regeneration command sequence is given below.

The following older command regenerates comparisons under the original saved judgments, for historical verification only. It does not regenerate the current rescored tables:

```
PYTHONPATH=src OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python3 -m moe_exp.moe_identity_guiding.bootstrap_saved \
  --bootstrap-workers 8 --bootstrap-replicates 5000 --bootstrap-seed 42
```

Each complete pair receives a separate `comparison_bootstrap.json`; incomplete pairs are skipped. Intervals resample source problems within benchmarks, preserve paired attempts, and use percentile endpoints. Random streams are indexed by replicate, so worker count does not change the estimates. Single-problem strata do not receive uncertainty intervals.

J.1 Completed run and independent checks

The numerical source root is `results/correlation_pipeline/reasoning-vllm-v1`. Under it, `analysis/unsloth--Qwen3.5-35B-A3B/correlations.json` records completed analysis of 4,647 attempts. Source hashes are saved in `report/correlation_tables_`

`sources.json`; the audit separately records the analysis SHA-256. Companion CSVs contain trace, problem, and expert features. The `views-v1` directory contains full reasoning, seven classes, ten position bins, overflow, and coverage summaries. Saved analysis uses 500 bootstrap replicates, seed 42, minimum expert support of four traces, and at most 128 ordered expert pairs.

The position-reference corpus has 4,647 traces and mean reasoning length 12,498.062 tokens. Its SHA-256 is `99b71a50bf12f39ecec05f9034b3b07ff57236d2e53a7343f01caa791e108c1e`. View population version 2 pools all reasoning tokens for full and position views, and available tagged sentence tokens for class views. All attempts, including limit hits, are retained with pairwise omission of missing features.

Independent audit.. `report/audit_tagged_correlation_run.py` matches all generation records to replay fingerprints, verifies available annotation fingerprints and sentence offsets, checks selected layers and compact values, and validates inclusion, correctness propagation, token-limit flags, and view coverage. It independently recalculates saved binary and trace-level metric-pair coefficients across 20 scopes from CSV observations (52,199 coefficients). It additionally checks 984 problem-mean Spearman and within-problem difference estimates for whole continuation and full reasoning. It also checks reported expert maxima and their continuous-metric associations. Counts and numerical discrepancies are recorded in `report/correlation_run_audit.json`.

These checks do not independently rejudge mathematical answers or recompute model activations, bootstrap intervals, or BH adjustments.

Regenerating the report.. From the repository root, using Python with NumPy, SciPy, and Matplotlib:

```
python3 report/audit_tagged_correlation_run.py \
  > /tmp/correlation_run_audit.json
cp /tmp/correlation_run_audit.json report/correlation_run_audit.json
MPLCONFIGDIR=/tmp/moe-report-matplotlib \
  python3 report/generate_correlation_tables.py
cd report
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The renderer writes coverage, correctness, class, lexical-event, expert, and repeated-attempt tables, a compact plot atlas, and standalone PDF/PNG figures. The atlas replaces the large numeric appendix in the compiled report. The renderer reuses saved feature correlations and additionally computes descriptive sampled-label counts, adjacent-sentence transitions, class-share associations, and binned outcomes from fingerprinted inputs. These new summaries and bin counts/ranges are saved in `report/descriptive_plot_statistics.json`; router-value intervals use 500 source-problem bootstrap resamples with seed 42. No model is fitted. Original experiment outputs are unchanged.

Scope of reproducibility.. Mutable local Docker tags and checkpoint names do not fix all underlying binaries. Exact reproduction also requires resolved model revisions, image digests, package versions, and the matching code revision. Saved contracts and source fingerprints identify this report's numerical population.

Three-model report update.. Run `python3 report/generate_multimodel_correlations.py` from the repository root, then rebuild with the latexmk command above. This reads the Qwen run above and the corresponding `reasoning-vllm-v1` runs under `results/correlation_pipeline/gemma-nvfp4-nf4` and `results/correlation_pipeline/gpt-oss-20b`. The generator verifies coverage and recomputes reported aggregate coefficients from saved CSVs, then writes comparison tables, figures, and both additional model atlases. Source hashes and check results are recorded separately in `report/multimodel_correlation_sources.json` and `report/multimodel_correlation_audit.json`.

Stratified annotation refresh.. Gemma and GPT-OSS use the September 21 stratified export, imported on September 22. The active annotation directories preserve the source, sampling, and

import manifests and the failed-response record. The report checks the annotation hashes against every available replay checkpoint, including unlabelled traces, and compares class features in the analysis CSVs with the checkpoint values. It verifies 100,000 valid GPT-OSS labels and 99,999 valid Gemma labels, with one Gemma sentence missing and its trace marked partial. These checks and source fingerprints are included in the three-model audit and source manifest. `report/stratified_label_results.py`, invoked by the three-model generator, renders annotation counts, sentence composition, and class-conditioned confidence plots. The imported sampling manifests describe these labels; the older local error-weighted sampling directories do not.

Outcome-conditioned class transitions.. Run `python3 report/generate_class_transitions.py` before rebuilding the report. This reads the three models' active sentence annotations and saved attempt correctness, checks unique sentence indices and one labelled attempt per source problem, and counts only originally adjacent labelled pairs. Raw counts, mean counts per labelled attempt (including zero-pair attempts), and denominators are saved both pooled and by benchmark in `report/class_transition_statistics.json`; input SHA-256 hashes are recorded in `report/class_transition_sources.json`. No inference is required.

Expert-selection audit.. Run `python3 report/audit_expert_selection.py` to reconstruct saved selection sets and compare lift ranking with an activation-rate difference under the same problem-balanced weights. This uses completed full-sampling guided manifests and writes `report/expert_selection_audit.json`, including source hashes, per-expert statistics, selected IDs, and overlap counts, plus the report table. It does not recompute token-pooled paper scores, change policies, or run inference.

Interrupted Qwen intervention.. Run `python3 report/analyze_partial_guiding.py` to regenerate the separate partial-results subsection. It checks the interrupted manifest, unique completed IDs, matched baseline records, generation contracts, and hook diagnostics. Coverage, paired accuracy, output length, limit hits, and source hashes are saved in `report/partial_guiding_statistics.json`. It leaves the experiment interrupted and resumable, excludes missing attempts, and does not insert partial results into the complete-run tables.

Regenerating rescored intervention results.. Install the pinned packages from `report/rescoring_requirements.txt` in an isolated environment. From the repository root run:

```
PYTHONPATH=src python3 report/rescore_guiding.py
python3 report/generate_guiding_sensitivity.py
PYTHONPATH=src python3 report/generate_current_results.py
python3 report/analyze_partial_guiding.py
python3 report/audit_expert_selection.py
```

The scoring pass validates paired generation contracts and preserves raw outputs. Its sidecar records original and revised judgments, extracted answers, methods, source hashes, package versions, and new bootstrap comparisons. The weighting and report generators verify raw hashes and read revised decisions; the historical tables are archived in `report/guiding_legacy_scores.json`. The selection audit intentionally retains the frozen policies and their original calibration labels.

Qualitative failure inspection.. Run `python3 report/inspect_paper_failures.py` to reproduce the 22-case sample and text diagnostics from saved generations. The seed, termination strata, sampling unit, paired IDs, and source hashes are saved in `report/paper_failure_inspection.json`. Case-level assistant review notes are in `report/paper_failure_review.json`; they are interpretive annotations, not outputs of a fitted classifier or independent human labels. The qualitative sample uses historical judgments to preserve its original selection frame; current numerical tables use the offline rescoring sidecar.

Appendix K: Aggregate correlation plot atlas

Each page fixes a token scope. Lower triangles show metric–metric Spearman ρ; the last column (Acc) shows point-biserial r with whole-attempt correctness. A dash means undefined or unavailable, not zero. Panels are descriptive; no significance threshold is implied. Sample sizes and saved intervals remain in the fingerprinted source JSON. Legacy lexical targets are omitted. Detailed numeric tables remain in `correlation_tables.tex` as a separate generated artifact.

Table K.1: Common metric notation; $k = 8$ for this model.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap
Ch	Character count
St	Step count

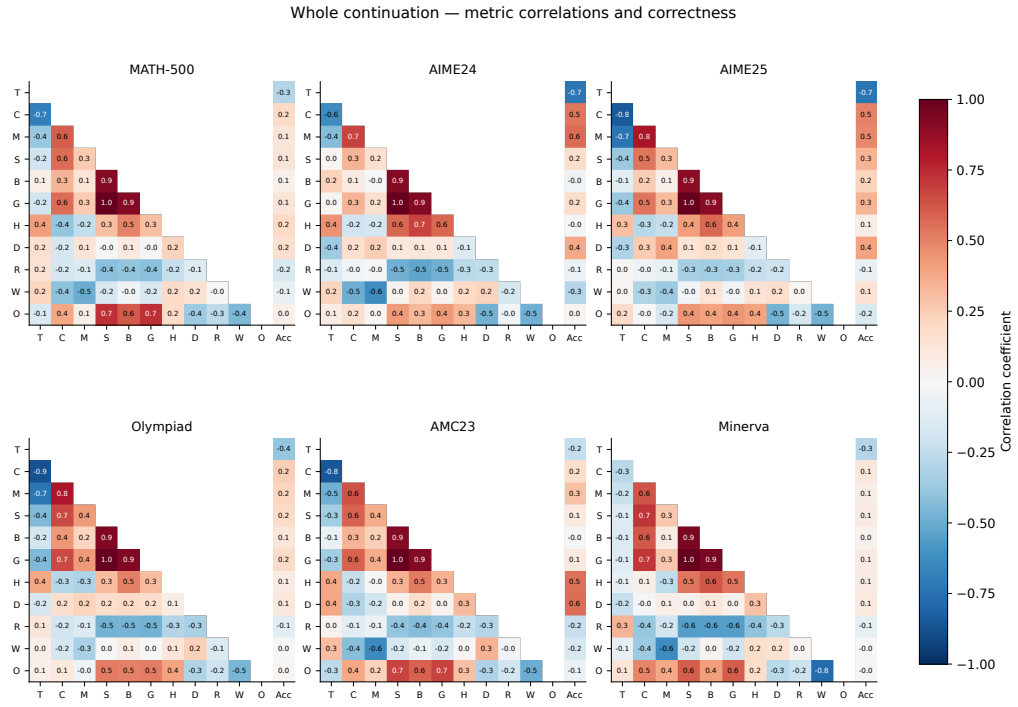


Figure K.1
Whole continuation: metric relationships and correctness. Codes are defined in Table K.1.

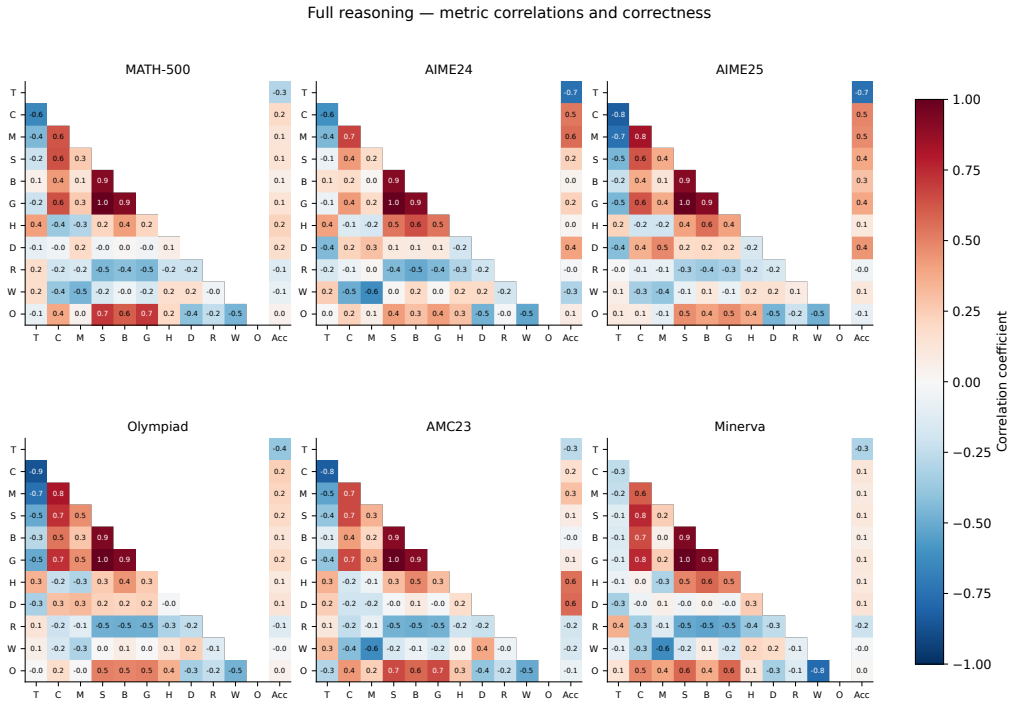


Figure K.2
Full reasoning: metric relationships and correctness. Codes are defined in Table K.1.

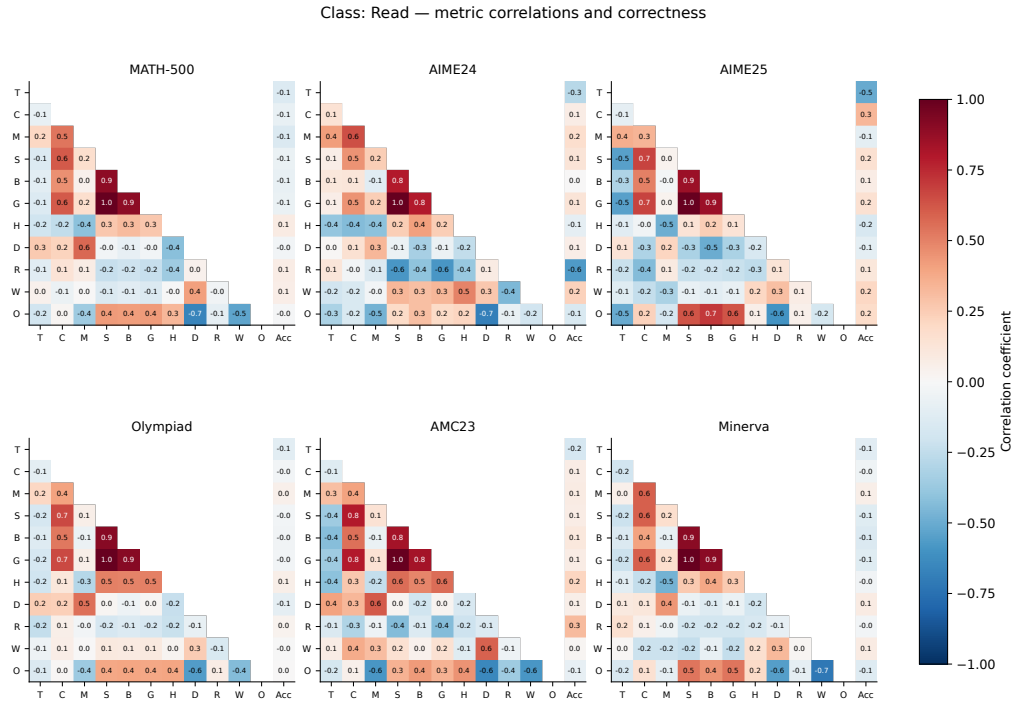


Figure K.3
Class: Read: metric relationships and correctness. Codes are defined in Table K.1.

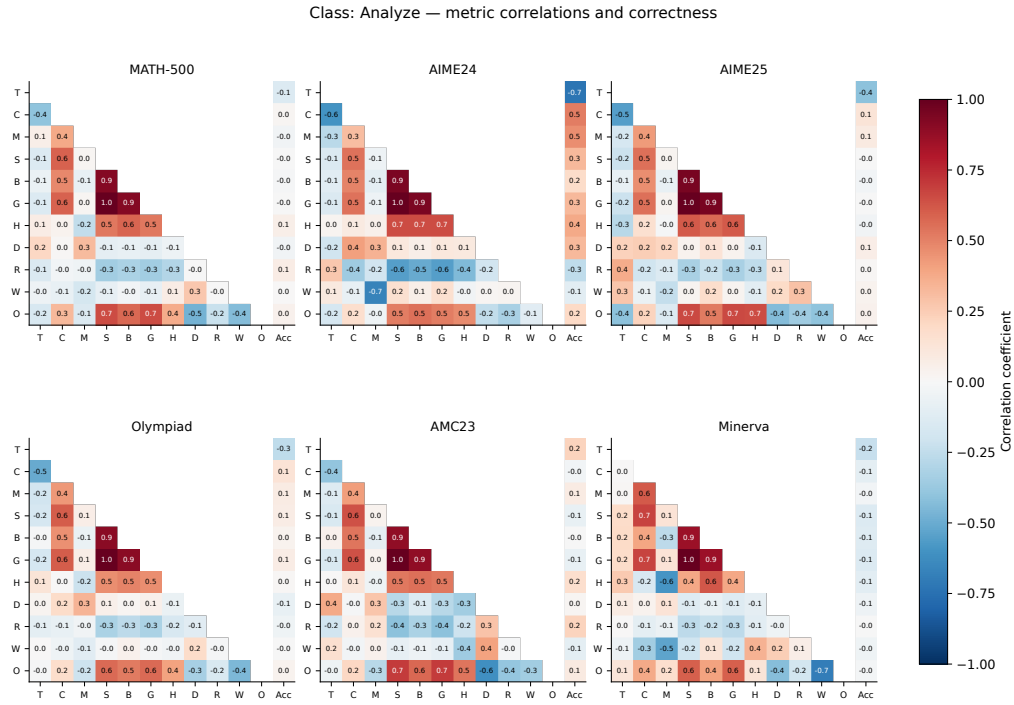


Figure K.4
Class: Analyze: metric relationships and correctness. Codes are defined in Table K.1.

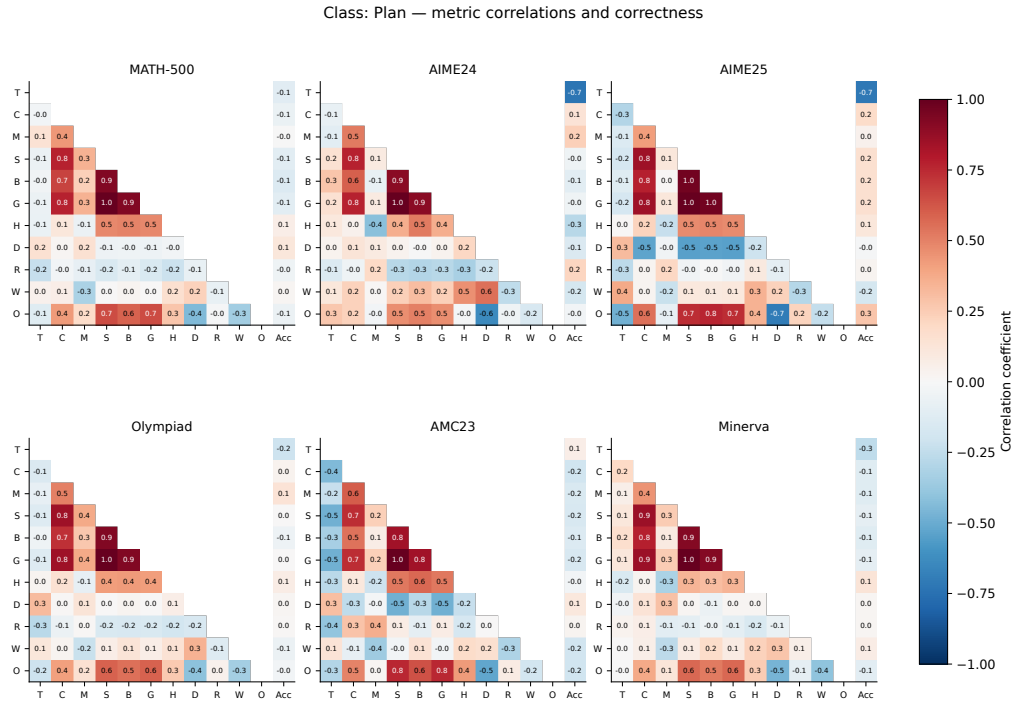


Figure K.5
Class: Plan: metric relationships and correctness. Codes are defined in Table K.1.

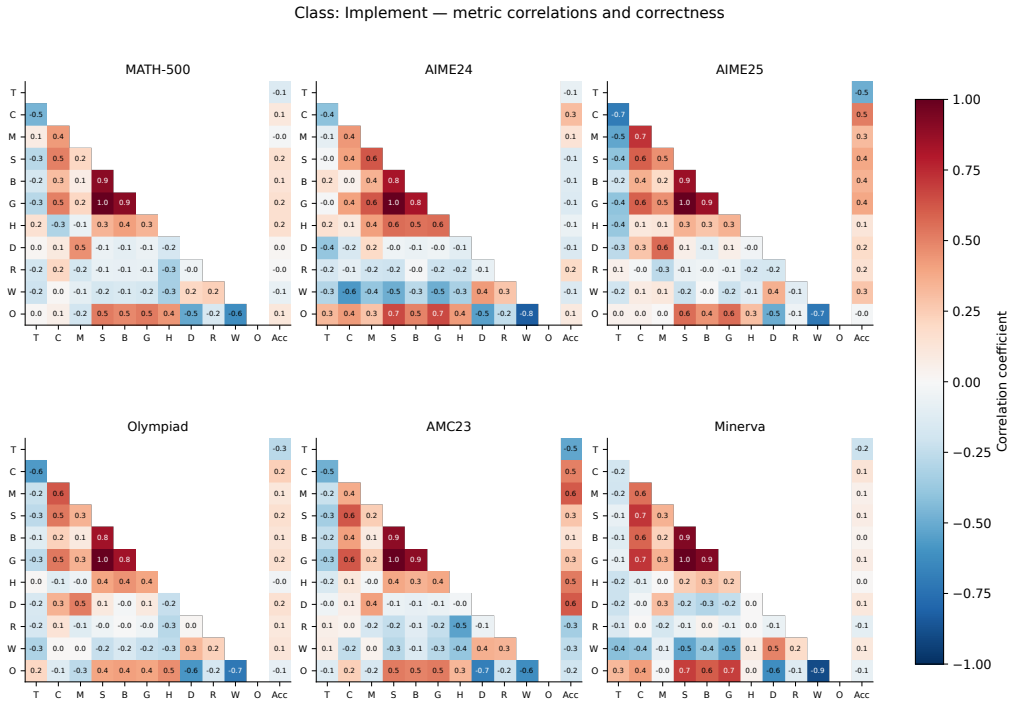


Figure K.6
Class: Implement: metric relationships and correctness. Codes are defined in Table K.1.

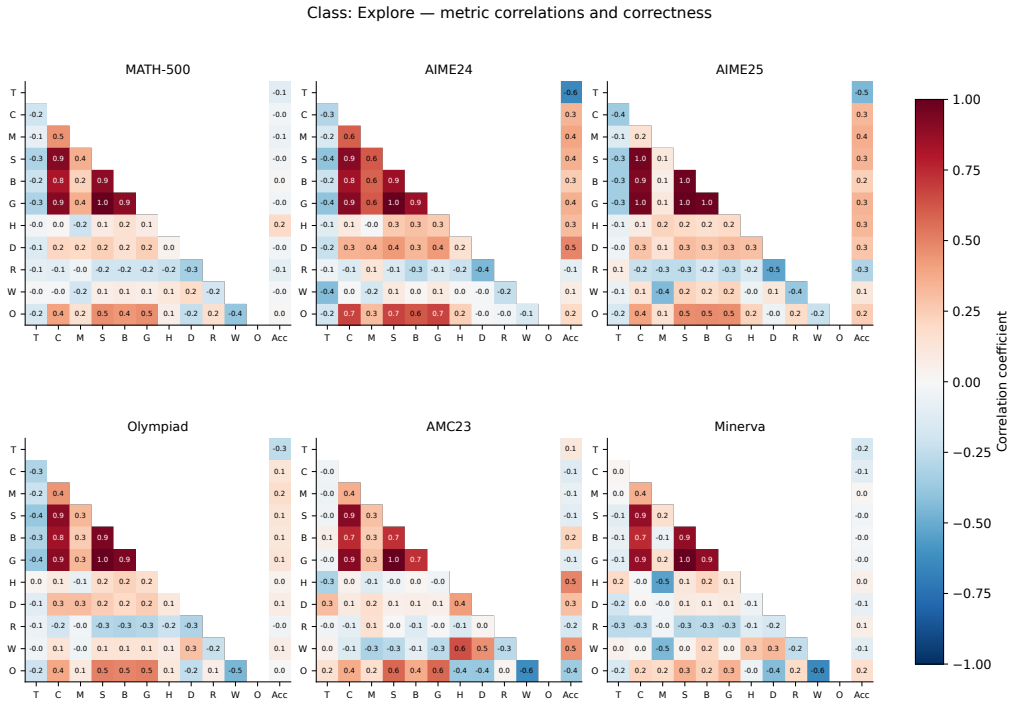


Figure K.7
Class: Explore: metric relationships and correctness. Codes are defined in Table K.1.

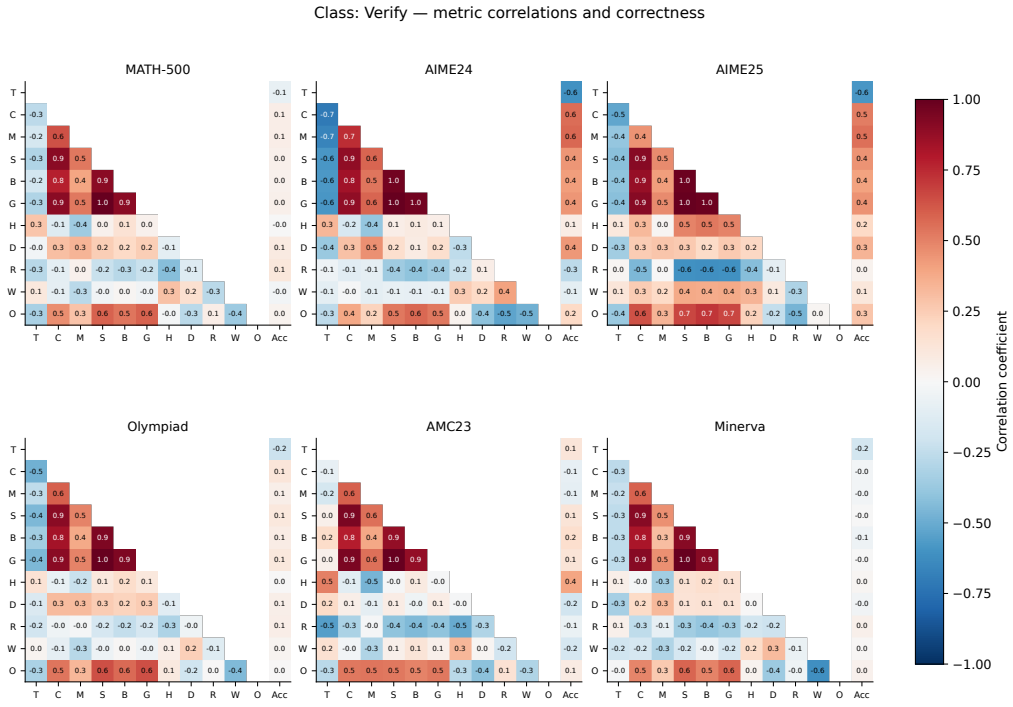


Figure K.8
Class: Verify: metric relationships and correctness. Codes are defined in Table K.1.

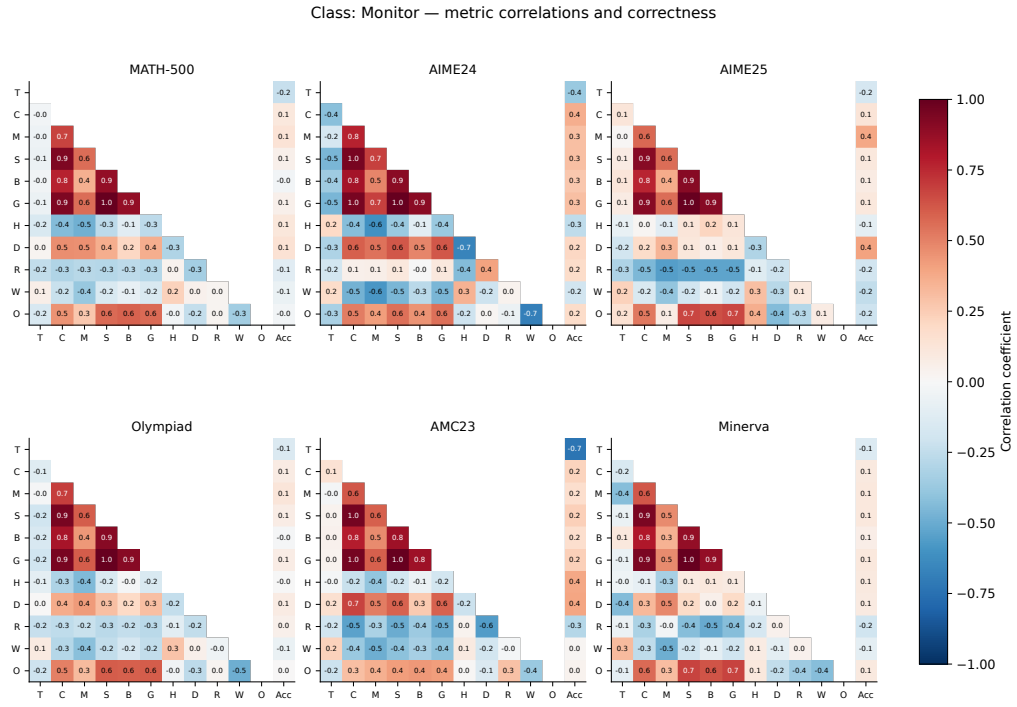


Figure K.9
Class: Monitor: metric relationships and correctness. Codes are defined in Table K.1.

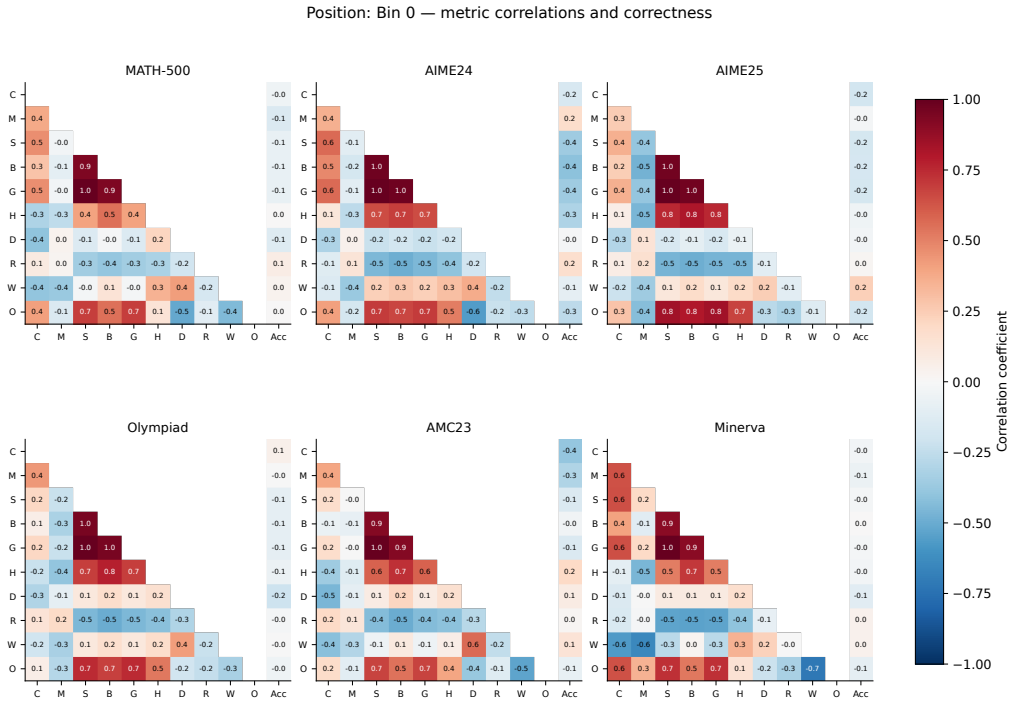


Figure K.10
Position: Bin 0: metric relationships and correctness. Codes are defined in Table K.1.

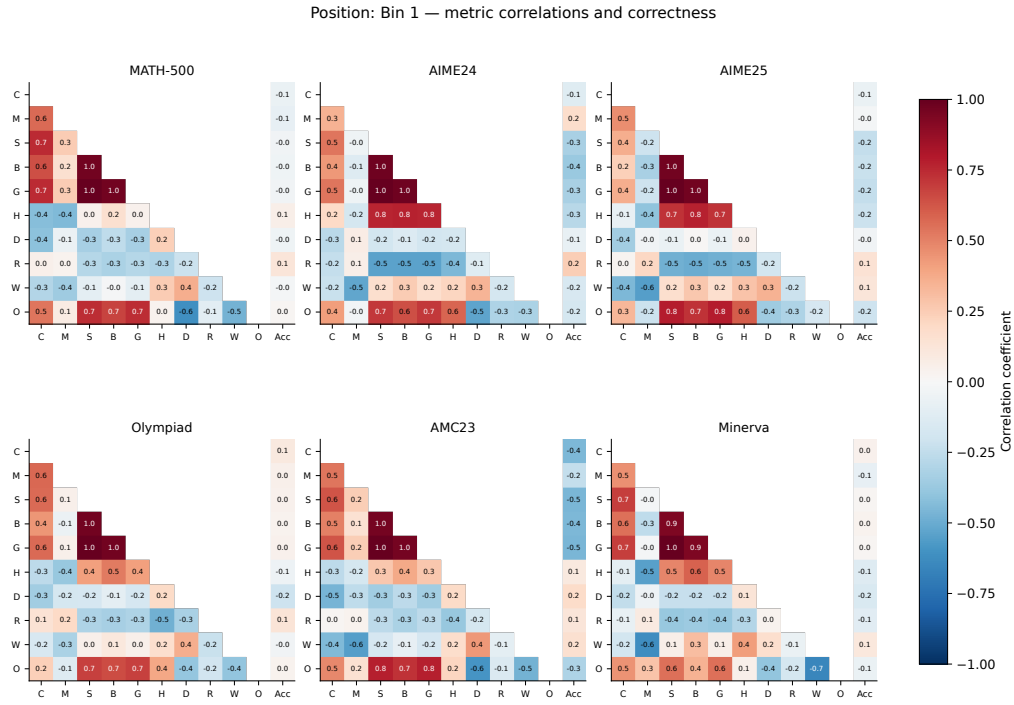


Figure K.11
Position: Bin 1: metric relationships and correctness. Codes are defined in Table K.1.

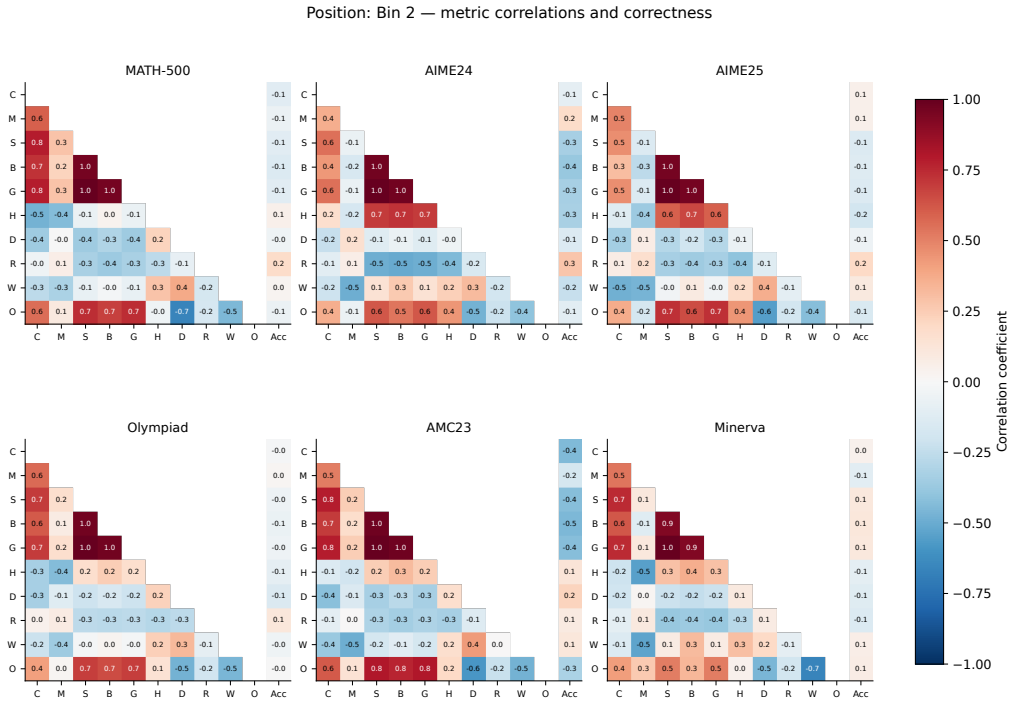


Figure K.12
Position: Bin 2: metric relationships and correctness. Codes are defined in Table K.1.

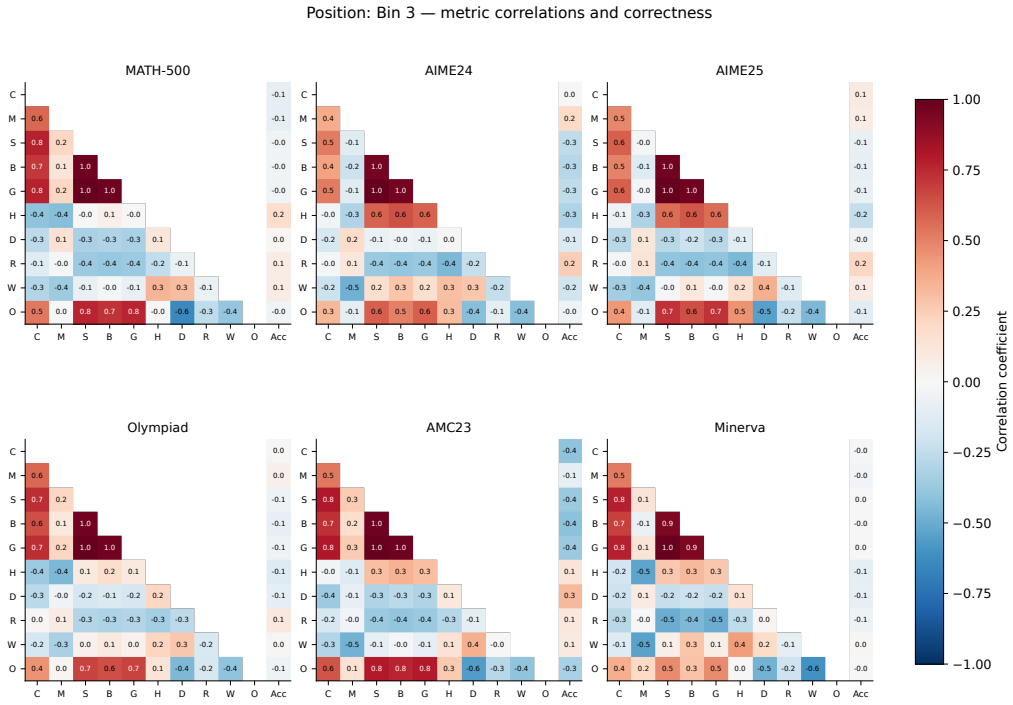


Figure K.13
Position: Bin 3: metric relationships and correctness. Codes are defined in Table K.1.

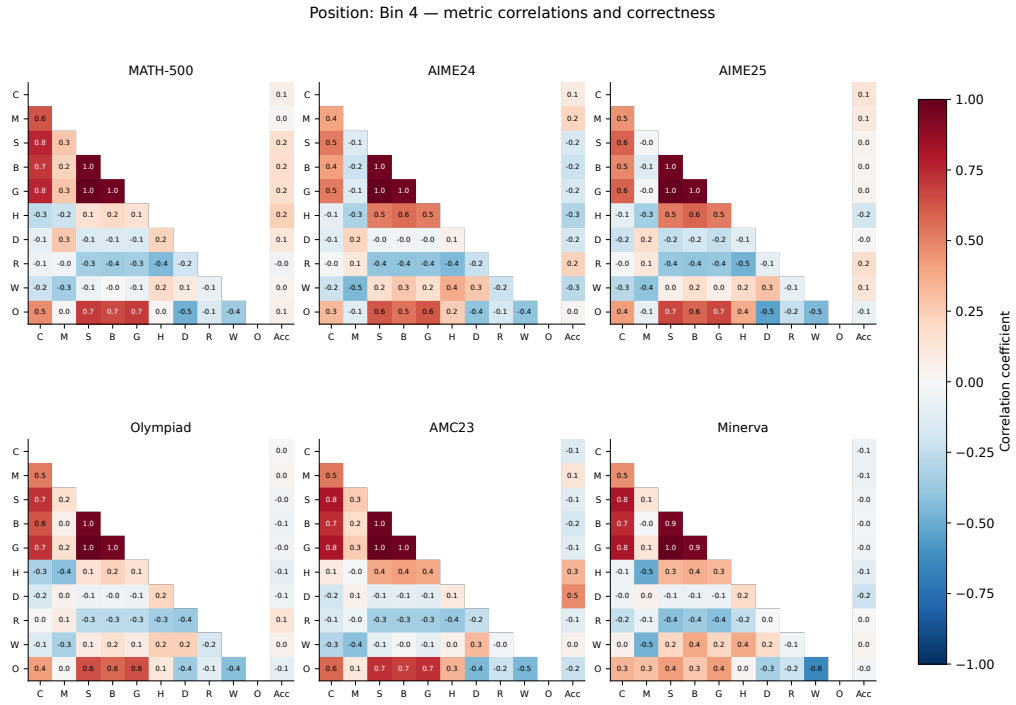


Figure K.14
Position: Bin 4: metric relationships and correctness. Codes are defined in Table K.1.

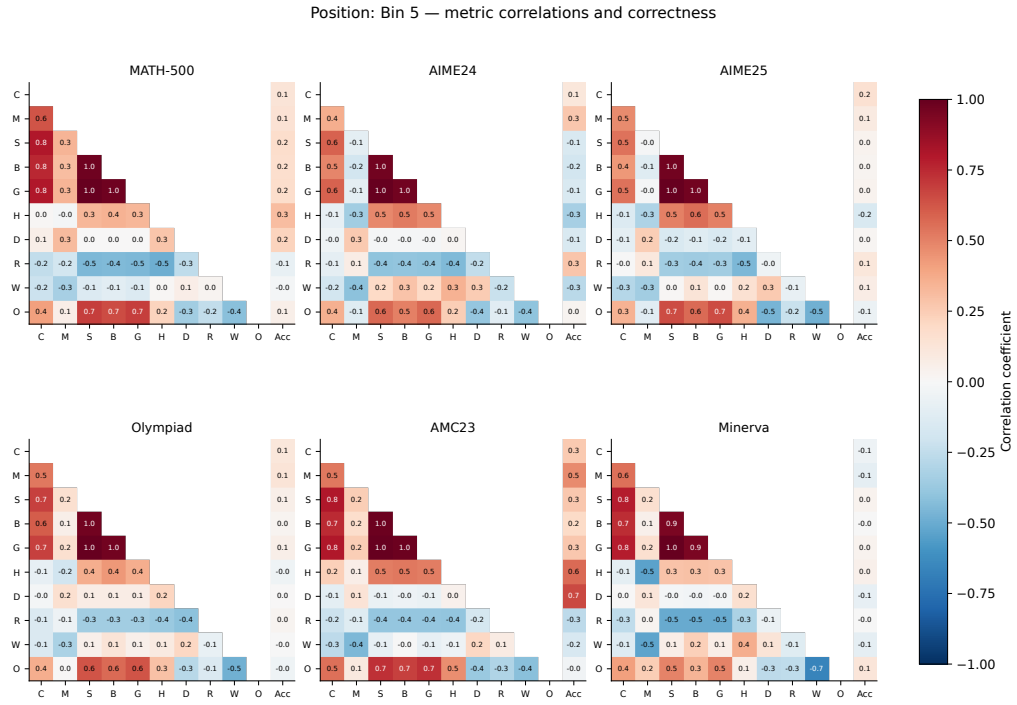


Figure K.15
Position: Bin 5: metric relationships and correctness. Codes are defined in Table K.1.

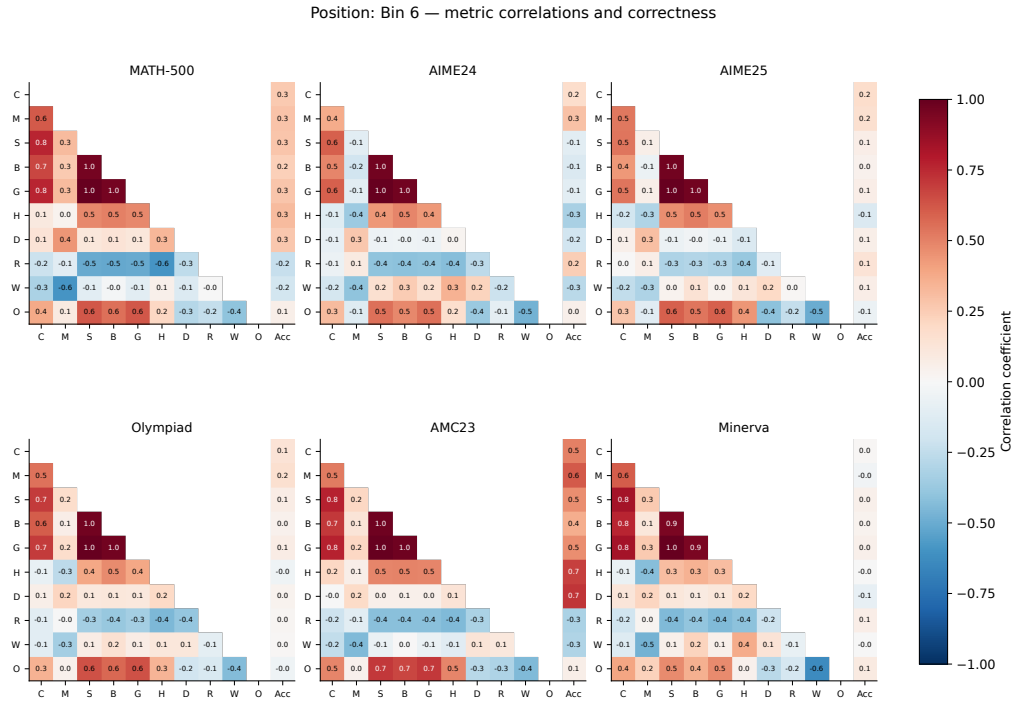


Figure K.16
Position: Bin 6: metric relationships and correctness. Codes are defined in Table K.1.

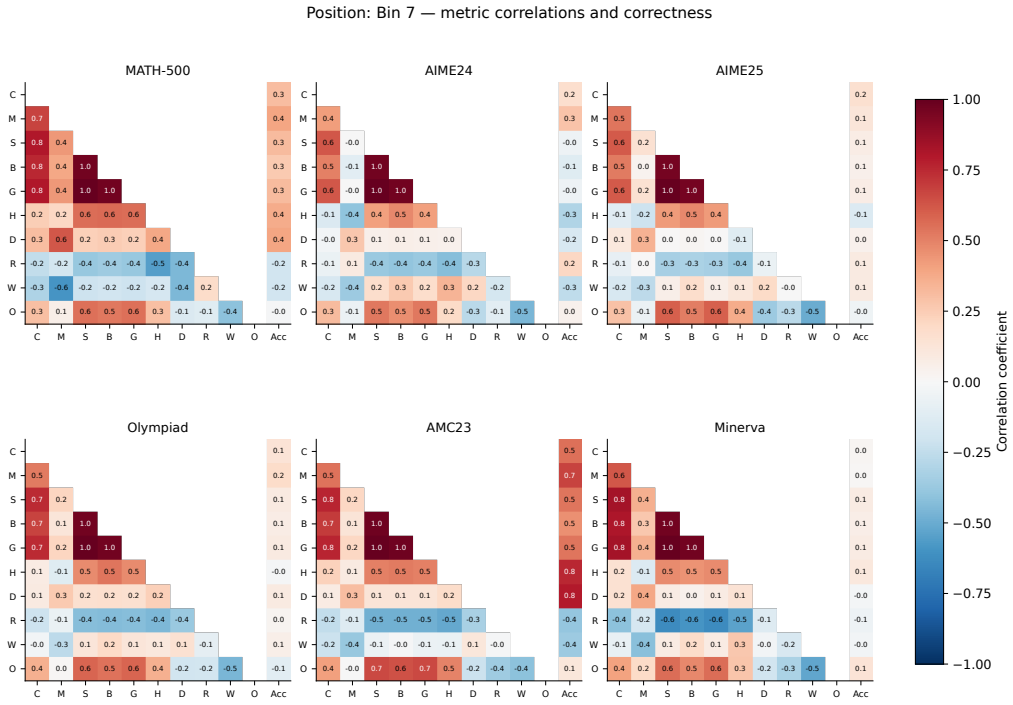


Figure K.17
Position: Bin 7: metric relationships and correctness. Codes are defined in Table K.1.

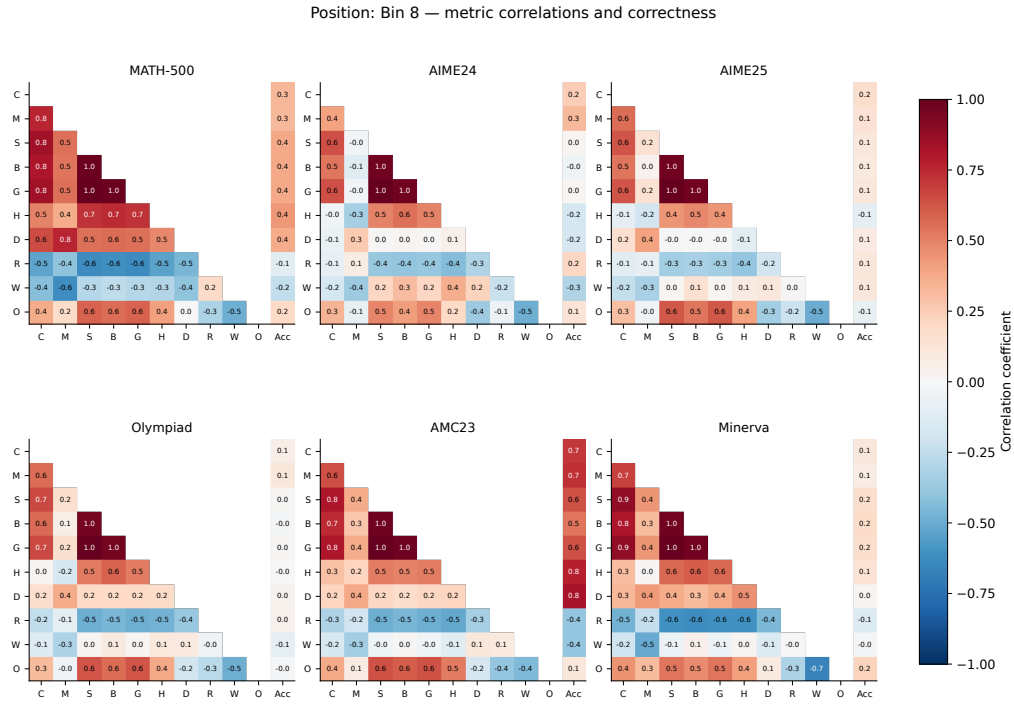


Figure K.18
Position: Bin 8: metric relationships and correctness. Codes are defined in Table K.1.

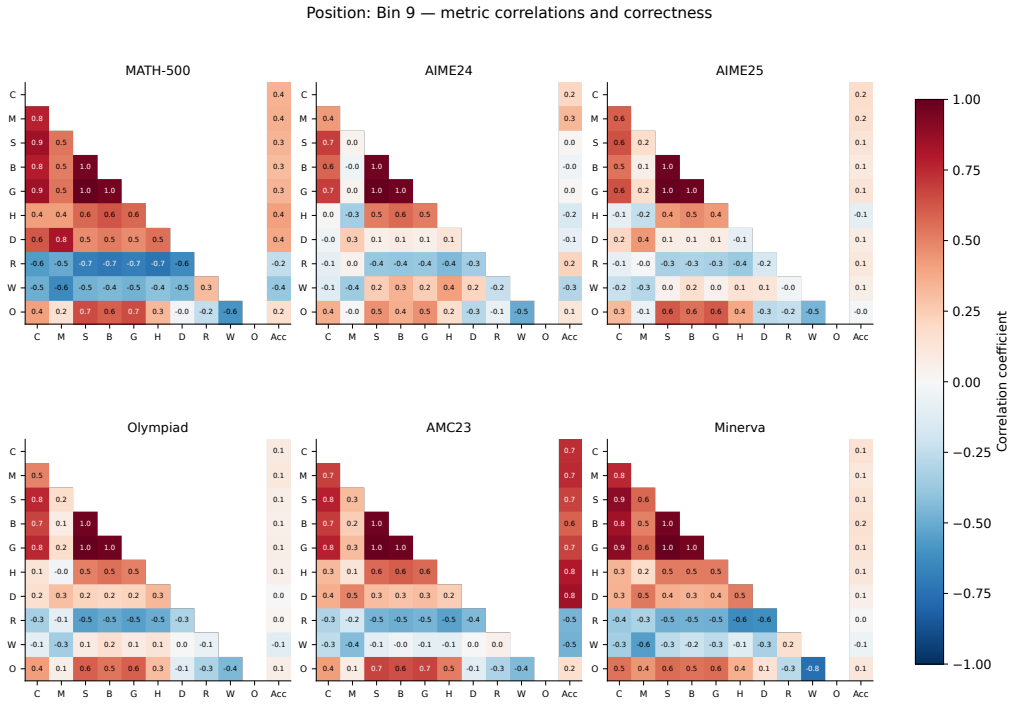


Figure K.19
Position: Bin 9: metric relationships and correctness. Codes are defined in Table K.1.

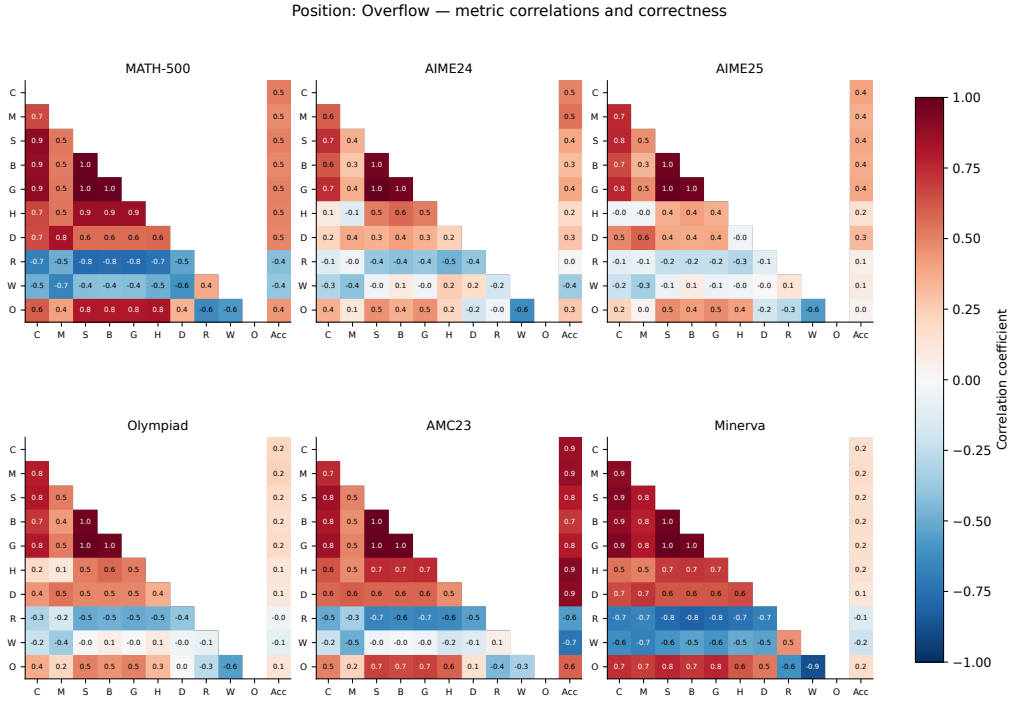


Figure K.20
Position: Overflow: metric relationships and correctness. Codes are defined in Table K.1.

Appendix L: Gemma and GPT-OSS correlation plot atlas

Lower triangles show metric–metric Spearman ρ; Acc shows point-biserial r with correctness. Missing or undefined values are dashes, not zero. Token-bearing counts are upper bounds on pairwise finite counts. Each model has whole-continuation, full-reasoning, seven class, and eleven position scopes. Qwen plots appear in the preceding atlas. These are descriptive screens without significance selection.

Table L.1: Metric key for all models.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap

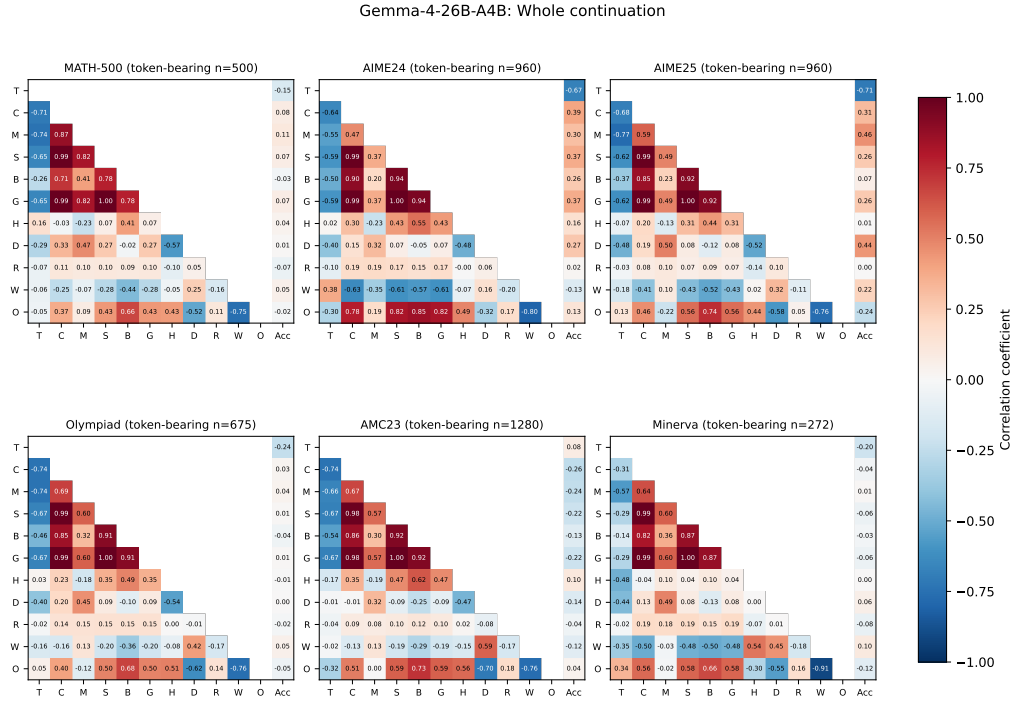


Figure L.1
Gemma-4-26B-A4B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

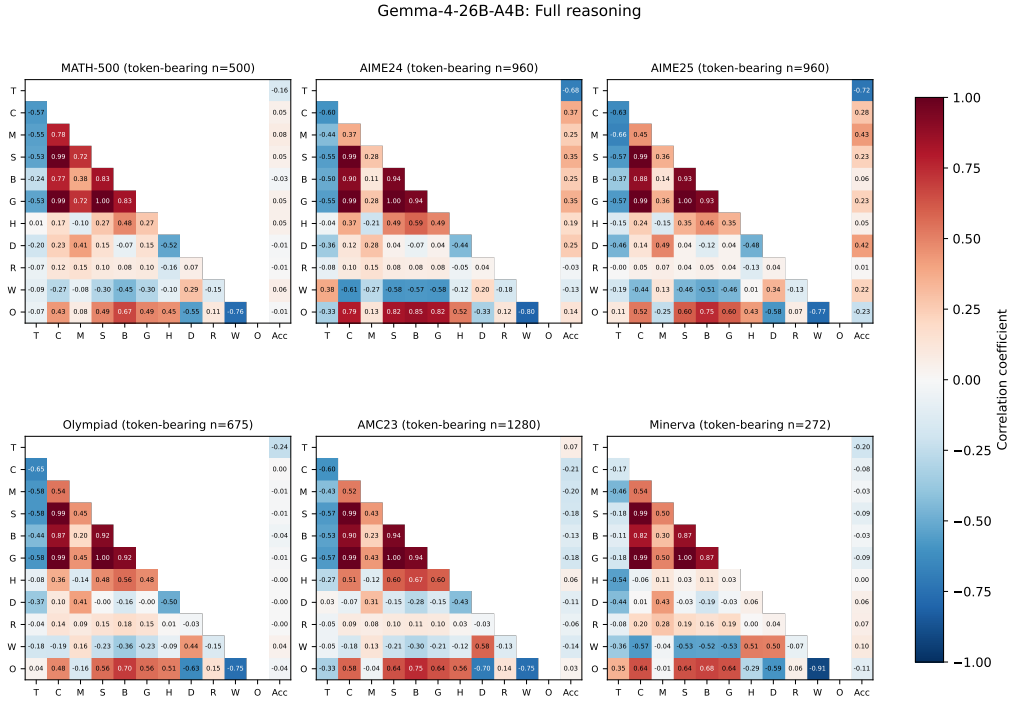


Figure L.2
Gemma-4-26B-A4B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

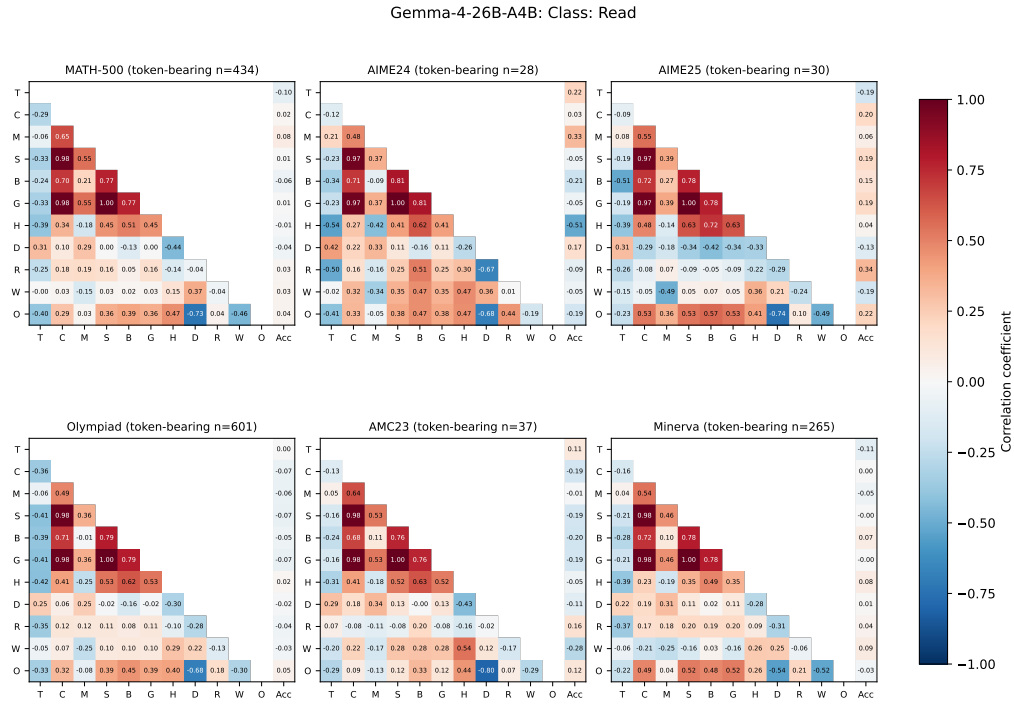


Figure L.3
Gemma-4-26B-A4B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

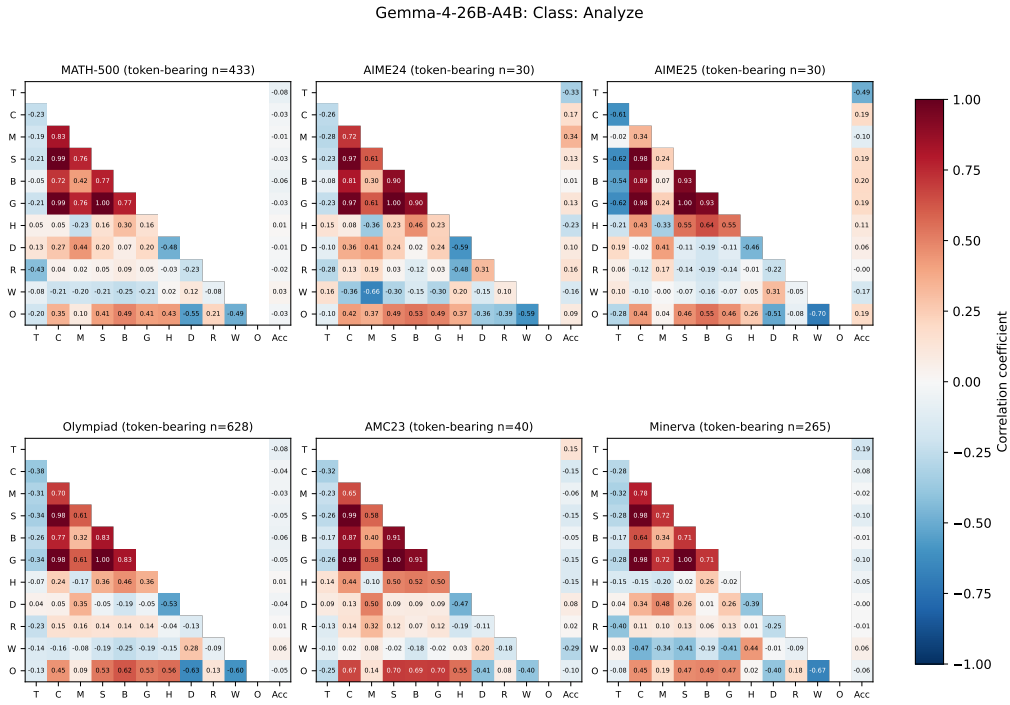


Figure L.4
Gemma-4-26B-A4B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

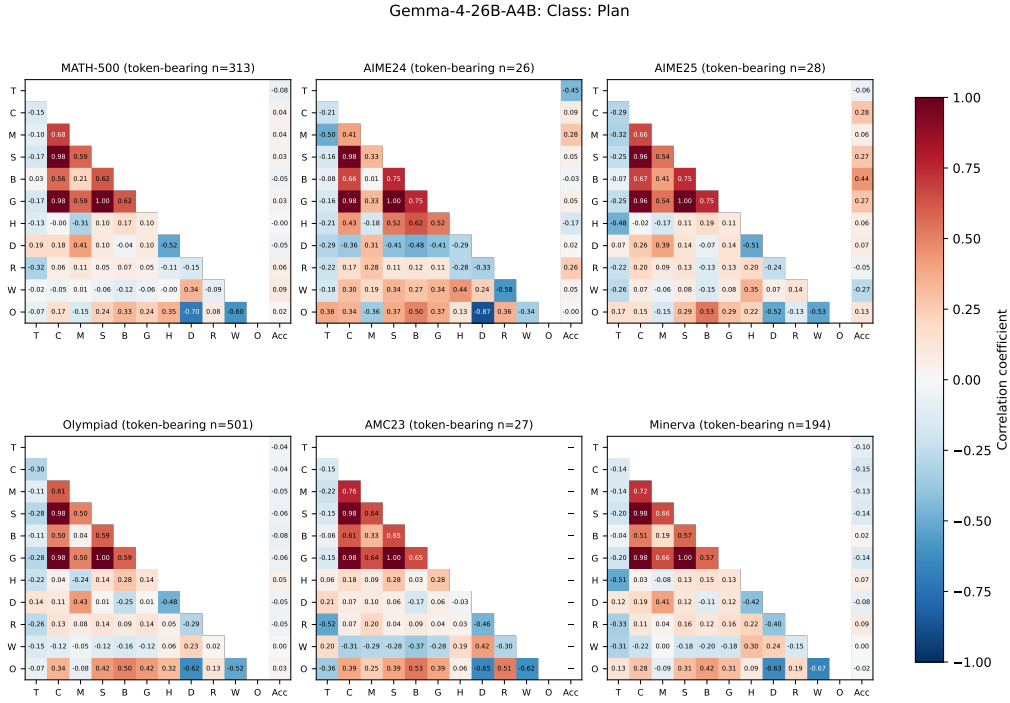


Figure L.5
Gemma-4-26B-A4B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

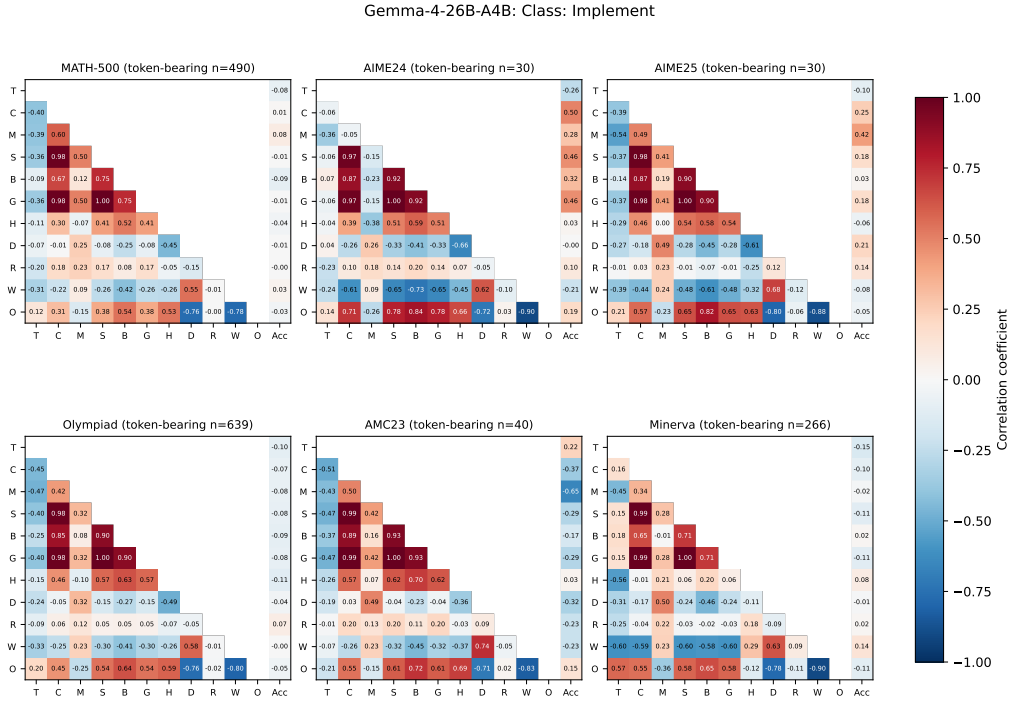


Figure L.6
Gemma-4-26B-A4B / Class: Implement. Metric correlations and correctness; model-specific cohorts and tokenization.

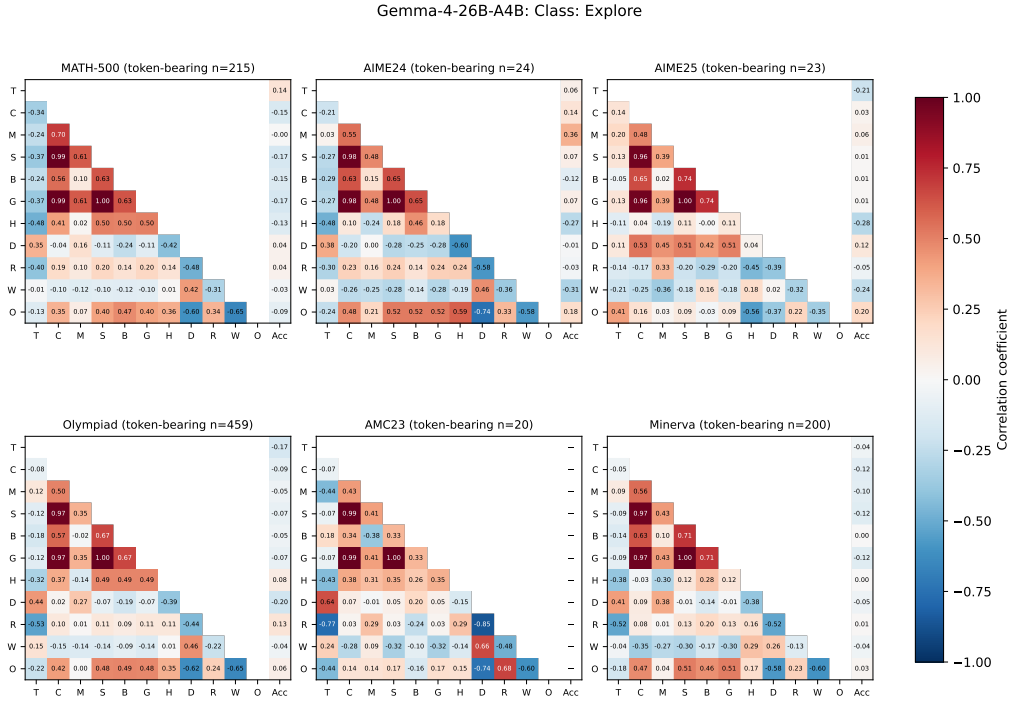


Figure L.7
Gemma-4-26B-A4B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

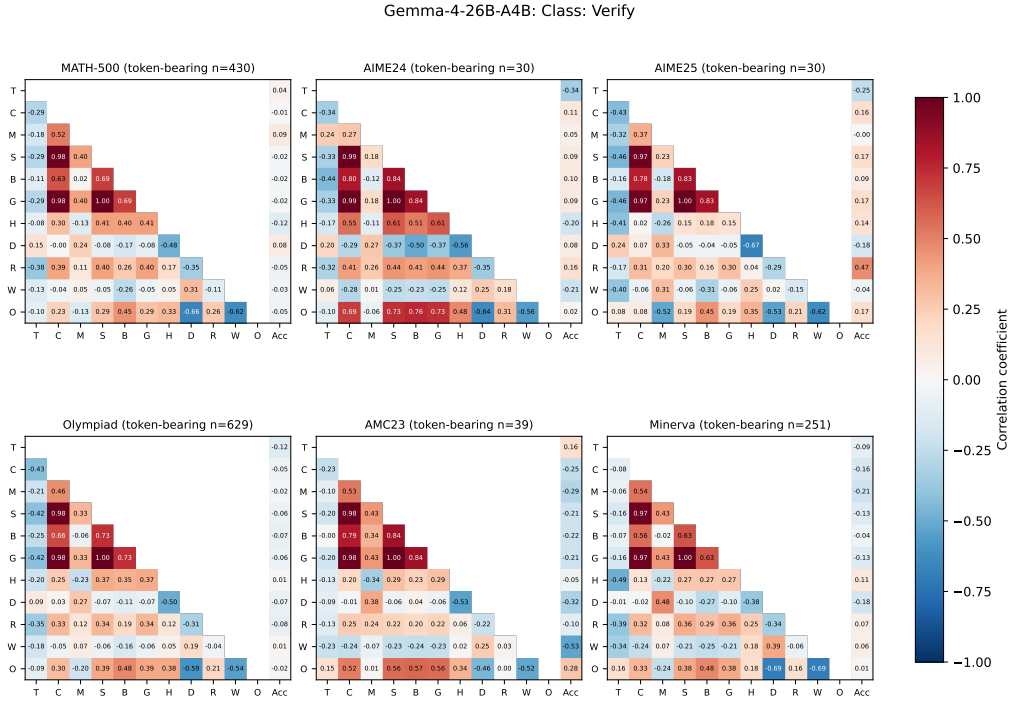


Figure L.8
Gemma-4-26B-A4B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

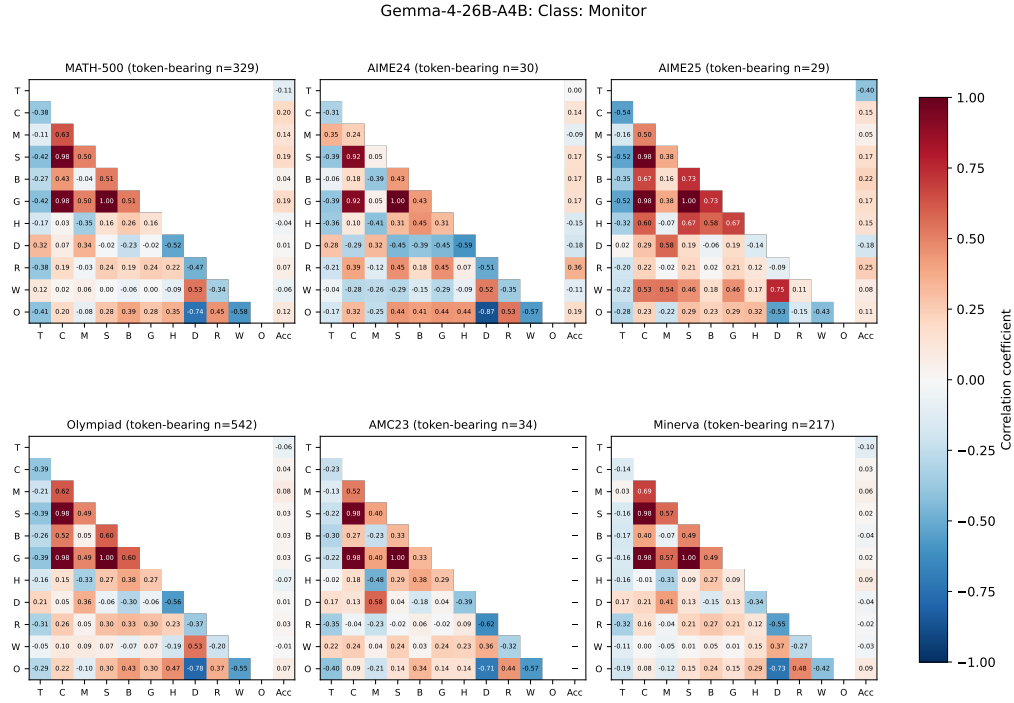


Figure L.9
Gemma-4-26B-A4B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

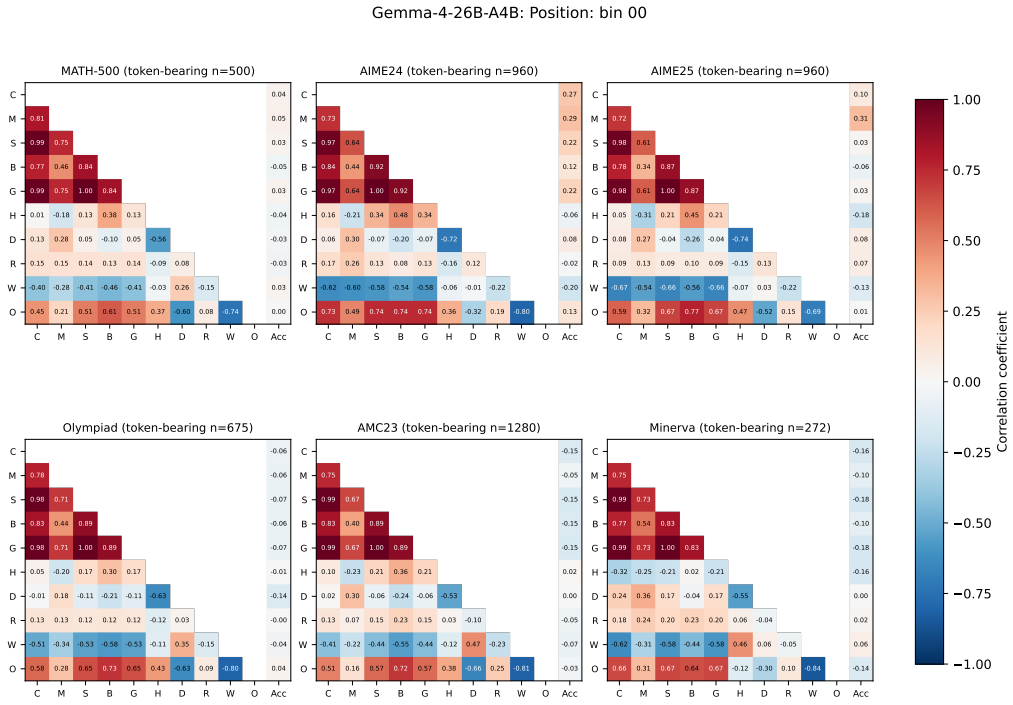


Figure L.10
Gemma-4-26B-A4B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

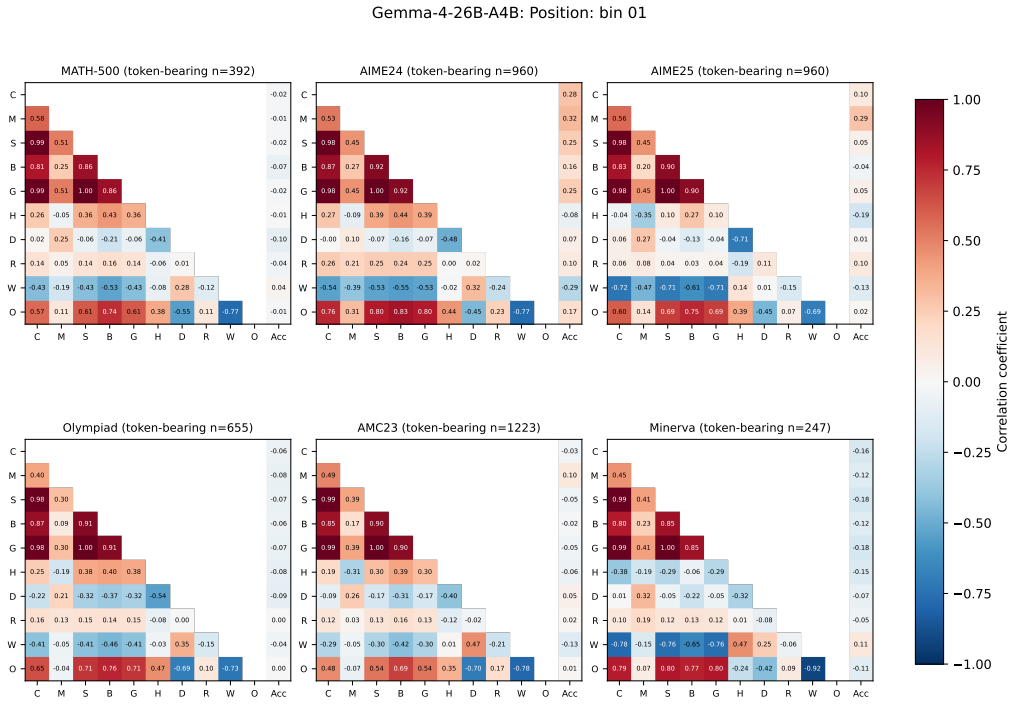


Figure L.11
Gemma-4-26B-A4B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.

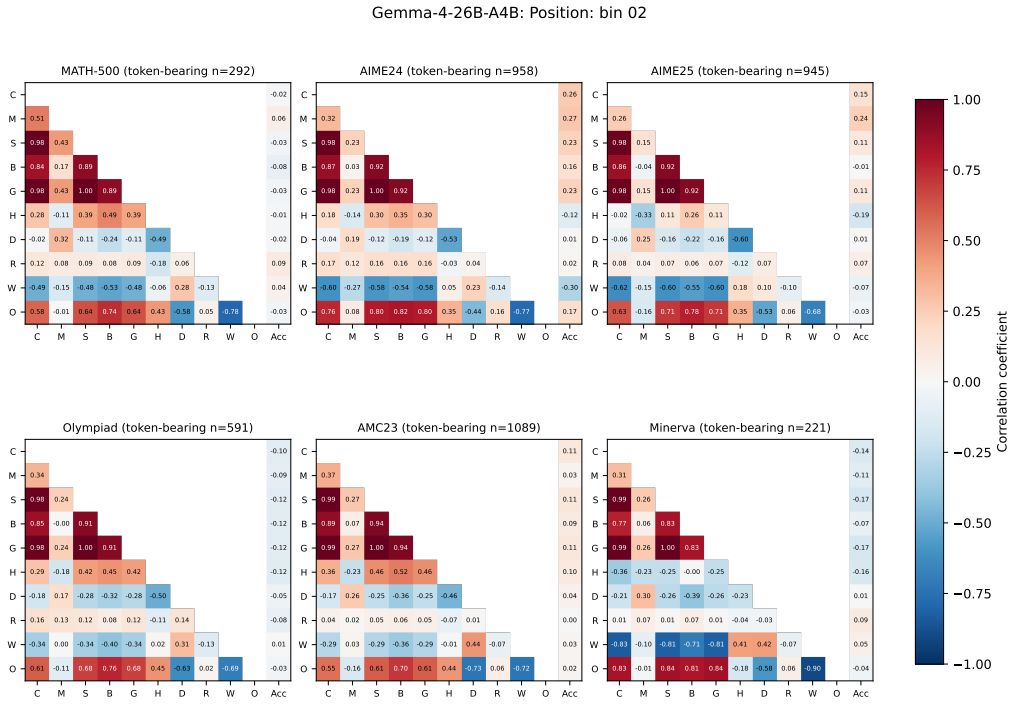


Figure L.12
Gemma-4-26B-A4B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

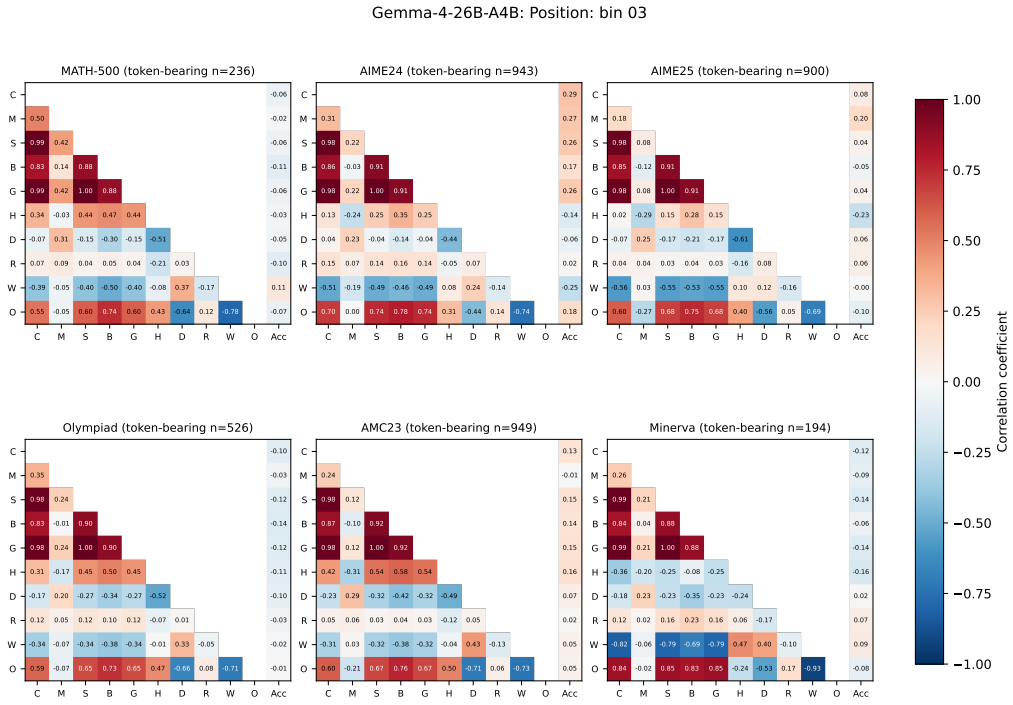


Figure L.13
Gemma-4-26B-A4B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

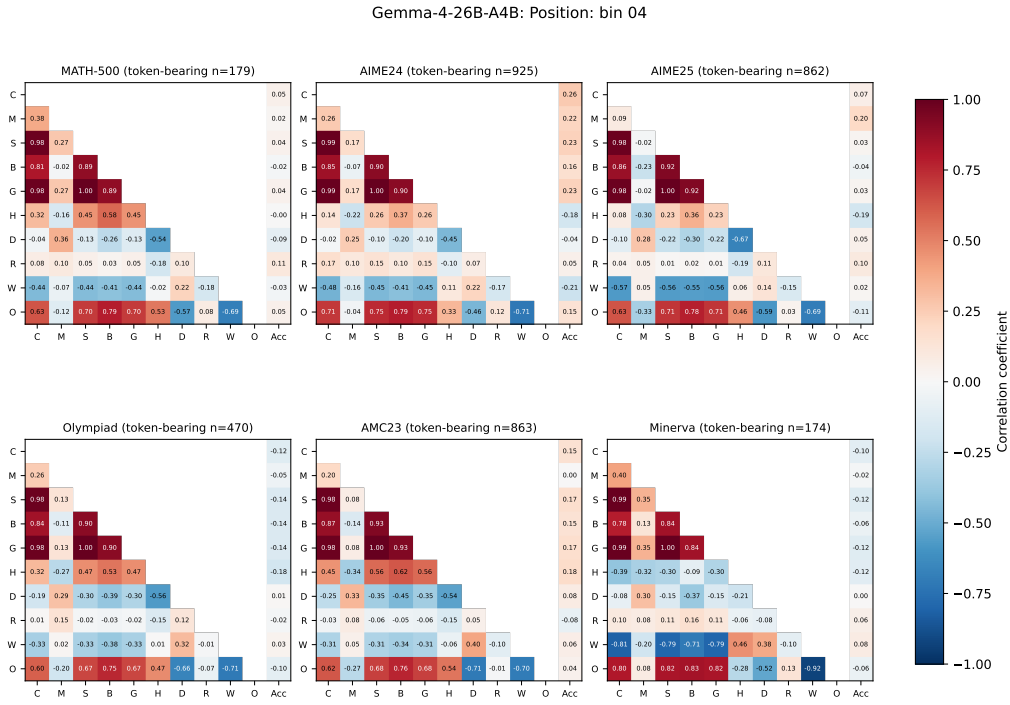


Figure L.14
Gemma-4-26B-A4B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

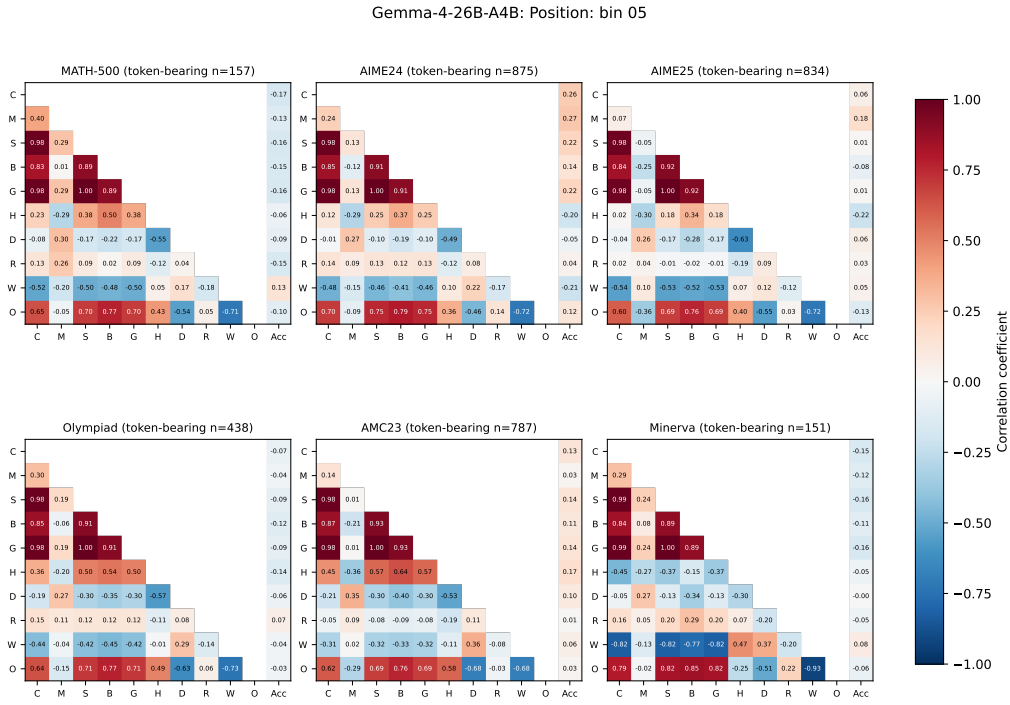


Figure L.15
Gemma-4-26B-A4B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

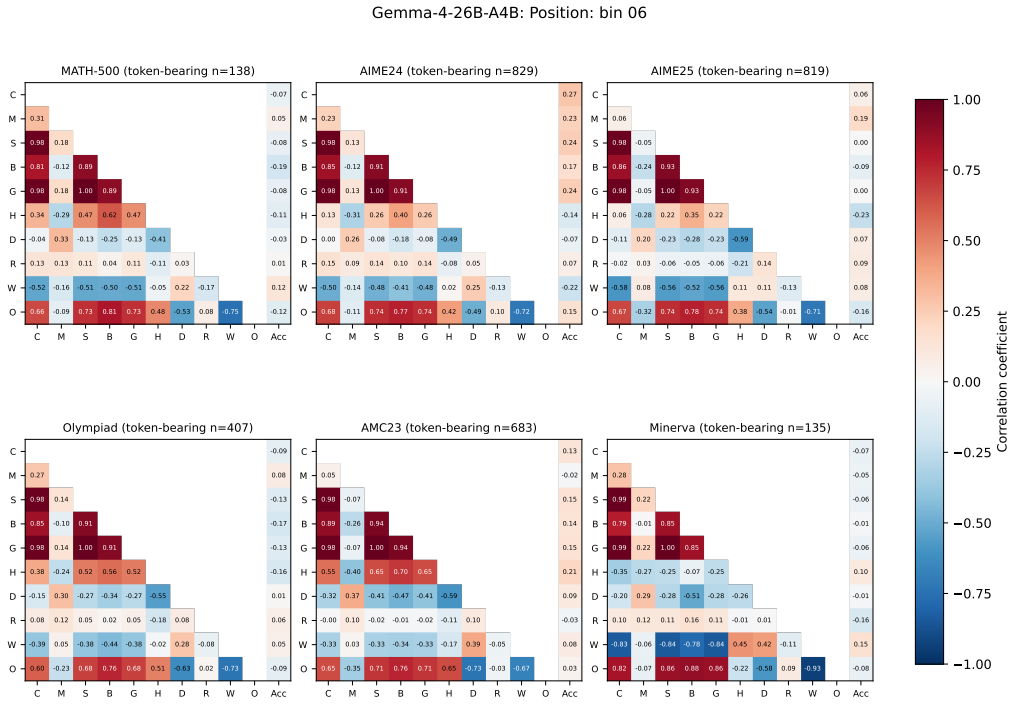


Figure L.16
Gemma-4-26B-A4B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

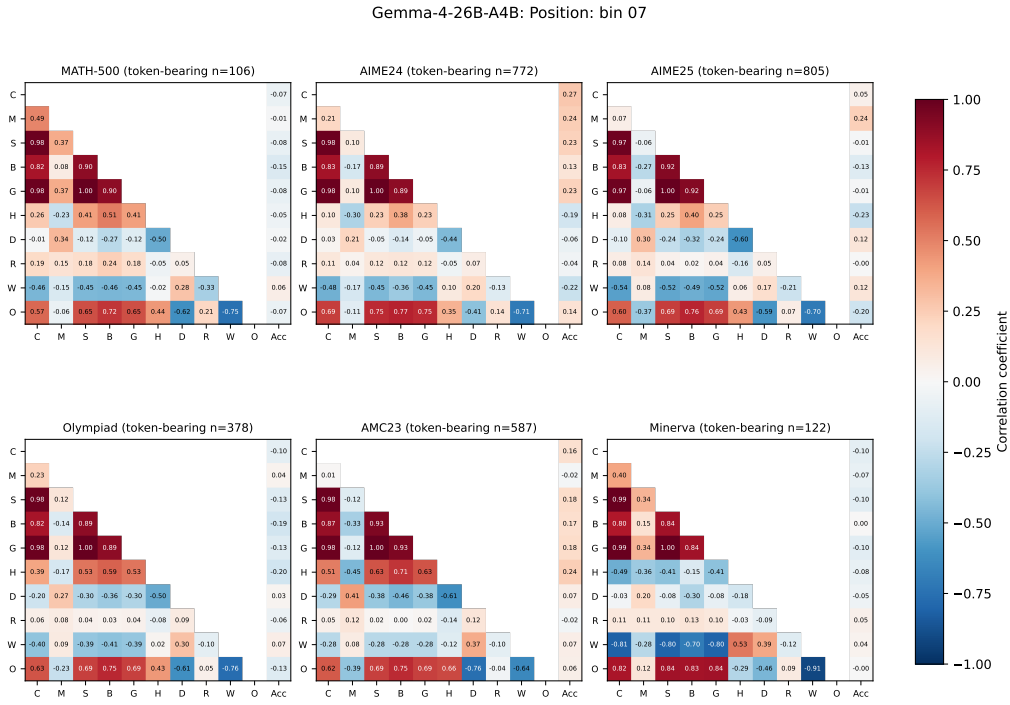


Figure L.17
Gemma-4-26B-A4B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

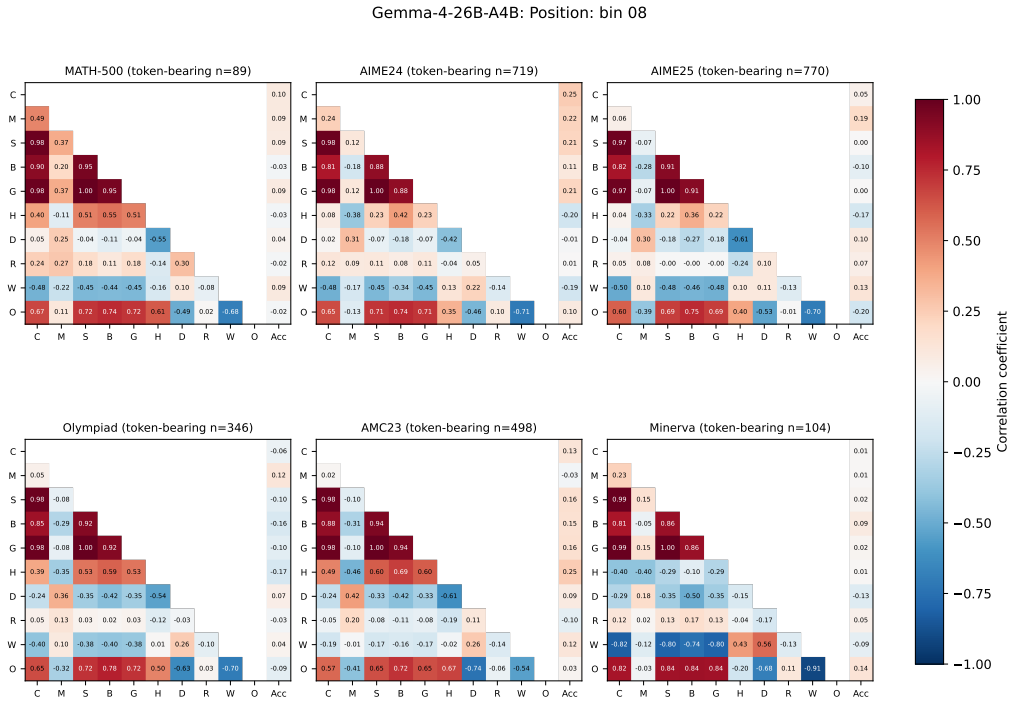


Figure L.18
Gemma-4-26B-A4B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

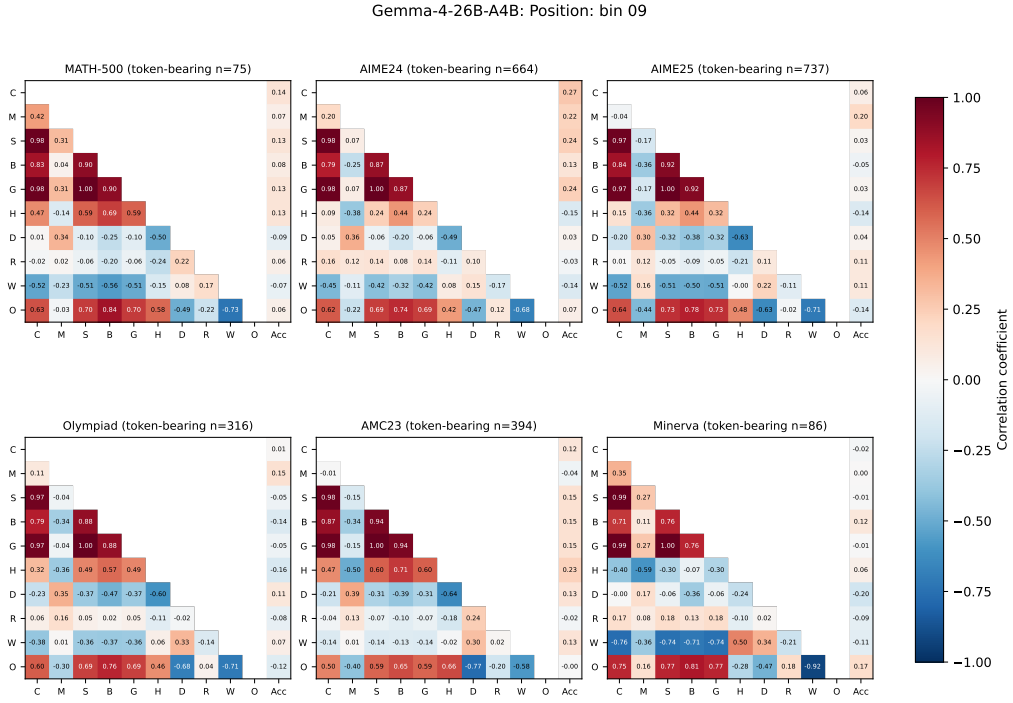


Figure L.19
Gemma-4-26B-A4B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

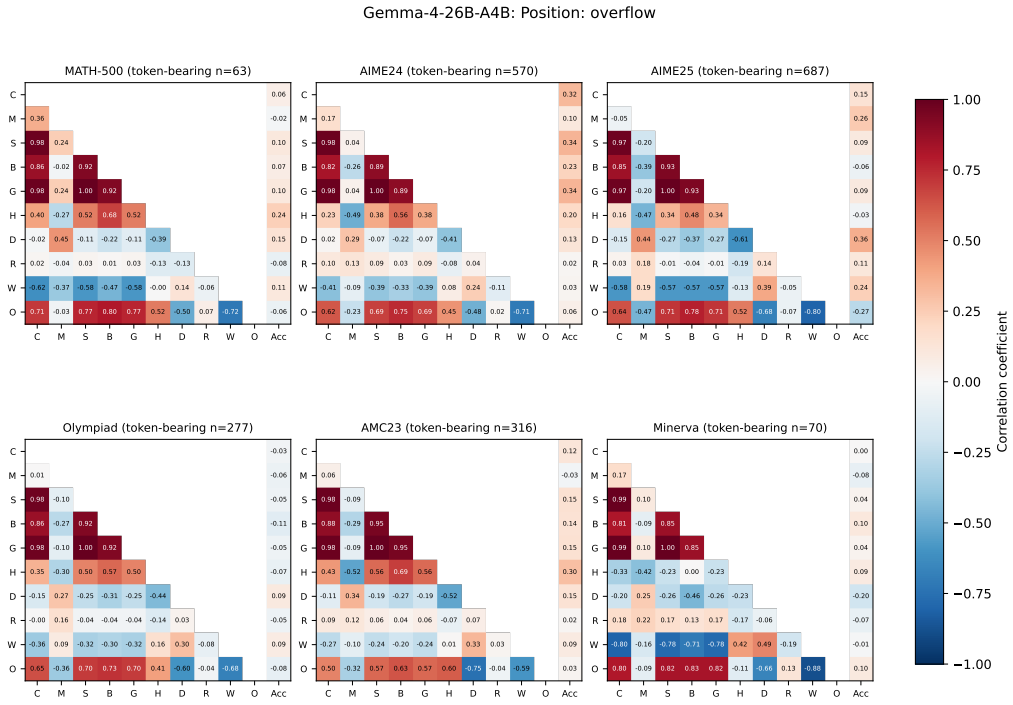


Figure L.20
Gemma-4-26B-A4B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

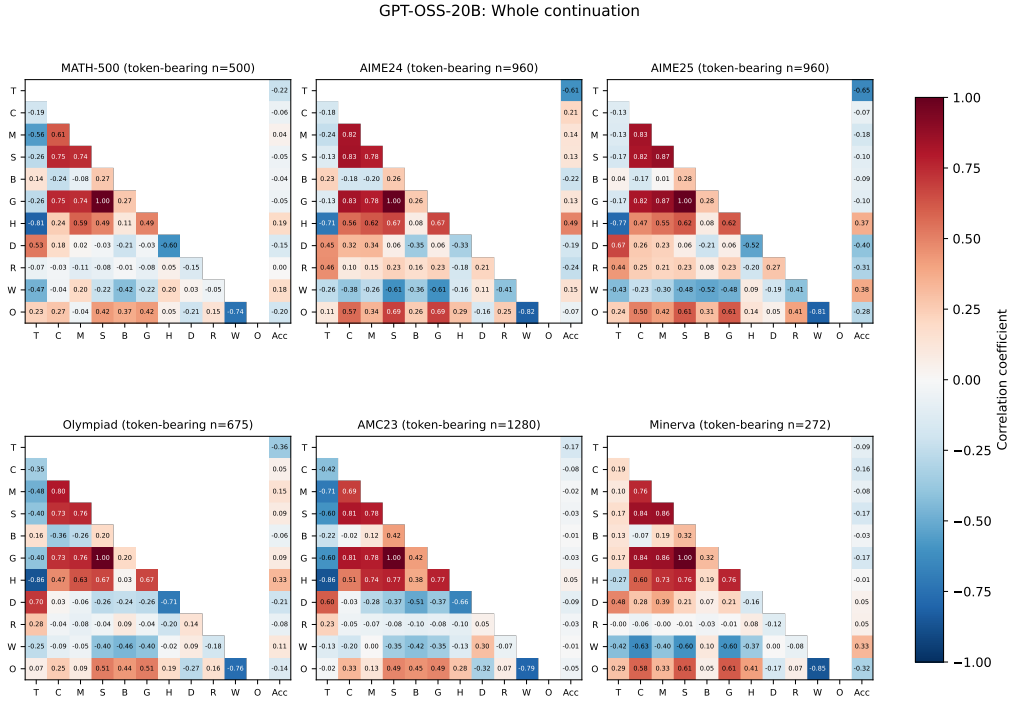


Figure L.21
GPT-OSS-20B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

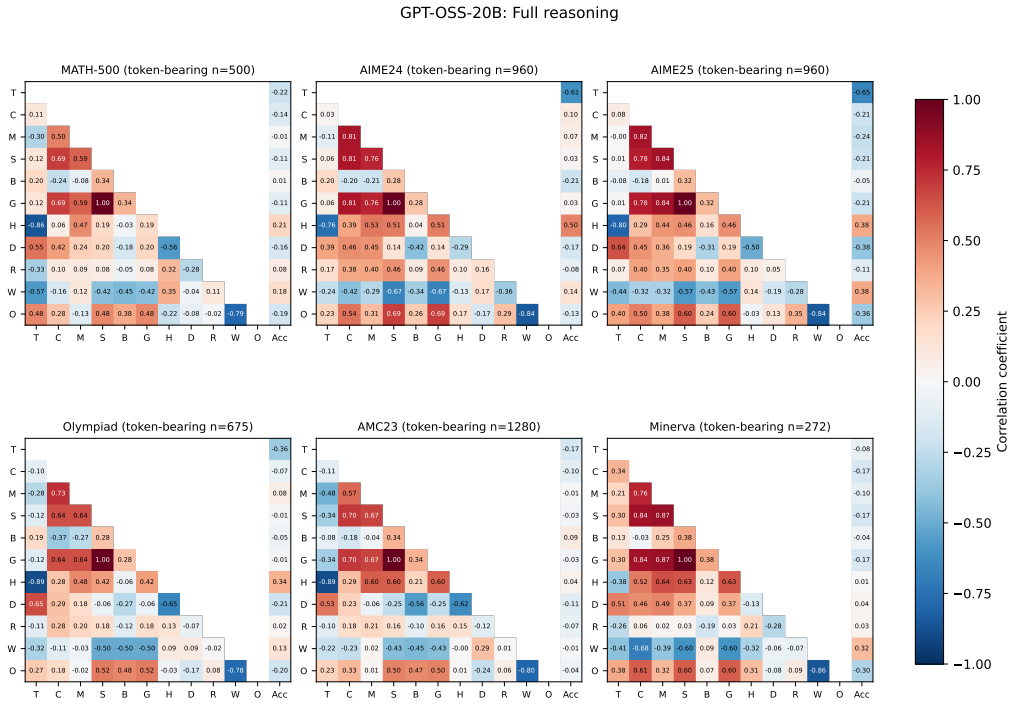


Figure L.22
GPT-OSS-20B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

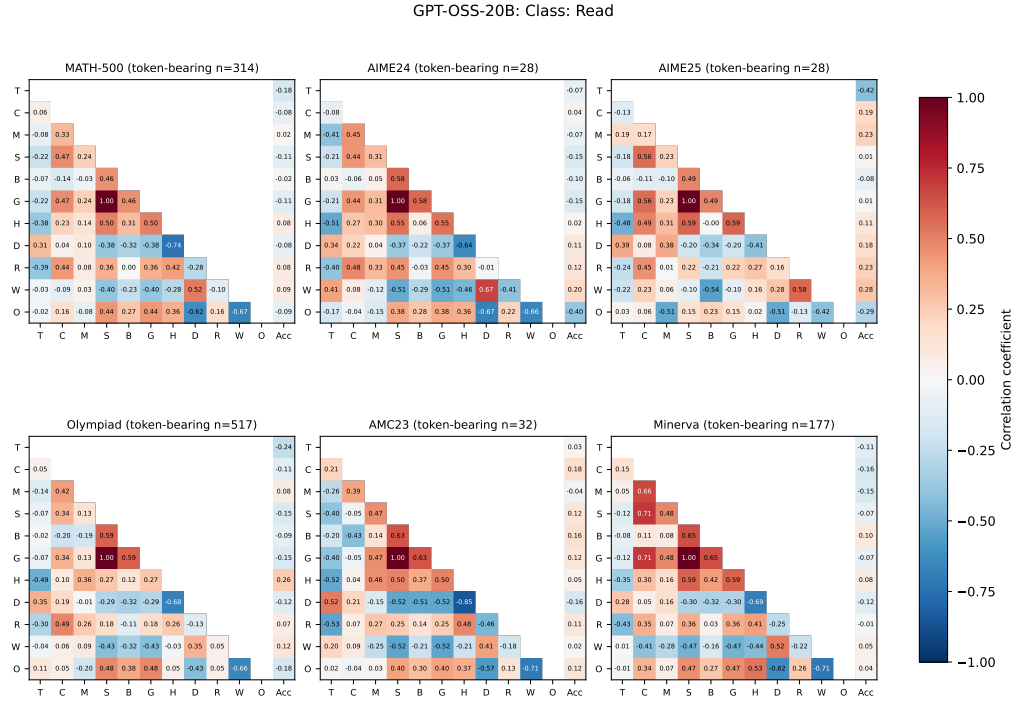


Figure L.23
GPT-OSS-20B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

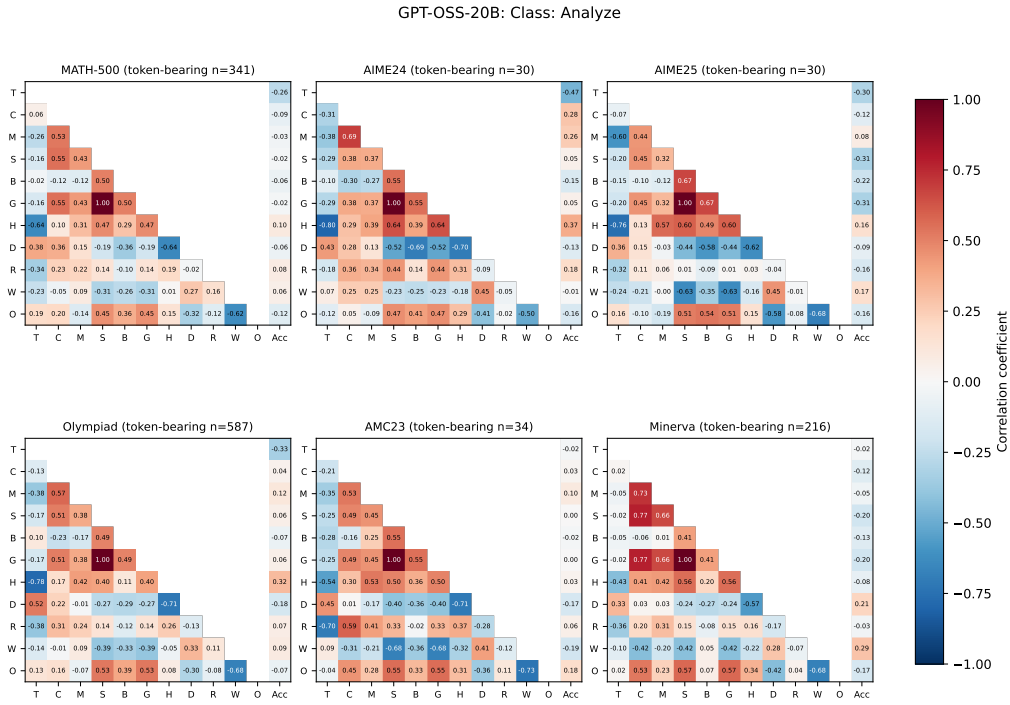


Figure L.24
GPT-OSS-20B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

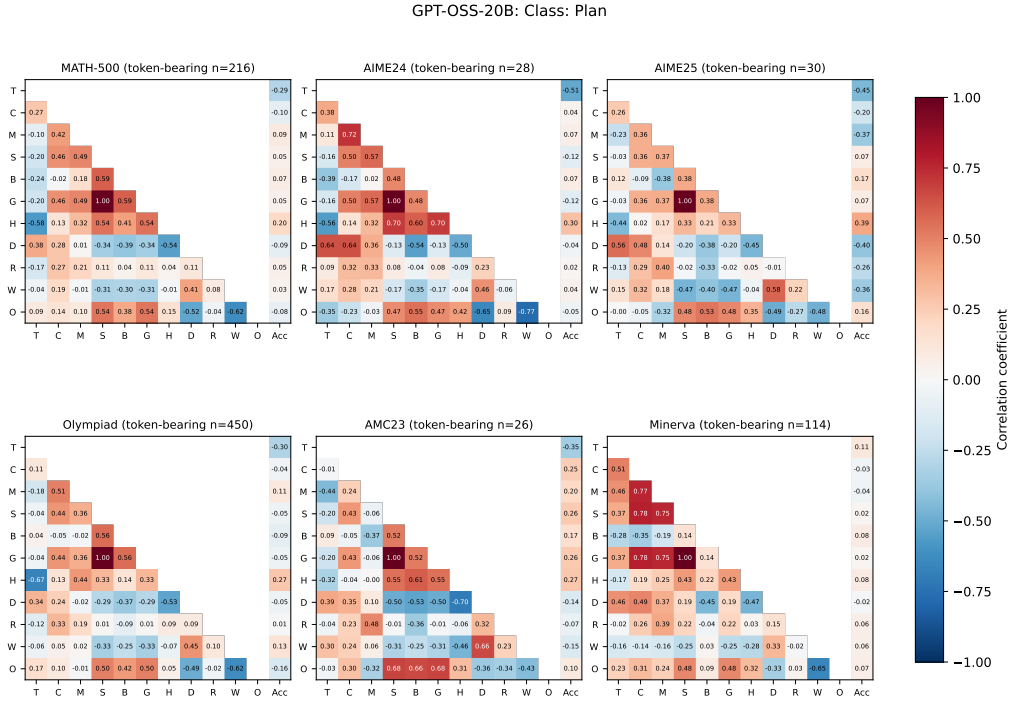


Figure L.25
GPT-OSS-20B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

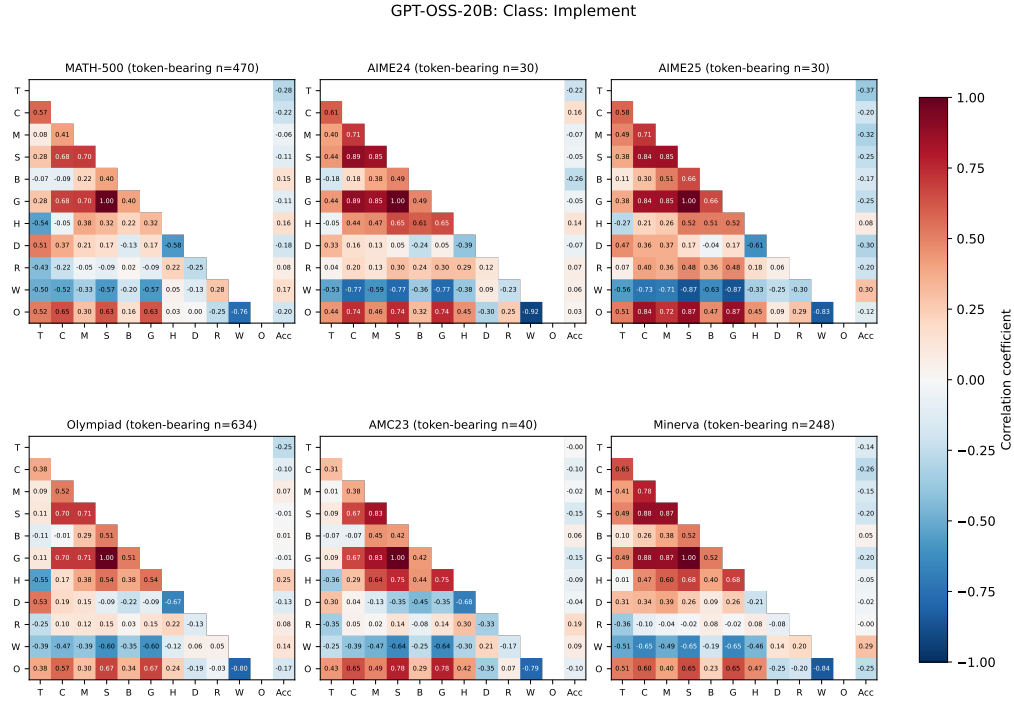


Figure L.26
GPT-OSS-20B / Class: Implement. Metric correlations and correctness; model-specific cohorts and tokenization.

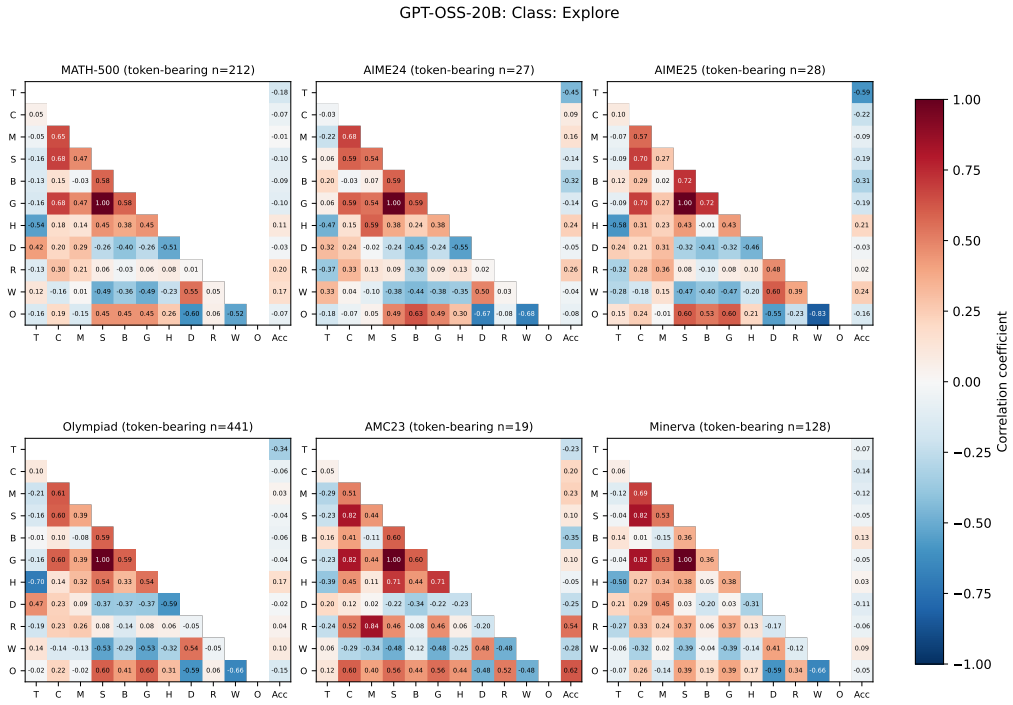


Figure L.27
GPT-OSS-20B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

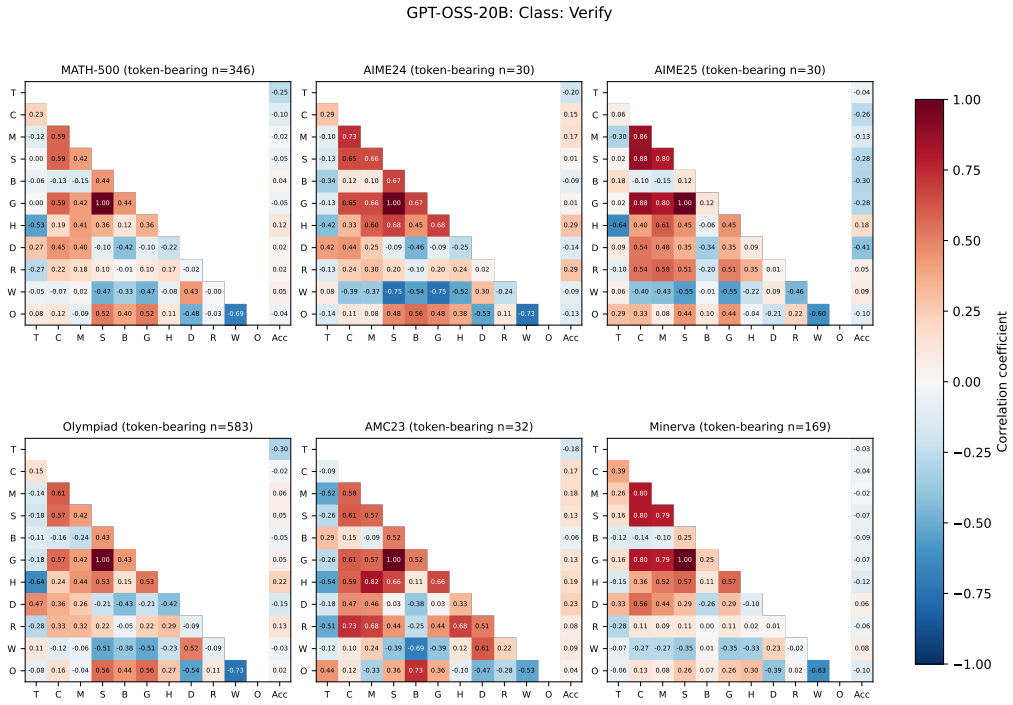


Figure L.28
GPT-OSS-20B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

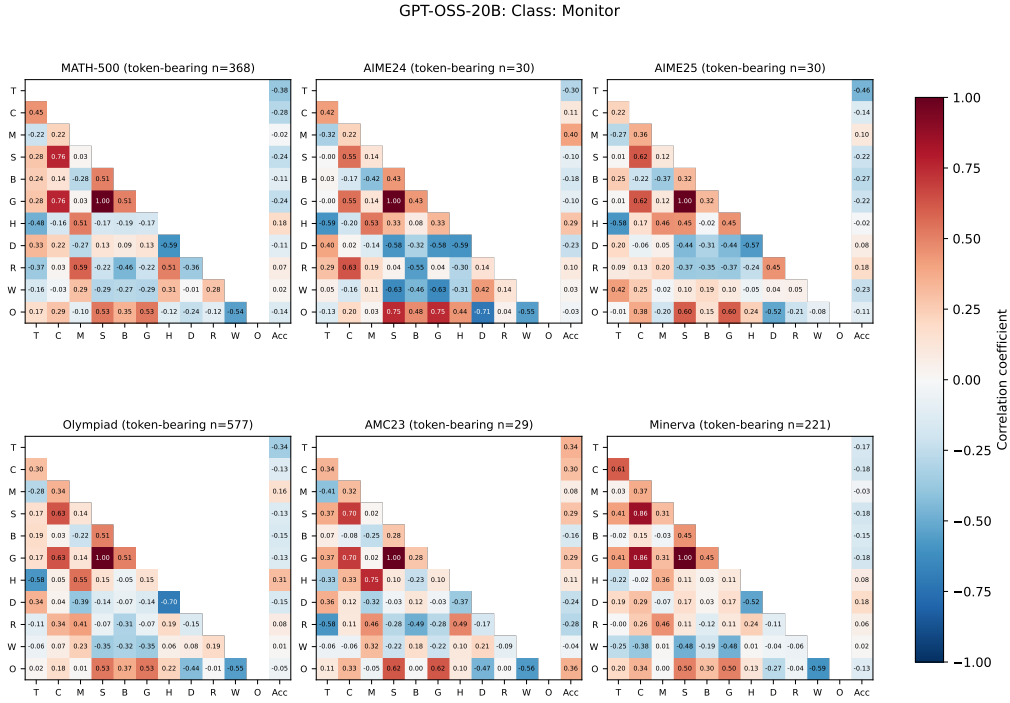


Figure L.29
GPT-OSS-20B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

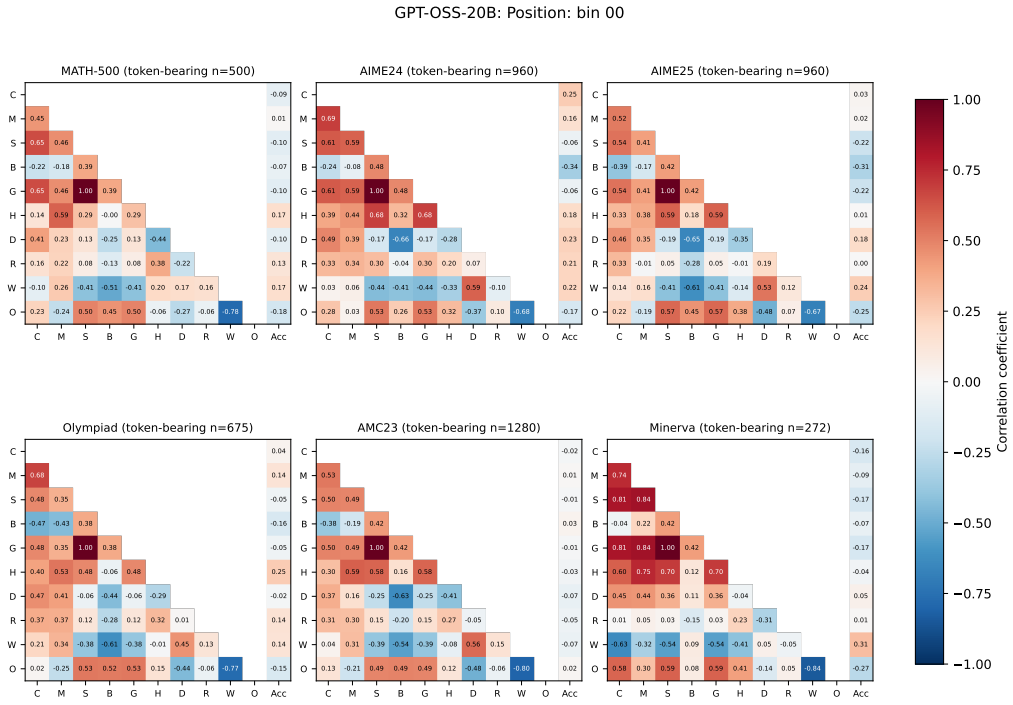


Figure L.30
GPT-OSS-20B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

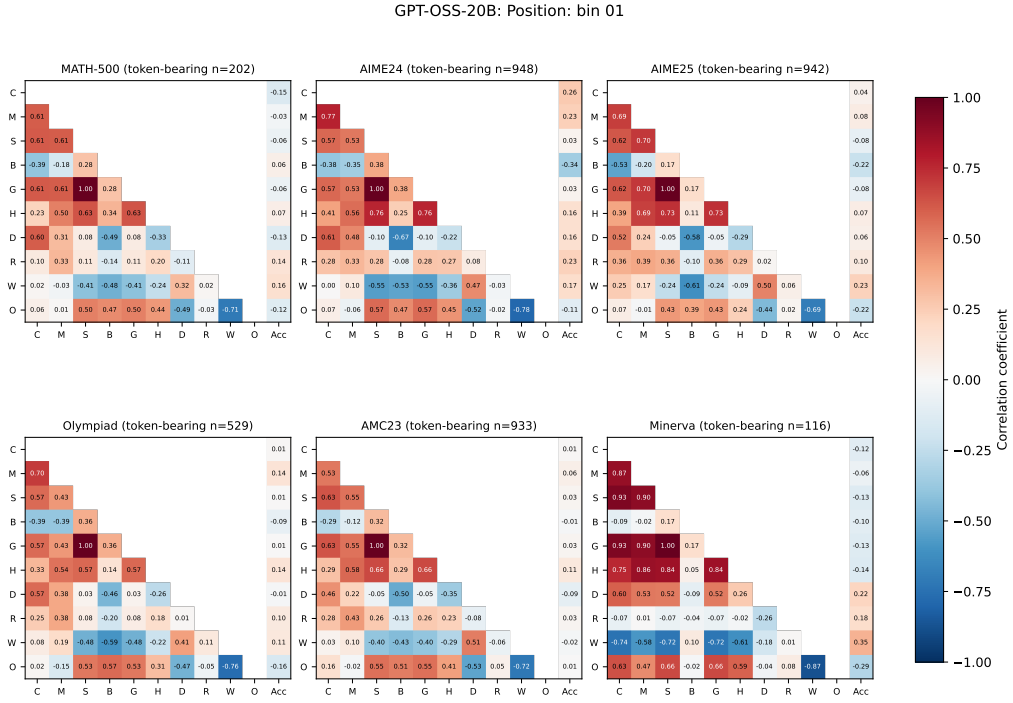


Figure L.31
GPT-OSS-20B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.

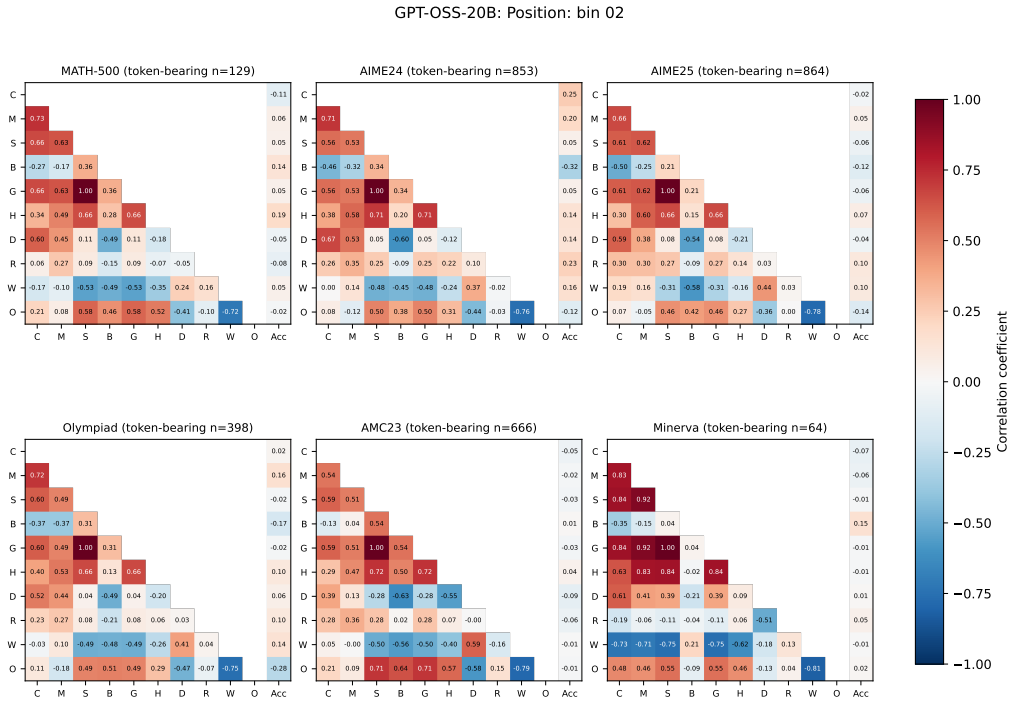


Figure L.32
GPT-OSS-20B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

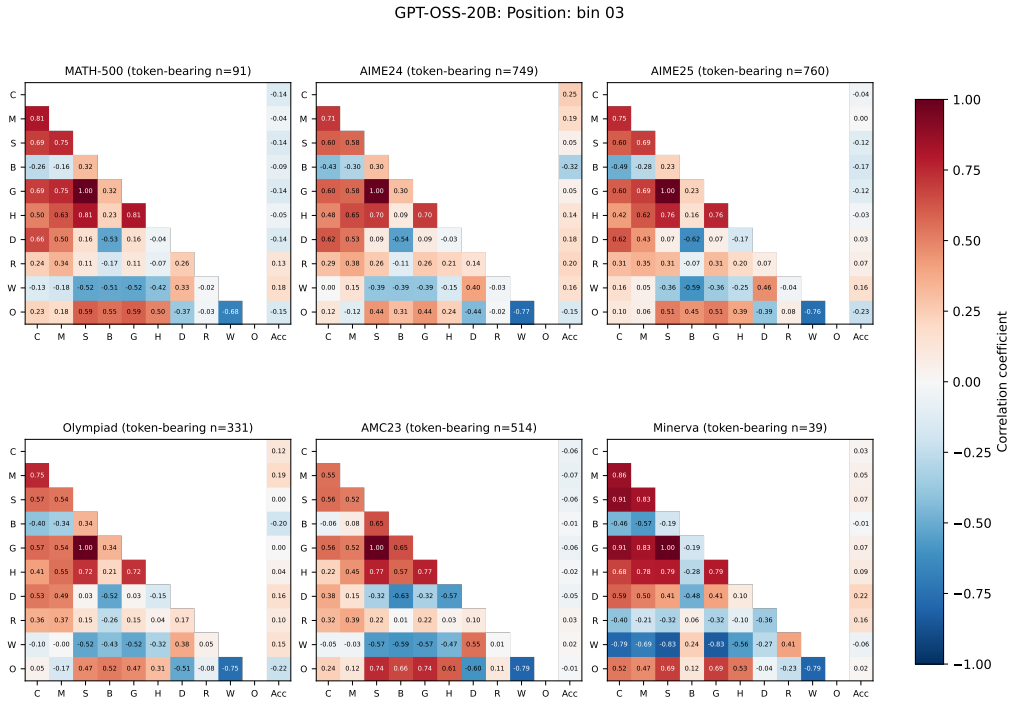


Figure L.33
GPT-OSS-20B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

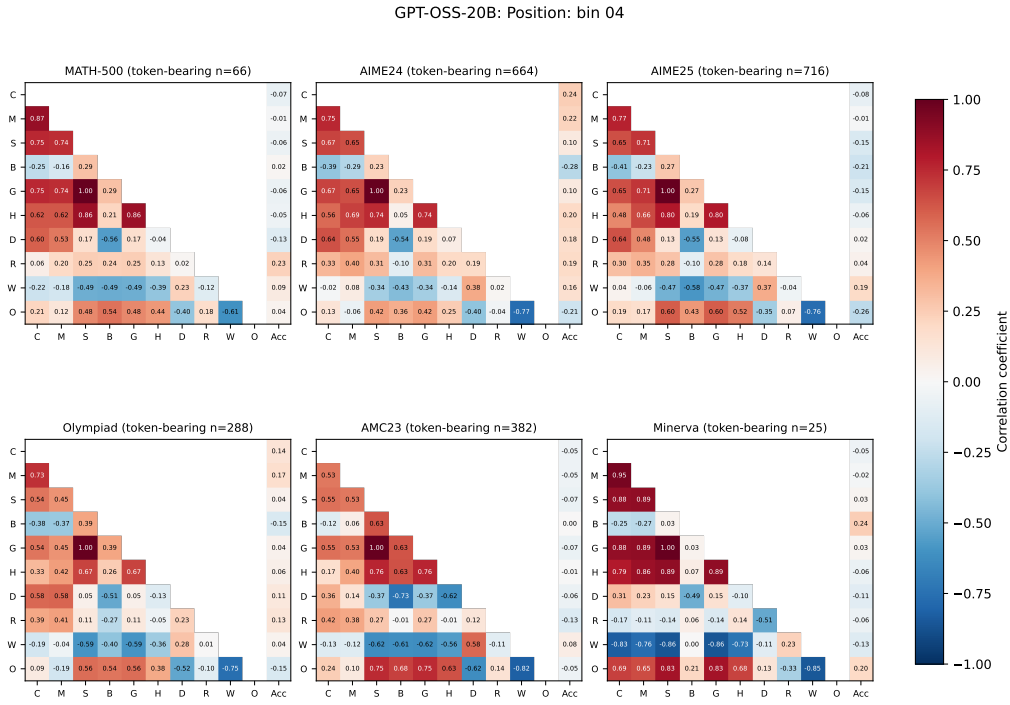


Figure L.34
GPT-OSS-20B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

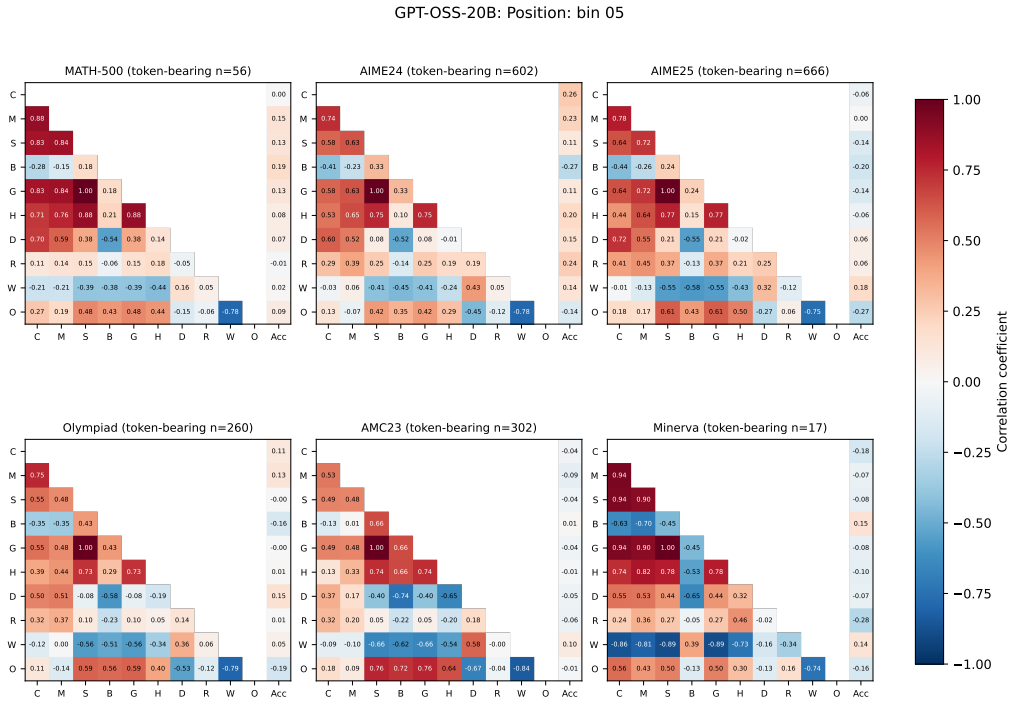


Figure L.35
GPT-OSS-20B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

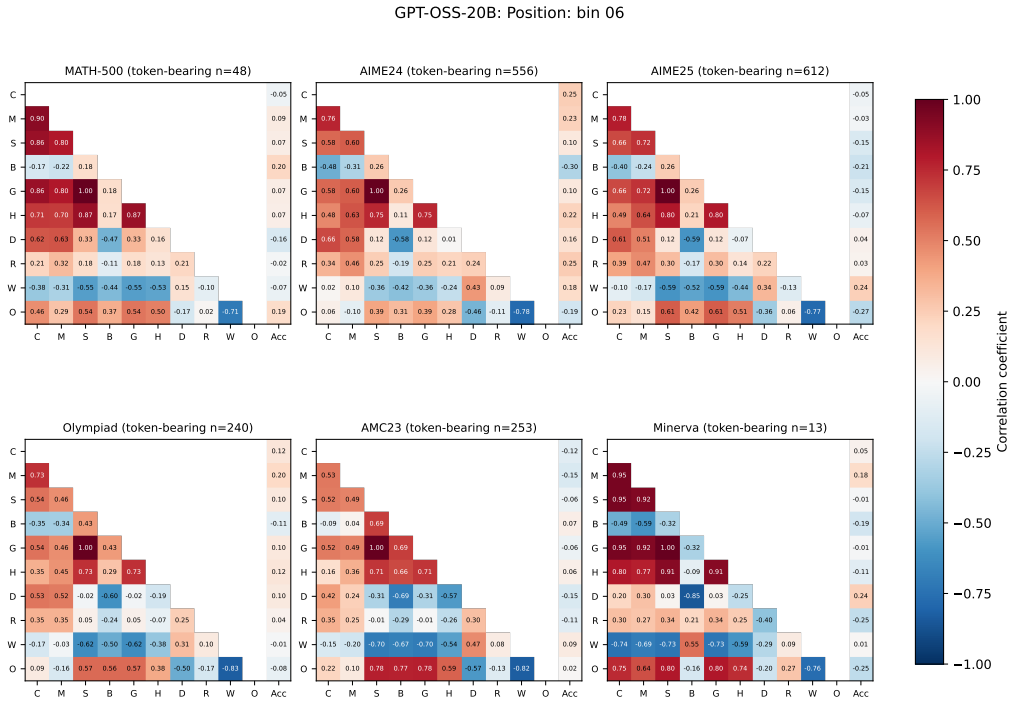


Figure L.36
GPT-OSS-20B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

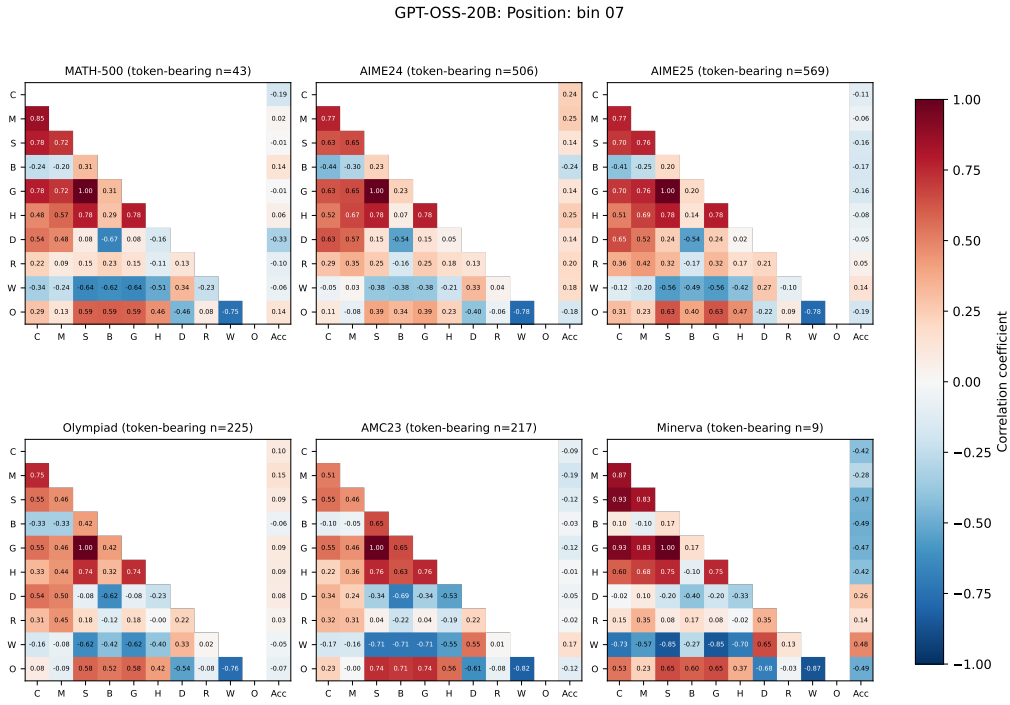


Figure L.37
GPT-OSS-20B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

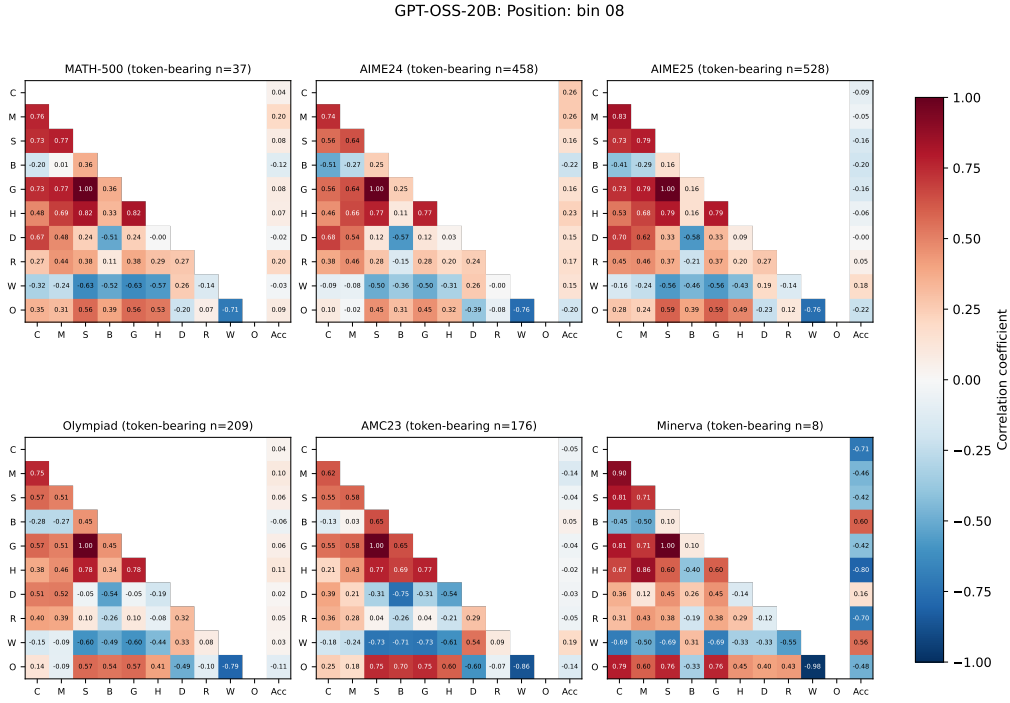


Figure L.38
GPT-OSS-20B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

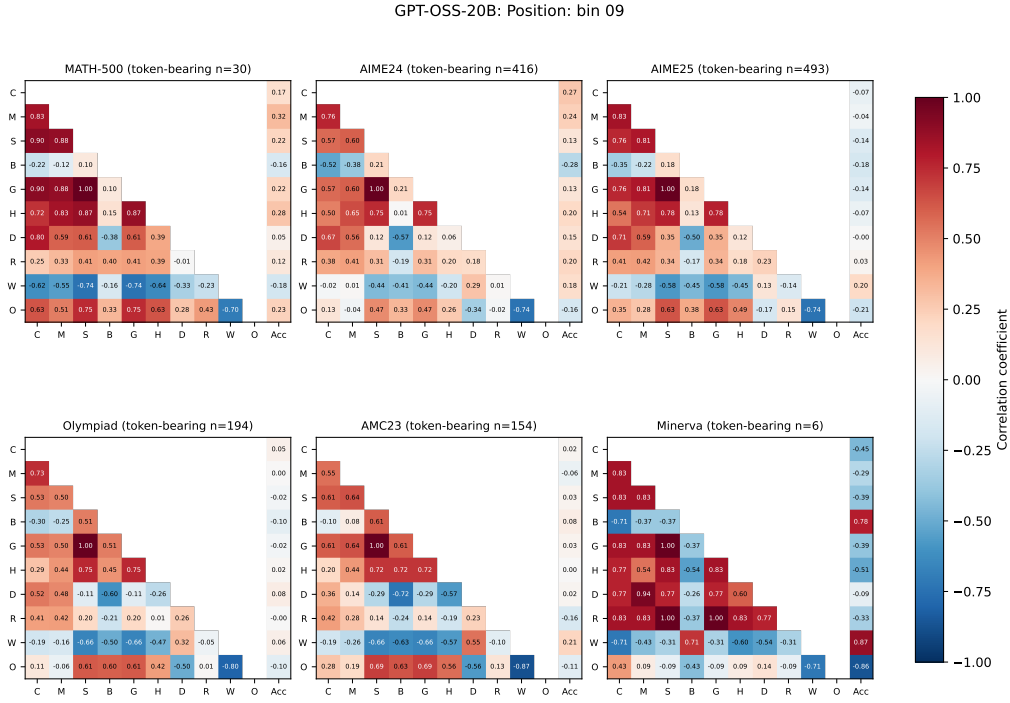


Figure L.39
GPT-OSS-20B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

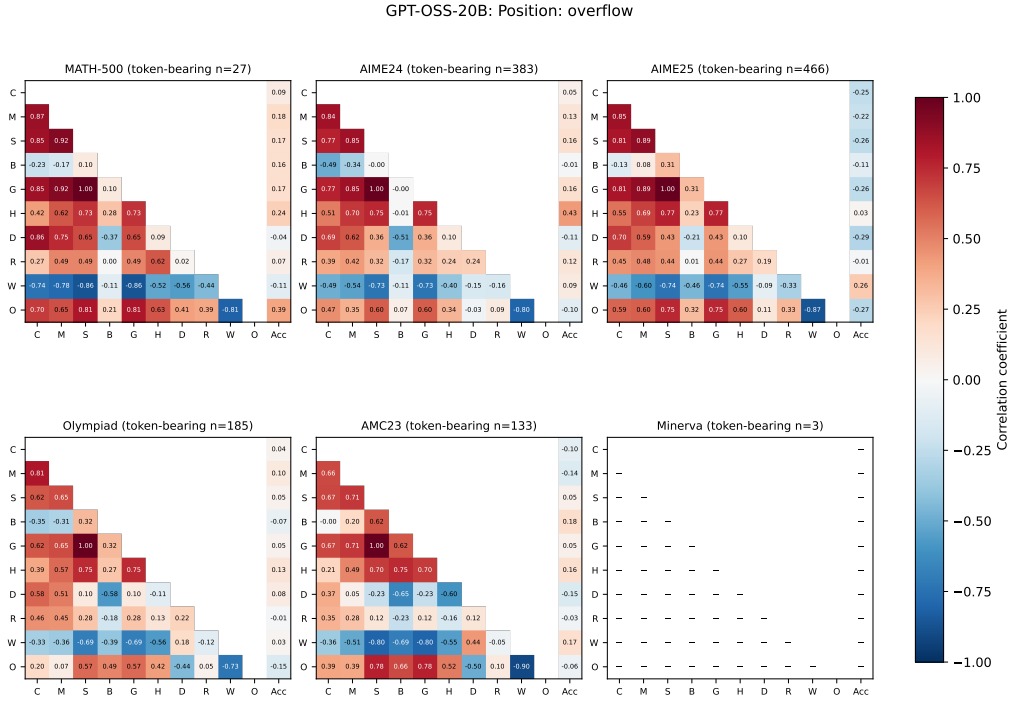


Figure L.40
GPT-OSS-20B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

Campaign v3 observed closeout.. The seven-model terminal report retains 243747 configured cells. Scientific coverage is incomplete. Failed and unavailable cells remain explicit. Paired accuracy, token-cost and causal contrasts, unknown bounds, support, and source-bound PDF/SVG figures are saved with the campaign. Memory values require matching full-context physical qualification; an empty panel establishes no measured savings. Intervals crossing zero do not establish equivalence.

Appendix M: Continuation launch observation, 14 October 2026

This dated observation describes implementation and ownership. It does not add an intervention result to the findings above. The separately sealed `research-campaign-v3-20261014-r5-r2` continuation preserves the seven models, original evaluation families, decoding, controls, qualification tolerances and registered comparison families. Its source-seal prefix is `e5fdce1548eda0ac`.

All 102 regression contracts passed. A replay of the actual imported state certified seven saved-sequence measurement manifests, 69,459 distinct non-REAP memory executions and 346 compact count shapes, in addition to seven native qualification shapes. It used the real CPU implementations for identity joins, strict scoring, original native-context samples and forecasting. The graph traversal prohibited scheduler submissions. The original failed import check and interrupted validation remain preserved; one bounded deterministic recovery completed successfully. Their combined validation expenditure was 4.612 local accounting hours. No scientific generation or reader draw was repeated.

The verified import retains all seven repaired derivations and the complete 81-pair GPT and 77-pair Nemotron coordination banks. The other five models require 741 original coordination pairs. Four compatible REAP qualifications are reused; the three affected verdicts require within-execution adjudication on their original saved sequences and tolerances. Routing arrays, weights, norms and reconstruction diagnostics must describe the same execution. Cross-execution routing equality is not a qualification requirement.

In the first subsequent engineering closeout, GPT's native memory fixture qualified at the full 131,072-token context. Its corrected REAP reconstruction still exceeded the unchanged tolerances and remains a method limitation; its own router bank is usable. Nemotron's corrected REAP reconstruction qualified. These verdicts describe measured implementation fidelity, without establishing an expert-removal accuracy result or a steering effect.

Import job 60454550 completed successfully. Replacement October 15 snapshot 60454552 and controller 60454554 with both failure owners 60454557 and 60454559 were accepted before predecessor snapshot 60226466 was retired with zero runtime. Subsequent work belongs to the immutable submission and handoff journals. Non-REAP memory qualification and retained pricing proceed independently of REAP collection and steering calibration. Each retained observation is scored and exported locally before the next shape advances.

Full production still requires complete measured admission, including all generation and reader roles, scoring, preparation, coordination, storage and bounded recovery. The 25% margin stays inside available entitlement under the original 42,050-hour ceiling; WORK and private FAST keep separate 10-GiB reserves. An unaffordable complete study stops before production with its shortfall. At this launch observation, new accuracy–resident-memory curves and steering effects remain unmeasured. Durable reports, scientific completion, a verified manuscript/PDF and delivery to each remote destination are separate obligations.